

On the Cartan Graphs of Nichols Algebras over Coquasi-Hopf Algebras

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Abstract

In this paper, we continue the study of the reflection theory of Nichols algebras over coquasi-Hopf algebras with bijective antipode. We prove that for a tuple of finite-dimensional simple Yetter-Drinfeld modules admitting all reflections, the associated semi-Cartan graph is actually a Cartan graph. Furthermore, we provide equivalent conditions for the finite-dimensionality of the corresponding Nichols algebra. Finally, we show that such a Cartan graph is indeed invariant under specific braided monoidal equivalences. As an application, we examine Nichols algebras of diagonal type over coquasi-Hopf algebras, proving that they yield isomorphic Cartan graphs originating from Nichols algebras of diagonal type over Hopf algebras.

1 Introduction

1.1 Notation

Throughout this paper, unless otherwise stated, we adopt the following notation.

- $\mathbb{I}$: A finite index set $\{1, 2, \dots, \theta\}$ for an integer $\theta \geq 2$.
- H : Coquasi-Hopf algebra with bijective antipode. ${}^H_H\mathcal{YD}$: The category of Yetter-Drinfeld modules over H , sometimes denoted by $\mathcal{C}$ for simplicity.
- Let $M = (M_1, M_2, \dots, M_\theta)$ be a θ -tuple, where $M_i \in {}^H_H\mathcal{YD}$ is finite-dimensional and simple for each $i \in \mathbb{I}$. By abuse of notation, we define $\mathcal{B}(M) := \mathcal{B}(M_1 \oplus M_2 \cdots \oplus M_\theta)$.
- $\mathcal{G}$: A semi-Cartan graph. $\mathcal{W}(\mathcal{G})$: The Weyl groupoid over $\mathcal{G}$.
- $\mathbb{k}$: An algebraically closed field of characteristic zero.

1.2 From Hopf algebras to coquasi-Hopf algebras

The classification of finite-dimensional pointed Hopf algebras has been a central theme in Hopf algebra theory, largely driven by the lifting method proposed by Andruskiewitsch and Schneider [2, 3, 5]. A cornerstone of this program is the study of Nichols algebras. In the case of diagonal type, the complete classification was achieved by Heckenberger through the theory of Weyl groupoids and arithmetic root systems [11, 12].

Parallel to the developments in the Hopf setting, the theory of coquasi(quasi)-Hopf algebras, introduced by Drinfeld in the context of the Knizhnik-Zamolodchikov equations, has gained significant attention due to its deep connections with conformal field theory and modular tensor categories. Significant progress has also been made in the classification of coquasi-Hopf algebras. To date, finite-dimensional coradically graded pointed coquasi-Hopf algebras

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over abelian groups have been completely classified [17,16,18]. However, it seems that the classification of coquasi-Hopf algebras over non-abelian groups is impossible without our theory.

1.3 Cartan graphs and their Weyl groupoids

The theory of Weyl groupoids and arithmetic root systems was further systematically developed, giving rise to the concepts of semi-Cartan graphs and generalized root systems. While these concepts are independent of the theory of Nichols algebras, their deep connection to Nichols algebras has been rigorously established [1,14]. The most celebrated theorem states that a tuple of simple Yetter-Drinfeld modules M over a Hopf algebra with bijective antipode gives rise to a semi-Cartan graph, provided that it admits all reflections. This machinery has provided several powerful criteria to determine the finite-dimensionality of corresponding Nichols algebras

However, the axioms of a semi-Cartan graph are insufficiently strong to directly yield PBW basis. Heckenberger and Schneider demonstrated that a finite semi-Cartan graph originating from the aforementioned tuple M is a Cartan graph [13]. Subsequently, they removed the restriction and proved that the tuple $\mathcal{G}(M)$ is always a Cartan graph [15]. Therefore, the theory of Cartan graphs over ordinary Hopf algebras is now well-established.

1.4 The Cartan graph of a Nichols algebra over coquasi-Hopf algebra with bijective antipode

Let M be a θ -tuple. If M admits all reflections, we proved

$$\mathcal{G}(M) = (\mathbb{I}, \mathcal{X}, r, (A^X)_{X \in \mathcal{X}})$$

is a semi-Cartan graph in [19, 20] (see Section 2.6 for details). The primary goal of this paper is to prove that $\mathcal{G}(M)$ is actually a Cartan graph.

To verify the remaining axioms of a Cartan graph, sequences of graded right coideal subalgebras of Nichols algebras and their compatibility with reflections are studied. By [20, Corollary 4.3], there exists a braided tensor equivalence for each $i \in \mathbb{I}$.

$$\Omega_i, : \mathcal{B}(M_i) \mathcal{YD}(\mathcal{C})_{\text{rat}} \longrightarrow \mathcal{B}(M_i^*) \mathcal{YD}(\mathcal{C})_{\text{rat}}. \quad (1.1)$$

Now suppose M admits the i -th reflection. This equivalence induces the following isomorphism of Hopf algebras, which establishes a strong structural connection between $\mathcal{B}(M)$ and $\mathcal{B}(R_i(M))$.

$$\Theta_i : \mathcal{B}(R_i(M)) \cong \Omega_i \left(\mathcal{B}(M)^{\text{co } \mathcal{B}(M_i)} \right) \# \mathcal{B}(M_i^*). \quad (1.2)$$

In this paper, from the above isomorphism, we obtain an algebra isomorphism in ${}^H_H \mathcal{YD}$,

$$T_i^M : L_i^{R_i(M)} = {}^{\text{co } \mathcal{B}(M_i^*)}(\Omega_i(K_i^M) \# \mathcal{B}(M_i^*)) \xrightarrow{\sim} \mathcal{B}(M)^{\text{co } \mathcal{B}(M_i)} = K_i^M.$$

The isomorphisms T_i^M play the role of the Lusztig automorphisms. In [4,7], the author used T_i^M to construct the PBW basis of Nichols algebras of diagonal type and determine their generators and relations. In this paper, they can be used to construct right coideal subalgebras in $\mathcal{B}(M)$ stepwisely. Furthermore, using T_i^M , we provide a new factorization of $\mathcal{B}(M)$, which leads to our main result.

Theorem 1.1. (Theorem 4.6) *With the above notation, if M admits all reflections, then $\mathcal{G}(M)$ is a Cartan graph.*

Furthermore, we establish that the finiteness of $\mathcal{G}(M)$ is equivalent to the tensor decomposability of $\mathcal{B}(M)$. Meanwhile, we provide a criterion for the finite-dimensionality of Nichols algebras over coquasi-Hopf algebras.

Theorem 1.2. (Theorem 5.9) *The following statements are equivalent.*

- (1) $\mathcal{B}(M)$ is finite-dimensional.
- (2) M admits all reflections, $\mathcal{G}(M)$ is finite and $\mathcal{B}(P_i)$ is finite-dimensional for all $[P] \in \mathcal{X}$ and $i \in \mathbb{I}$.

1.5 A characterization of isomorphic Cartan graphs

A natural question arises: is the Cartan graph associated with Yetter-Drinfeld modules an invariant? We demonstrate that such a Cartan graph is indeed invariant under specific braided monoidal equivalences. Let H, H' be two coquasi-Hopf algebras with bijective antipodes. We fix a braided monoidal equivalence:

$$F : {}^H_H\mathcal{YD} \longrightarrow {}^{H'}_{H'}\mathcal{YD}.$$

We prove the following theorem from a categorical perspective:

Theorem 1.3. (Theorem 6.2) *Under the above assumptions, the tuple M admits all reflections if and only if $F(M)$ does. Furthermore, $\mathcal{G}(M)$ is isomorphic to $\mathcal{G}(F(M))$ as Cartan graphs.*

As an application, we investigate Nichols algebras $\mathcal{B}(M)$ of diagonal type over coquasi-Hopf algebras. These Nichols algebras have been classified. According to the method in [16], the authors associated $\mathcal{B}(M)$ with a Nichols algebra $\mathcal{B}(M')$ of diagonal type over a Hopf algebra. With the Cartan graph theory over coquasi-Hopf algebra now established, we prove that the tuple M admits all reflections if and only if M' does and $\mathcal{G}(M)$ is isomorphic to $\mathcal{G}(M')$. Consequently, we deduce that the set of isomorphism classes of Cartan graphs arising from diagonal-type Nichols algebras over coquasi-Hopf algebras coincides exactly with the set of isomorphism classes arising from the Hopf algebra setting.

1.6 Organization of the Paper

The paper proceeds as follows. Section 2 collects necessary preliminaries on Cartan graphs, Weyl groupoids, and coquasi-Hopf algebras. In Section 3, we develop the structural theory of one-sided coideal subalgebras within braided Hopf algebras, which serves as our primary algebraic tool. Section 4 contains the core proof that $\mathcal{G}(M)$ is a Cartan graph. In Section 5, we introduce tensor decomposable Nichols algebras and prove the finite-dimensionality criterion. Section 6 concludes the paper by establishing the invariance of Cartan graphs under braided monoidal equivalences and applies this to Nichols algebras of diagonal type over coquasi-Hopf algebras.

2 Preliminaries

In this section, we briefly recall the fundamental definitions and fix the notation for coquasi-Hopf algebras and Yetter-Drinfeld modules. For a more detailed exposition and proofs of standard properties, we refer the reader to [19].

2.1 Cartan graphs and root systems

We start from the definition of semi-Cartan graphs, which plays an important role in this paper, we follow the notation in [15]. One may refer to that book for detailed proofs and various examples.

Definition 2.1. Let $\mathbb{I}$ be a non-empty finite set, $\mathcal{X}$ a non-empty set, $r : \mathbb{I} \times \mathcal{X} \rightarrow \mathcal{X}$, $A : \mathbb{I} \times \mathbb{I} \times \mathcal{X} \rightarrow \mathbb{Z}$ maps. For all $i, j \in \mathbb{I}$ and $X \in \mathcal{X}$ we write $r_i(X) = r(i, X)$, $a_{ij}^X = A(i, j, X)$ and $A^X = (a_{ij}^X)_{i, j \in \mathbb{I}} \in \mathbb{Z}^{\mathbb{I} \times \mathbb{I}}$. The quadruple $\mathcal{G} = \mathcal{G}(\mathbb{I}, \mathcal{X}, (r_i)_{i \in \mathbb{I}}, (A^X)_{X \in \mathcal{X}})$ is called a semi-Cartan graph if for all $X \in \mathcal{X}$, the matrix A^X is a generalized Cartan matrix, and the following axioms hold.

(CG1) For all $i \in \mathbb{I}$, the map r_i satisfies $r_i^2 = \text{id}_{\mathcal{X}}$.

(CG2) For all $i \in \mathbb{I}$, $X \in \mathcal{X}$, A^X and $A^{r_i(X)}$ have the same i -th row.

Kac-Moody Lie algebras, Kac-Moody Lie superalgebras and Nichols algebras over coquasi-Hopf algebras provide an abundant class of examples of semi-Cartan graphs [20].

Given semi-Cartan graphs, we can construct Weyl groupoids, and define the sets of real roots. Given a semi-Cartan graph, its Weyl groupoid is defined as follows.

Definition 2.2. We denote by $\mathcal{D}(\mathcal{X}, \text{End}(\mathbb{Z}^{\mathbb{I}}))$ to be the category with objects $\text{Ob } \mathcal{D}(\mathcal{X}, \text{End}(\mathbb{Z}^{\mathbb{I}})) = \mathcal{X}$, and morphisms

$$\text{Hom}(X, Y) = \{(Y, f, X) \mid f \in \text{End}(\mathbb{Z}^{\mathbb{I}})\},$$

where the composition of morphisms is defined by

$$(Z, g, Y) \circ (Y, f, X) = (Z, gf, X), \text{ for all } X, Y, Z \in \mathcal{X}, f, g \in \text{End}(\mathbb{Z}^{\mathbb{I}}).$$

Let $\alpha_i, 1 \leq i \leq \theta$ be the standard basis of $\mathbb{Z}^{\mathbb{I}}$, and

$$s_i^X \in \text{Aut}(\mathbb{Z}^{\mathbb{I}}), s_i^X(\alpha_j) = \alpha_j - a_{ij}^X \alpha_i, \text{ for all } j.$$

We call the smallest subcategory of $\mathcal{D}(\mathcal{X}, \text{End}(\mathbb{Z}^{\mathbb{I}}))$ which contains all morphisms $(r_i(X), s_i^X, X)$ with $i \in \mathbb{I}, X \in \mathcal{X}$ the Weyl groupoid of $\mathcal{G}$, denoted by $\mathcal{W}(\mathcal{G})$.

Now let $X, Y \in \mathcal{G}$, $\omega \in \text{Hom}(X, Y)$. The following definition is parallel to that in Lie algebra. We call

$$l(\omega) = \min\{k \mid \omega = s_{i_1} \cdots s_{i_k}, i_1, \dots, i_k \in \mathbb{I}\}$$

the length of ω and the decomposition $(i_1, \dots, i_k)$ is called the reduced decomposition of ω . Here we omit the subscript for simplicity

For each $X \in \mathcal{X}$, the set of real roots of $\mathcal{G}$ at X is defined as follows:

$$\Delta^{X, \text{re}} = \{\omega(\alpha_i) \in \mathbb{Z}^{\mathbb{I}} \mid \omega \in \text{Hom}(\mathcal{W}(\mathcal{G}(M)), X), i \in \mathbb{I}\}.$$

We say $\mathcal{G}$ is a finite Cartan graph if $\Delta^{X, \text{re}}$ is finite for each $X \in \mathcal{X}$.

The elements

$$\Delta_+^{X, \text{re}} = \Delta^{X, \text{re}} \cap \mathbb{N}_0^{\mathbb{I}}, \Delta_-^{X, \text{re}} = \Delta^{X, \text{re}} \cap -\mathbb{N}_0^{\mathbb{I}}$$

are called positive and negative real roots, respectively. For any $X \in \mathcal{X}$ and $i, j \in \mathbb{I}$, let

$$m_{ij}^X = |\Delta^{X, \text{re}} \cap (\mathbb{N}_0 \alpha_i + \mathbb{N}_0 \alpha_j)|.$$

With the above notations, we can give the definition of a Cartan graph.

Definition 2.3. We say that a semi-Cartan graph $\mathcal{G}$ is a **Cartan graph** if the following hold.

(CG3) For any $X \in \mathcal{X}$, the set $\Delta^{X, \text{re}} = \Delta_+^{X, \text{re}} \cup \Delta_-^{X, \text{re}}$.

(CG4) For any $i \neq j \in \mathbb{I}$ and $X \in \mathcal{X}$, if the cardinality $m_{ij}^X := |\Delta^{X, \text{re}} \cap (\mathbb{N}_0 \alpha_i + \mathbb{N}_0 \alpha_j)|$ is finite, then $(r_i r_j)^{m_{ij}^X}(X) = X$.

There are many examples showing that a semi-Cartan graph may not satisfy (CG3) or (CG4), see [15, Example 9.1.26, Example 9.2.3].

We are going to introduce root systems over semi-Cartan graphs.

Definition 2.4. Let $\mathcal{G} = \mathcal{G}(\mathbb{I}, \mathcal{X}, r, A)$ be a semi-Cartan graph. For all $X \in \mathcal{X}$, let $(R^X)_{X \in \mathcal{X}}$ be a subset of $\mathbb{Z}^{\mathbb{I}}$ with the following properties.

- (1) $0 \notin R^X$ and $\alpha_i \in R^X$ for all $X \in \mathcal{X}$ and $i \in \mathbb{I}$.
- (2) $R^X \subseteq \mathbb{N}_0^{\mathbb{I}} \cup -\mathbb{N}_0^{\mathbb{I}}$ for all $X \in \mathcal{X}$.
- (3) For any $X \in \mathcal{X}$ and $i \in \mathbb{I}$, $s_i^X(R^X) = R^{r_i(X)}$.
- (4) If $i, j \in \mathbb{I}$ and $X \in \mathcal{X}$ such that $i \neq j$ and m_{ij}^X in Definition 2.3 is finite, then $(r_i r_j)^{m_{ij}^X}(X) = X$.

Then we say that the pair $(\mathcal{G}, (R^X)_{X \in \mathcal{X}})$ is a **root system** over $\mathcal{G}$. A root system over $\mathcal{G}$ is said to be **reduced** if for all $X \in \mathcal{X}$ and $\alpha \in R^X$ the roots α and $-\alpha$ are the only rational multiples of α in R^X . A root system is said to be **finite** if for all $X \in \mathcal{X}$, R^X is finite.

If $\mathcal{G}$ is a Cartan graph, the pair $(\mathcal{G}, (\Delta^{X, re})_{X \in \mathcal{X}})$ is automatically a reduced root system over $\mathcal{G}$. Furthermore, if $(R^X)_{X \in \mathcal{X}}$ is finite, then $R^X = \Delta^{X, re}$ for each $X \in \mathcal{X}$.

Remark 2.5. It was proved that there is an equivalence between a semi-Cartan graph with a reduced root system $(\mathcal{G}, (R^X)_{X \in \mathcal{X}})$ and a crystallographic Tits arrangement. A crystallographic Tits arrangement consists of a pair $(\mathcal{A}, T)$, where $\mathcal{A}$ is a set of linear hyperplanes in $\mathbb{R}^{\mathbb{I}}$, and T is a Tits cone in $\mathbb{R}^{\mathbb{I}}$, satisfying some additional conditions. One may refer to [9] for details.

2.2 Reduced sequences and an equivalent form of Cartan graphs

One of our purposes in this paper is to show reflections of Nichols algebras over coquasi-Hopf algebras with bijective antipode will give rise to Cartan graphs, which means (CG3) and (CG4) hold automatically in our settings. However, it is difficult to directly verify these two axioms, and therefore equivalent conditions are required.

Let $\mathcal{G} = \mathcal{G}(\mathbb{I}, \mathcal{X}, r, A)$ be a semi-Cartan graph. Let $X \in \mathcal{X}, l \geq 0$, and $\kappa = (i_1, \dots, i_l) \in \mathbb{I}^l$. For all $1 \leq k \leq l$ let

$$\beta_k^{X, \kappa} = \text{id}_X s_{i_1} \cdots s_{i_{k-1}}(\alpha_{i_k}),$$

and let

$$\Lambda^X(\kappa) = \{\beta_k^{X, \kappa} \mid 1 \leq k \leq l\}.$$

Definition 2.6. We say that κ is **X -reduced** if for any $1 \leq k < l$,

$$\alpha_{i_k} \notin \Lambda^{r_{i_k} \cdots r_{i_1}(X)}(i_{k+1}, \dots, i_l).$$

The integer l is called the **length** of κ .

Remark 2.7. (1) Let $X \in \mathcal{X}, l \geq 2$ and $\kappa = (i_1, \dots, i_l) \in \mathbb{I}^l$. If $i_j = i_{j+1}$ for some $1 \leq j < l$, then

$$\alpha_{i_j} \in \Lambda^{r_{i_j} \cdots r_{i_1}(X)}(i_{j+1}, \dots, i_l),$$

and hence κ is not X -reduced.

(2) A sequence $\kappa = (i_1, \dots, i_l)$ is X -reduced if and only if $(i_2, \dots, i_l)$ is $r_{i_1}(X)$ -reduced and $\alpha_{i_1} \notin \Lambda^{r_{i_1}(X)}(i_2, \dots, i_l)$.

(3) κ is X -reduced if and only if $\beta_p^{X, \kappa} \neq -\beta_q^{X, \kappa}$ for any $1 \leq p < q \leq l$.

Now we consider a special case.

Definition 2.8. Let $i, j \in \mathbb{I}$ with $i \neq j$ and $\kappa = (i_k)_{k \geq 1} = (i, j, i, \dots)$ with $i_k = i$ if k is odd and $i_k = j$ if k is even. Let $X \in \mathcal{X}$. We call any X -reduced sequence with entries in $\{i, j\}$ and starting with i a **beginning of κ** . Let κ_{ij}^X be the longest X -reduced beginning of κ , if it exists, and κ otherwise. We write $\bar{m}_{ij}^X$ for the length of κ_{ij}^X .

With the help of the notation above, we are in the position to introduce axioms characterizing Cartan graphs.

(CG3') For any $X \in \mathcal{X}$ and any X -reduced sequence γ , $\Lambda^X(\gamma) \subseteq \mathbb{N}_0^\mathbb{I}$.

(CG4') For any $X \in \mathcal{X}$ and any $i, j \in \mathbb{I}$ with $i \neq j$ and $\bar{m}_{ij}^X < \infty$, we have

$$(r_i r_j)^{\bar{m}_{ij}^X}(X) = X, \quad \text{id}_X(s_i s_j)^{\bar{m}_{ij}^X}(\alpha_k) = \alpha_k$$

for all $k \in \mathbb{I}$.

The following lemma is important for our purposes.

Lemma 2.9. [15, Corollary 9.2.20] *For any semi-Cartan graph $\mathcal{G}$, the following are equivalent.*

- (1) $\mathcal{G}$ is a Cartan graph.
- (2) $\mathcal{G}$ satisfies (CG3') and (CG4').

We collect some lemmas needed later, which can be found in [15, Section 9.2].

Lemma 2.10. *Assume that $|\mathbb{I}| \geq 2$. Let $\mathcal{G} = \mathcal{G}(\mathbb{I}, \mathcal{X}, r, A)$ be a semi-Cartan graph, $X \in \mathcal{X}$, $i, j \in \mathbb{I}$ with $i \neq j$*

- (1) *Assume $\kappa = (i, j, i, j, \dots)$ is a X -reduced sequence beginning with κ_{ij}^X with length $m \geq 2$. If $\bar{m}_{ji}^{r_j(X)} = m$, then $\beta_m^{X, \kappa} = \alpha_j$.*
- (2) *Assume $\mathcal{G}$ satisfies (CG3'). Then $\bar{m}_{ij}^{r_k(X)} = \bar{m}_{ji}^{r_k(X)} = \bar{m}_{ij}^X$ for $k \in \{i, j\}$.*
- (3) *Assume $\kappa = (i, j, i, j, \dots)$, $\bar{m}_{ij} < \infty$ and $\mathcal{G}$ satisfies (CG3'), then $\beta_1^{X, \kappa} = \alpha_i$, $\beta_{\bar{m}_{ij}}^{X, \kappa} = \alpha_j$, $\text{id}_X(s_i s_j)^{\bar{m}_{ij}^X}(\alpha_i) = \alpha_i$ and $\text{id}_X(s_i s_j)^{\bar{m}_{ij}^X}(\alpha_j) = \alpha_j$.*

The following lemma can be found in [15, Section 9.3], which is used to give a new criterion to determine the finite-dimensionality of a Nichols algebra.

Lemma 2.11. (1) *Let $\mathcal{G}$ be a Cartan graph, and $\omega = \text{id}_X s_{i_1} \cdots s_{i_l} \in \text{Hom}(\mathcal{W}(\mathcal{G}), X)$ be a reduced decomposition of ω . Then $\kappa = (i_1, \dots, i_l)$ is X -reduced.*

(2) *Let $\mathcal{G}$ be a finite Cartan graph and $(i_1, \dots, i_n)$ be a reduced decomposition of longest element $\omega_0 \in \text{Hom}(\mathcal{W}(\mathcal{G}), X)$, then*

$$\Delta_+^{X, \text{re}} = \Lambda^X(\kappa).$$

2.3 Coquasi-Hopf algebras

We follow the standard definition of a coquasi-Hopf algebra as introduced in [10].

Definition 2.12. *A coquasi-Hopf algebra H is a coalgebra (H, Δ, ϵ) equipped with a compatible quasi-algebra structure and an antipode $(\mathcal{S}, \alpha, \beta)$. Namely, there exist:*

- *Two coalgebra homomorphisms:*

$$m : H \otimes H \rightarrow H, \quad a \otimes b \mapsto ab,$$

$$\mu : \mathbb{k} \rightarrow H, \quad \lambda \mapsto \lambda 1_H,$$

- *A convolution-invertible map $\Phi : H^{\otimes 3} \rightarrow \mathbb{k}$ called an associator,*
- *A coalgebra antimorphism $\mathcal{S} : H \rightarrow H$,*
- *Two linear functions $\alpha, \beta : H \rightarrow \mathbb{k}$*

such that for all $a, b, c, d \in H$ the following equalities hold:

$$a_1(b_1c_1)\Phi(a_2, b_2, c_2) = \Phi(a_1, b_1, c_1)(a_2b_2)c_2, \quad (2.1)$$

$$1_H a = a = a 1_H, \quad (2.2)$$

$$\Phi(a_1, b_1, c_1 d_1) \Phi(a_2 b_2, c_2, d_2) = \Phi(b_1, c_1, d_1) \Phi(a_1, b_2 c_2, d_2) \Phi(a_2, b_3, c_3), \quad (2.3)$$

$$\Phi(a, 1_H, b) = \varepsilon(a) \varepsilon(b), \quad (2.4)$$

$$\mathcal{S}(a_1) \alpha(a_2) a_3 = \alpha(a) 1_H, \quad a_1 \beta(a_2) \mathcal{S}(a_3) = \beta(a) 1_H, \quad (2.5)$$

$$\Phi(a_1, \mathcal{S}(a_3), a_5) \beta(a_2) \alpha(a_4) = \Phi^{-1}(\mathcal{S}(a_1), a_3, \mathcal{S}(a_5)) \alpha(a_2) \beta(a_4) = \varepsilon(a). \quad (2.6)$$

Throughout this paper, we use the Sweedler sigma notation $\Delta(a) = a_1 \otimes a_2$ for the co-product and $a_1 \otimes a_2 \otimes \cdots \otimes a_{n+1}$ for the result of the n -iterated application of Δ on a . We say H has a bijective antipode if $\mathcal{S}$ is bijective.

We now turn our attention to the Yetter-Drinfeld module category structure. The definition of the Yetter-Drinfeld module category ${}^H_H\mathcal{YD}$ over an arbitrary coquasi Hopf-algebra H was already given in [8]. Since it plays a crucial role in our paper, we recall this definition.

Definition 2.13. [8, Definition 3.1] *Let H be a coquasi-Hopf algebra with associator Φ . A left-left Yetter-Drinfeld module over H is a triple $(V, \delta_V, \triangleright)$ such that:*

- (V, δ_V) is a left comodule of H and we denote $\delta_V(v)$ by $v_{-1} \otimes v_0$ as usual;
- $\triangleright : H \otimes V \rightarrow V$ is a $\mathbb{k}$ -linear map satisfying for all $h, l \in H$ and $v \in V$:

$$(hl) \triangleright v = \frac{\Phi(h_2, (l_2 \triangleright v_0)_{-1}, l_3)}{\Phi(h_1, l_1, v_{-1}) \Phi((h_3 \triangleright (l_2 \triangleright v_0)_0)_{-1}, h_4, l_4)} (h_3 \triangleright (l_2 \triangleright v_0)_0)_0, \quad (2.7)$$

$$1_H \triangleright v = v, \quad (2.8)$$

$$(h_1 \triangleright v)_{-1} h_2 \otimes (h_1 \triangleright v)_0 = h_1 v_{-1} \otimes (h_2 \triangleright v_0). \quad (2.9)$$

A morphism $f : (V, \delta_V, \triangleright) \rightarrow (V', \delta_{V'}, \triangleright')$ is a colinear map $f : (V, \delta_V) \rightarrow (V', \delta_{V'})$ such that $f(h \triangleright v) = h \triangleright' f(v)$ for all $h \in H$.

The category ${}^H_H\mathcal{YD}$ is a $\mathbb{k}$ -linear braided monoidal abelian category over the field $\mathbb{k}$. The unit object of ${}^H_H\mathcal{YD}$ is $\mathbb{k}$, which is regarded as an object in ${}^H_H\mathcal{YD}$ via trivial structures. For $V, W \in {}^H_H\mathcal{YD}$, the tensor product of Yetter-Drinfeld modules is defined by:

$$(V, \delta_V, \triangleright) \otimes (W, \delta_W, \triangleright) = (V \otimes W, \delta_{V \otimes W}, \triangleright),$$

where $\delta_{V \otimes W}$ is given by

$$\delta_{V \otimes W}(v \otimes w) = v_{-1} w_{-1} \otimes v_0 \otimes w_0,$$

and

$$h \triangleright (v \otimes w) = \frac{\Phi(h_1, v_{-1}, w_{-2}) \Phi((h_2 \triangleright v_0)_{-1}, (h_4 \triangleright w_0)_{-1}, h_5)}{\Phi((h_2 \triangleright v_0)_{-2}, h_3, w_{-1})} (h_2 \triangleright v_0)_0 \otimes (h_4 \triangleright w_0)_0. \quad (2.10)$$

The braiding $c_{V, W} : V \otimes W \rightarrow W \otimes V$ is given by:

$$c_{V, W}(v \otimes w) = (v_{-1} \triangleright w) \otimes v_0. \quad (2.11)$$

Lemma 2.14. [20, Lemma 2.5] *Let H be a coquasi-Hopf algebra with bijective antipode. The category ${}^H_H\mathcal{YD}$ is braided monoidal isomorphic to the right center of the category of left H -comodules $\mathcal{Z}_r({}^H\mathcal{M})$.*

Bosonizations and coinvariants are fundamental constructions that build new Hopf algebras. These operations for coquasi-Hopf algebras first appeared in [8]. We omit these details.

Lemma 2.15. [20, Lemma 2.7] *Let H be a coquasi-Hopf algebra. Suppose R is a Hopf algebra in ${}^H_H\mathcal{YD}$. With appropriate operations, $M := R \otimes H$ can be turned into a coquasi-Hopf algebra. We denote by $R\#H$.*

Now suppose N and H are both coquasi-Hopf algebras. Furthermore, assume there exist morphisms of coquasi-Hopf algebras

$$\pi : N \rightarrow H \text{ and } \sigma : H \rightarrow N$$

such that $\pi\sigma = \text{id}_N$.

Lemma 2.16. [20, Lemma 2.9] *Under above assumptions, $L := N^{\text{co } H}$ is a Hopf algebra in ${}^H_H\mathcal{YD}$. We emphasize that the map satisfying Equation (2.8) hold by*

$$\text{ad} : H \otimes L \rightarrow L, h \otimes l \mapsto \text{ad}(h)(l).$$

Remark 2.17. For any monoidal category $\mathcal{C}$ and Hopf algebra $A \in \mathcal{C}$, the category ${}^A_A\mathcal{YD}(\mathcal{C})$ of Yetter-Drinfeld module over A in $\mathcal{C}$ is well-defined. One may refer to [15, Section 3.4] for more details.

2.4 Monoidal equivalence between Yetter-Drinfeld module categories related by a dual pair

In this subsection, we recall an important braided monoidal equivalence

$$(\Omega, \beta) : {}^B_B\mathcal{YD}(\mathcal{C})_{\text{rat}} \rightarrow {}^A_A\mathcal{YD}(\mathcal{C})_{\text{rat}}$$

constructed in [20, Section 3].

Let A and B be locally finite $\mathbb{N}_0$ -graded Hopf algebras in $\mathcal{C} = {}^H_H\mathcal{YD}$ with bijective antipodes related by a non-degenerate Hopf pairing ω satisfying

$$\omega(A(n), B(m)) = 0, \text{ if } n \neq m.$$

We briefly review the relevant definitions.

Remark 2.18. (1) Let $R = \bigoplus_{n \geq 0} R(n)$ be an $\mathbb{N}_0$ -graded Hopf algebra in $\mathcal{C}$. A left module (X, λ_X) over R in $\mathcal{C}$ is called rational if for any $x \in X$, there exists a natural number n_0 such that $\lambda(R(n) \otimes x) = 0$ for all $n \geq n_0$. We denote the category of left rational modules over R in $\mathcal{C}$ by ${}_R\mathcal{C}_{\text{rat}}$. A left Yetter-Drinfeld module X in ${}_R\mathcal{YD}(\mathcal{C})$ is called rational if X is rational as a module of R in $\mathcal{C}$. We denote the corresponding category by ${}_R\mathcal{YD}(\mathcal{C})_{\text{rat}}$. This category was proved to be braided monoidal in [20, Lemma 3.3].

(2) We have a braided monoidal equivalence

$$\Pi : {}^R_R\mathcal{YD}(\mathcal{C}) \cong {}^{R\#H}_{R\#H}\mathcal{YD},$$

where $\mathcal{C} = {}^H_H\mathcal{YD}$ by [20, Remark 2.16]. Now restricting to rational Yetter-Drinfeld modules, we denote the image of Π by ${}^{R\#H}_{R\#H}\mathcal{YD}_{\text{rat}}$, which is a monoidal full subcategory of ${}^{R\#H}_{R\#H}\mathcal{YD}$.

The following lemma is necessary when constructing Ω . We omit certain structural details here for brevity. We use the notation $\bar{\mathcal{C}}$ to denote the reverse category of $\mathcal{C}$.

Lemma 2.19. (1) [20, Theorem 3.11] *The functor*

$$\Gamma : \overline{{}^B_B\mathcal{YD}(\mathcal{C})_{\text{rat}}} \longrightarrow {}^{A^{\text{cop}}}_{A^{\text{cop}}}\mathcal{YD}(\bar{\mathcal{C}})_{\text{rat}}, \quad V \mapsto V, \quad (2.12)$$

and the morphisms f are mapped onto f , is an equivalence of braided monoidal categories.

(2) [15, Theorem 3.4.15] We have a braided monoidal equivalence:

$$(F_{rl}, \alpha) : \mathcal{YD}(\mathcal{C})_{A, \text{rat}}^A \rightarrow {}^A\mathcal{YD}(\mathcal{C})_{\text{rat}}, V \mapsto V \quad (2.13)$$

$$\alpha_{X,Y} = c_{Y,X}^{\mathcal{YD}(\mathcal{C})_A^A} \circ \bar{c}_{X,Y}, \quad (2.14)$$

where morphisms f are mapped onto f .

(3) [15, Theorem 3.4.16] We have a braided strict monoidal equivalence:

$$F_{lr} : {}^{A^{\text{cop}}}_{A^{\text{cop}}} \mathcal{YD}(\bar{\mathcal{C}})_{\text{rat}} \rightarrow \overline{\mathcal{YD}(\mathcal{C})_{A, \text{rat}}^A}, V \mapsto V, \quad (2.15)$$

where morphisms f are mapped onto f .

The construction of the functor Ω relies on the interplay between the equivalences established above, as illustrated in the following diagram:

$$\begin{array}{ccc} \overline{{}^B_B \mathcal{YD}(\mathcal{C})_{\text{rat}}} & \xrightarrow[\cong]{\Gamma} & {}^{A^{\text{cop}}}_{A^{\text{cop}}} \mathcal{YD}(\bar{\mathcal{C}})_{\text{rat}} \\ \downarrow \Omega & & \downarrow F_{lr} \\ \overline{{}^A_A \mathcal{YD}(\mathcal{C})_{\text{rat}}} & \xleftarrow{(F_{rl}, \alpha)} & \overline{\mathcal{YD}(\mathcal{C})_{A, \text{rat}}^A} \end{array}$$

Theorem 2.20. [20, Theorem 3.16] The following functor is a braided monoidal equivalence:

$$(\Omega, \beta) : {}^B_B \mathcal{YD}(\mathcal{C})_{\text{rat}} \rightarrow {}^A_A \mathcal{YD}(\mathcal{C})_{\text{rat}}, \text{ where } (V, \lambda_B, \delta_B) \mapsto (V, \lambda_A, \delta_A),$$

where morphisms f are mapped onto f . Here we omit the definition of λ_A, δ_A for simplicity. The monoidal structure is given by

$$(\Omega, \beta) : {}^B_B \mathcal{YD}(\mathcal{C})_{\text{rat}} \rightarrow {}^A_A \mathcal{YD}(\mathcal{C})_{\text{rat}}, \beta_{X,Y} = c_{Y,X}^{{}^B_B \mathcal{YD}(\mathcal{C})} \circ \bar{c}_{X,Y}.$$

2.5 Nichols algebras

Let $\mathcal{C}$ be an abelian braided monoidal category. We first recall the definition of Nichols algebras $\mathcal{B}(V)$ in $\mathcal{C}$, where $V \in \mathcal{C}$ is an arbitrary object.

We denote by

$$T(V) = \bigoplus_{n \geq 0} V^{\otimes n} = \bigoplus_{n \geq 0} (\cdots ((V \otimes V) \otimes V) \cdots) \otimes V$$

the tensor algebra in $\mathcal{C}$ generated freely by V . The tensor algebra $T(V)$ is naturally a graded Hopf algebra in $\mathcal{C}$.

Definition 2.21. Let $V \in \mathcal{C}$. The Nichols algebra of V is defined to be the quotient Hopf algebra

$$\mathcal{B}(V) := T(V)/I(V).$$

Here, $I(V)$ is the largest Hopf ideal of $T(V)$ contained in $\bigoplus_{n \geq 2} T^n(V)$.

To facilitate further analysis, we assume $\mathcal{C}$ is a $\mathbb{k}$ -linear braided monoidal abelian category. In this setting, the following equivalent definition of the Nichols algebra is more convenient.

Definition 2.22. [21, Definition 2.4] For a given object $V \in \mathcal{C}$, the Nichols algebra $\mathcal{B}(V)$ is the unique Hopf algebra in $\mathcal{C}$ that satisfies the following conditions:

- (1) The Hopf algebra $\mathcal{B}(V)$ is graded by the non-negative integers.
- (2) The zeroth component of the grading satisfies $\mathcal{B}(V)_0 = \mathbb{k}$.
- (3) The first component of the grading satisfies $\mathcal{B}(V)_1 = V$, and $\mathcal{B}(V)$ is generated by V as an algebra in $\mathcal{C}$.
- (4) The subobject of primitive elements of $\mathcal{B}(V)$ is V .

Now we return to the case of a coquasi-Hopf algebra H with a bijective antipode. Let V be a finite-dimensional object in ${}^H_H\mathcal{YD}$. There is a well-defined dual pair

$$\omega : \mathcal{B}(V^*) \otimes \mathcal{B}(V) \rightarrow \mathbb{k}.$$

One may refer to [20, Section 4.1] for more details.

Corollary 2.23. [20, Corollary 4.3] *For any finite-dimensional $V \in {}^H_H\mathcal{YD}$, there exists a braided monoidal equivalence:*

$$(\Omega_V, \beta_V) : {}_{\mathcal{B}(V)}^{\mathcal{B}(V)}\mathcal{YD}(\mathcal{C})_{\text{rat}} \longrightarrow {}_{\mathcal{B}(V^*)}^{\mathcal{B}(V^*)}\mathcal{YD}(\mathcal{C})_{\text{rat}}. \quad (2.16)$$

2.6 Reflection theory over coquasi-Hopf algebras

In this subsection, we recall the reflection of simple Yetter-Drinfeld modules; this construction is crucial for the present paper.

Definition 2.24. Let $\mathcal{F}_\theta$ denote the class of all θ -tuples $M = (M_1, \dots, M_\theta)$, where $M_1, \dots, M_\theta \in {}^H_H\mathcal{YD}$ are finite-dimensional simple Yetter-Drinfeld modules. If $M \in \mathcal{F}_\theta$, we define

$$\mathcal{B}(M) := \mathcal{B}(M_1 \oplus \dots \oplus M_\theta).$$

Two tuples $M, M' \in \mathcal{F}_\theta$ are called isomorphic, denoted $M \cong M'$, if $M_j \cong M'_j$ in ${}^H_H\mathcal{YD}$ for all j . The isomorphism class of $M \in \mathcal{F}_\theta$ is denoted by $[M]$.

For $1 \leq i \leq \theta$ and $M \in \mathcal{F}_\theta$, we say the tuple M admits the i -th reflection $R_i(M)$ if for all $j \neq i$ there is a natural number $m_{ij}^M \geq 0$ such that $\text{ad}(M_i)^{m_{ij}^M}(M_j)$ is a non-zero finite-dimensional subspace of $\mathcal{B}(M)$, and $(\text{ad } M_i)^{m_{ij}^M+1}(M_j) = 0$.

Assume M admits the i -th reflection. Then we set $R_i(M) = (R_i(M_1)_1, \dots, R_i(M)_\theta)$, where

$$R_i(M)_j = \begin{cases} M_i^*, & \text{if } j = i, \\ \text{ad}(M_i)^{m_{ij}^M}(M_j), & \text{if } j \neq i. \end{cases}$$

Having defined individual reflections, we now extend this notion to sequences of reflections, which will be important for our study of repeated reflections of tuples.

Definition 2.25. Let $M \in \mathcal{F}_\theta$, with each M_i being simple for $i \in \mathbb{I}$. For $l \in \mathbb{N}_0$ and $i_1, i_2, \dots, i_l \in \mathbb{I}$.

(1) We say M admits the reflection sequence $(i_1, i_2, \dots, i_l)$ if $l = 0$ or M admits the i_1 -th reflection and $R_{i_1}(M)$ satisfies the reflection sequence $(i_2, i_3, \dots, i_l)$.

(2) We say M admits all reflection sequences if M admits all reflection sequences $(i_1, i_2, \dots, i_l)$ for all $l \in \mathbb{N}_0$ and $i_1, i_2, \dots, i_l \in \mathbb{I}$.

Remark 2.26. [20, Lemma 5.9]

(1) Suppose $M \in \mathcal{F}_\theta$ and M admits the i -th reflection for each $i \in \mathbb{I}$. We define $a_{ii}^M = 2$ for all $1 \leq i \leq \theta$ and define $a_{ij}^M = -m_{ij}^M$. Then $(a_{ij}^M)_{i,j \in \mathbb{I}}$ is a generalized Cartan matrix.

(2) Suppose $M \cong N$ in $\mathcal{F}_\theta$. If M admits the i -th reflection for some $i \in \mathbb{I}$, then so does N . Furthermore, $R_i(M) \cong R_i(N)$ and $a_{ij}^M = a_{ij}^N$ for each $j \in \mathbb{I}$.

Let $M = (M_1, \dots, M_\theta) \in \mathcal{F}_\theta$, with each component M_i being simple for $i \in \mathbb{I}$. By [20, Corollary 4.3], we have such a braided tensor equivalence for each $i \in \mathbb{I}$.

$$\Omega_i : {}_{\mathcal{B}(M_i)}^{\mathcal{B}(M_i)}\mathcal{YD}(\mathcal{C})_{\text{rat}} \longrightarrow {}_{\mathcal{B}(M_i^*)}^{\mathcal{B}(M_i^*)}\mathcal{YD}(\mathcal{C})_{\text{rat}}. \quad (2.17)$$

which is induced by the dual pair $\omega_i : \mathcal{B}(M_i^*) \otimes \mathcal{B}(M_i) \rightarrow \mathbb{k}$.

We also proved in [20, Lemma 5.10] that M admitting the i -th reflection for some $i \in \mathbb{I}$ is equivalent to the condition that $\mathcal{B}(M)^{\text{co}\mathcal{B}(M_i)}$ belongs to ${}_{\mathcal{B}(M_i)\#H}^{\mathcal{B}(M_i)\#H}\mathcal{YD}_{\text{rat}}$. The most important result in [19,20] is as follows.

Theorem 2.27. [20, Theorem 5.12] *With the above assumptions on M , if M admits the i -th reflection, there exists an isomorphism of Hopf algebras in ${}^H_H\mathcal{YD}$:*

$$\Theta_i : \mathcal{B}(R_i(M)) \cong \Omega_i \left(\mathcal{B}(M)^{\text{co } B(M_i)} \right) \# \mathcal{B}(M_i^*). \quad (2.18)$$

Corollary 2.28. [20, Corollary 5.15] *Suppose M admits all reflections. We define the set*

$$\mathcal{X} = \{[P] \mid P \in \mathcal{F}_\theta(M)\},$$

and the map

$$r : \mathbb{I} \times \mathcal{X} \rightarrow \mathcal{X}, \quad i \times [X] \mapsto [R_i(X)].$$

Then

$$\mathcal{G}(M) = (\mathbb{I}, \mathcal{X}, r, (A^{[X]})_{[X] \in \mathcal{X}}),$$

where $A^{[X]} = (a_{ij}^X)_{i,j \in \mathbb{I}}$ for all $[X] \in \mathcal{X}$, is a semi-Cartan graph.

3 One-sided coideal subalgebras of braided Hopf algebras in ${}^H_H\mathcal{YD}$

In this section, we study one-sided coideal subalgebras of braided Hopf algebras in ${}^H_H\mathcal{YD}$. We construct bijections between these subalgebras under reflections. These results provide the main algebraic tools to decompose Nichols algebras and verify the Cartan graph axioms in the next section.

3.1 Some properties of one-sided coideal subalgebra

Definition 3.1. *Let X be a bialgebra in $\mathcal{C} = {}^H_H\mathcal{YD}$. A left (respectively, right) coideal subalgebra Y is a subobject of X , and an algebra in $\mathcal{C}$ such that the inclusion map $Y \subseteq X$ is an algebra morphism, and $\Delta(Y) \subseteq X \otimes Y$ (respectively, $\Delta(Y) \subseteq Y \otimes X$).*

Suppose A and P are Hopf algebras in $\mathcal{C}$. Let $\gamma : A \rightarrow P$ be an injective Hopf algebra morphism in $\mathcal{C}$, and assume there exists a surjective Hopf algebra morphism $\pi : P \rightarrow A$ such that $\pi \circ \gamma = \text{id}_A$. The triple (P, π, γ) is called a Hopf algebra triple.

By [20, Lemma 2.15], there exists a Hopf algebra $R = P^{\text{co } A}$ in the braided monoidal category ${}^A_A\mathcal{YD}(\mathcal{C})$ such that $P \cong R \# A$. Denoting the comultiplication of P by $\Delta_P : P \rightarrow P \otimes P$, $p \mapsto p_1 \otimes p_2$, there is a well-defined map

$$\vartheta : P \rightarrow R, \quad p \mapsto p_1 \mathcal{S}_A(\pi(p_2)). \quad (3.1)$$

Moreover, R is a subalgebra of P in $\mathcal{C}$ such that $\Delta_P(R) \subseteq P \otimes R$. The A -action on R is given by $\text{ad} : A \otimes R \rightarrow R$, satisfying

$$\text{ad}(a)(r) = \vartheta(ar), \quad (3.2)$$

$$\vartheta(pa) = \vartheta(p)\varepsilon(a) \quad (3.3)$$

for all $a \in A, r \in R$, and $p \in P$. The A -coaction $\delta : R \rightarrow A \otimes R$ is defined by $\delta(r) = \pi(r_1) \otimes r_2$. The inverse of the Hopf algebra isomorphism $P \rightarrow R \# A, p \mapsto \vartheta(p_1) \otimes \pi(p_2)$, is given by the multiplication map.

For our purposes, we introduce the following notation.

Definition 3.2. (1) *Let*

$$S_r^+(P) = \{E \mid E \subseteq P \text{ right coideal subalgebra in } \mathcal{C}, A \subseteq E\},$$

$$S_r(P, X) = \{E \mid E \subseteq P \text{ right coideal subalgebra in } \mathcal{C}, E \subseteq X\},$$

where $X \subseteq P$ is a subobject in ${}^H_H\mathcal{YD}$, and

$$T_r(P) = \{F \mid F \subseteq R \text{ subalgebra in } \mathcal{C}, \Delta_R(F) \subseteq F \otimes R, F \subseteq R \text{ is left } A\text{-submodule}\}.$$

(2) Let

$$S_l^+(P) = \{E \mid E \subseteq P \text{ left coideal subalgebra in } \mathcal{C}, A \subseteq E\},$$

$$S_l(P, X) = \{E \mid E \subseteq P \text{ left coideal subalgebra in } \mathcal{C}, E \subseteq X\},$$

where $X \subseteq P$ is a subobject in ${}^H_H\mathcal{YD}$, and

$$T_l(P) = \{F \mid F \subseteq R \text{ subalgebra in } \mathcal{C}, \Delta_R(F) \subseteq R \otimes F, F \subseteq R \text{ is left } A\text{-subcomodule}\}.$$

Lemma 3.3. (1) For all $E \in S_r^+(P)$, the multiplication map

$$(E \cap R) \otimes A \rightarrow E$$

is an isomorphism in ${}^H_H\mathcal{YD}$.

(2) The map $S_r^+(P) \rightarrow T_r(P)$, $E \mapsto E \cap R$, is bijective with inverse given by $T_r(P) \rightarrow S_r^+(P)$, $F \mapsto FA$.

(3) $S_l(P, R) = T_l(P)$.

Proof. (1) This multiplication map is well-defined since $A \subseteq E$. Now we consider the map

$$\psi : E \rightarrow (E \cap R) \otimes A, \quad x \mapsto \vartheta(x_1) \otimes \pi(x_2),$$

For $x \in E \cap R$, $a \in A$, we have

$$\psi(xa) = \vartheta(x_1 a_1) \otimes \pi(x_2 a_2) = \vartheta(x_1) \varepsilon(a_1) \otimes \varepsilon(x_2) a_2 = \vartheta(x) \otimes a = x \otimes a.$$

Thus, ψ is inverse to the multiplication map, making both maps bijective. The map

$$(E \cap R) \otimes A \rightarrow E$$

is a morphism in ${}^H_H\mathcal{YD}$ because it is induced by the isomorphism $R \otimes A \rightarrow P$.

(2) We first show that both maps are well-defined. Let $E \in S_r^+(P)$. Then $E \cap R \subseteq R$ is a subalgebra in $\mathcal{C}$. For all $x \in E \cap R$, we have $\Delta_P(x) = x_1 \otimes x_2 \in (E \otimes P) \cap (P \otimes R)$. Since E is a right coideal subalgebra of P , we have $\Delta_P(e) = e_1 \otimes e_2 \in E \otimes P$ for any $e \in E$. By definition, $\vartheta(e) = e_1 S_A(\pi(e_2))$. The term $S_A(\pi(e_2)) \in A$. Note that E is an algebra in ${}^H_H\mathcal{YD}$, then $e_1 S_A(\pi(e_2)) \in E \cdot A \subseteq E \cdot E \subseteq E$. Therefore $\vartheta(E) \subseteq E$, and we can deduce that

$$\Delta_R(x) = \vartheta(x_1) \otimes x_2 \in (E \otimes P) \cap (R \otimes R) = (E \cap R) \otimes R.$$

Let c be the braiding isomorphism in ${}^H_H\mathcal{YD}$. Since $c(A \otimes E) \subseteq E \otimes A$, we have

$$\Delta_P(ax) = (a_1 \otimes a_2)(x_1 \otimes x_2) \in AE \otimes AP \subseteq E \otimes P$$

for all $x \in E$, $a \in A$. A direct computation shows that for $x \in E \cap R$,

$$\text{ad}(a)(x) = \vartheta(ax) = (ax)_1 S_A(\pi((ax)_2)) \in EA \cap R = E \cap R$$

Thus, $E \cap R$ is an A -submodule of R . This shows the map $S_r^+(P) \rightarrow T_r(P)$, $E \mapsto E \cap R$, is well-defined.

Let $F \in T_r(P)$ and $y \in F$. Since $F \subseteq R$ and

$$\Delta_P(y) = \vartheta(y_1) \pi(y_2) \otimes y_3,$$

we obtain $\Delta_P(F) \subseteq FA \otimes R$. Thus, for all $y \in F$, $a \in A$, $\Delta_P(ya) = \Delta_P(y) \Delta_P(a) \in FA \otimes P$.

To show that $FA \subseteq P$ is a subalgebra, it suffices to prove $AF \subseteq FA$. For all $a \in A, y \in F$,

$$\Delta_P(ay) = \Delta_P(a)\Delta_P(y) \in A(FA) \otimes P.$$

Since $F \subseteq R$ is a A -submodule, we have

$$ax = \vartheta((ax)_1)\pi((ax)_2) \in \vartheta(A(FA))A \stackrel{(3.3)}{\subseteq} \vartheta((AF)A)A \subseteq \vartheta(AF)A \stackrel{(3.2)}{=} \text{ad}(A)(F)A \subseteq FA.$$

Therefore FA is a subalgebra of P and the map $T_r(P) \rightarrow S_r^+(P), F \mapsto FA$ is well-defined.

If $E \in S_r^+(P)$, the following composition is the identity map by (1):

$$S_r^+(P) \rightarrow T_r(P) \rightarrow S_r^+(P), E \mapsto E \cap R \mapsto (E \cap R)A.$$

If $F \in T_r(P)$, we have the composition:

$$T_r(P) \rightarrow S_r^+(P) \rightarrow S_r^+(P), F \mapsto FA \mapsto (FA) \cap R.$$

Since $F \subseteq R$, the multiplication maps $(FA \cap R) \otimes A \rightarrow (FA \cap R)A = FA$ and $F \otimes A \rightarrow FA$ are bijective. This yields $F = FA \cap R$. It follows that the two maps are mutually inverse bijections.

(3) Let $E \in S_l(P, R)$, then $E \subseteq R$ is a subalgebra in $\mathcal{C}$ and $\Delta_R(x) = \vartheta(x_1) \otimes x_2 \in R \otimes E$ for $x \in E$. Moreover, $\delta(x) = \pi(x_1) \otimes x_2 \in A \otimes E$, hence E is a A -subcomodule. This shows $E \in T_l(P)$.

Conversely, let $F \in T_l(P)$, then $F \subseteq P$ is a subalgebra automatically. For $x \in F$, $\Delta_P(x) = \vartheta(x_1)\pi(x_2) \otimes x_3 \in P \otimes F$, because $\Delta_R(x) \in R \otimes F$. Hence, $F \in S_l(P, R)$. $\square$

3.2 Comparing one-sided coideal subalgebras via the isomorphism T

Let K be a Hopf algebra in ${}^B_B\mathcal{YD}(\mathcal{C})_{\text{rat}}$ with a bijective antipode. We obtain a Hopf algebra triple $(P := K \# B, \pi, \gamma)$ in $\mathcal{C}$, where $\pi : P \rightarrow B$ and $\gamma : B \rightarrow P$. Let $\Omega(K)$ denote the corresponding Hopf algebra in ${}^A_A\mathcal{YD}(\mathcal{C})_{\text{rat}}$ obtained via the braided monoidal equivalence Ω . This yields another Hopf algebra triple $(Q := \Omega(K) \# A, \bar{\pi}, \bar{\gamma})$ in $\mathcal{C}$, where

$$\bar{\pi} : Q \rightarrow A, \bar{\gamma} : A \rightarrow Q.$$

We also obtain a Hopf algebra triple $(Q^{\text{cop}}, \bar{\pi}, \bar{\gamma})$ in the reverse braided category $\bar{\mathcal{C}}$. The antipodes of $\Omega(K)$ and Q are bijective because the antipode of K is bijective.

By definition, $\Omega(K) = Q^{\text{co}A}$, which consists of the right coinvariant elements with respect to the projection $\bar{\pi}$. Meanwhile, we define

$$L := (Q^{\text{cop}})^{\text{co}\bar{\pi}}.$$

Then L is an object in ${}^{A^{\text{cop}}}_{A^{\text{cop}}}\mathcal{YD}(\bar{\mathcal{C}})_{\text{rat}}$. The following proposition establishes the relation between L and K .

Proposition 3.4. *With the above notation, there is an isomorphism of objects in $\mathcal{C}$*

$$T : L \longrightarrow K, x \mapsto \mathcal{S}_K^{-1}\mathcal{S}_Q(x). \quad (3.4)$$

In fact, the morphism T is an isomorphism of Hopf algebras in ${}^{A^{\text{cop}}}_{A^{\text{cop}}}\mathcal{YD}(\bar{\mathcal{C}})_{\text{rat}}$

$$T : L \longrightarrow \Gamma(K^{\text{cop}}). \quad (3.5)$$

Proof. By Lemma 2.19 (2) and (3), $\Omega(K)^{\text{cop}} \in \overline{\mathcal{YD}(\mathcal{C})}_A^A$. Then we have

$$F_{lr} \circ F_{rl}^{-1}(\Omega(K)^{\text{cop}}) \in {}_A^A\text{cop} \mathcal{YD}(\overline{\mathcal{C}})_{\text{rat}}.$$

By [15, Theorem 3.10.6], there is an isomorphism of objects $T : L \rightarrow K$ in $\mathcal{C}$. This map is an isomorphism of Hopf algebras in ${}_A^A\text{cop} \mathcal{YD}(\overline{\mathcal{C}})_{\text{rat}}$:

$$T : L \longrightarrow F_{lr} \circ F_{rl}^{-1}(\Omega(K)^{\text{cop}})$$

satisfying $\bar{\gamma} \circ T^{-1} = \mathcal{S}_Q^{-1} \circ \bar{\gamma} \circ \mathcal{S}_{\Omega(K)}$.

Since Ω is a braided monoidal functor, we have $\Omega(K)^{\text{cop}} = \Omega(K^{\text{cop}})$. Hence

$$F_{lr} \circ F_{rl}^{-1}(\Omega(K)^{\text{cop}}) = F_{lr} \circ F_{rl}^{-1}(\Omega(K^{\text{cop}})) = \Gamma(K^{\text{cop}}).$$

Therefore, $T : L \rightarrow \Gamma(K^{\text{cop}})$ is a Hopf algebra isomorphism. For any $x \in K$, we compute $T^{-1}(x) = \mathcal{S}_Q^{-1} \circ \bar{\gamma} \circ \mathcal{S}_{\Omega(K)}(x) = \mathcal{S}_Q^{-1} \circ \bar{\gamma} \circ \mathcal{S}_K(x) = \mathcal{S}_Q^{-1} \circ \mathcal{S}_K(x)$ using the identification $\mathcal{S}_{\Omega(K)} = \mathcal{S}_K$. Thus, $T(x) = \mathcal{S}_K^{-1} \mathcal{S}_Q(x)$ for all $x \in L$. $\square$

Let $(P := K \# B, \pi, \gamma)$, $(Q := \Omega(K) \# A, \bar{\pi}, \bar{\gamma})$ and L be defined as above. Define

$$\bar{K} := {}^{\text{co} \bar{\pi}} Q.$$

to be the space of left coinvariant elements with respect to the projection $\bar{\pi}$. The isomorphism T induces bijection between the one-sided coideal subalgebras of P and Q .

Proposition 3.5. (1) *With the above notation, there is a bijection*

$$S_l(Q^{\text{cop}}, L) \longrightarrow S_r^+(P), E \mapsto T(E)B$$

with inverse given by $E \mapsto T^{-1}(E \cap K)$. Furthermore, the multiplication map

$$T(E) \otimes B \rightarrow T(E)B$$

is bijective for all $E \in S_l(Q^{\text{cop}}, L)$.

(2) *There is a bijection*

$$S_l(P, K) \longrightarrow S_l^+(Q), E \mapsto T^{-1}(E)A$$

with inverse given by $E \mapsto T(E \cap L)$. Furthermore, the multiplication map

$$T^{-1}(E) \otimes A \rightarrow T^{-1}(E)A$$

is bijective for all $E \in S_l(P, K)$.

Proof. (1) By definition, the object $\bar{K} \cong L$ in $\mathcal{C}$, hence

$$S_r(Q, \bar{K}) = S_l(Q^{\text{cop}}, L).$$

By Lemma 3.3 (3), $S_l(Q^{\text{cop}}, L) = T_l(Q^{\text{cop}})$. We now establish the bijection between $T_l(Q^{\text{cop}})$ and $S_r^+(P)$.

By definition,

$$T_l(Q^{\text{cop}}) = \{E \mid E \subseteq L \text{ subalgebra in } \overline{\mathcal{C}}, \Delta_L(E) \subseteq L \otimes E, E \text{ is } A^{\text{cop}}\text{-subcomodule}\}.$$

We apply the Hopf algebra isomorphism T , which is an A^{cop} -colinear map in $\overline{\mathcal{C}}$. For any $E \in T_l(Q^{\text{cop}})$, $T(E)$ is a A^{cop} -subcomodule of $\Gamma(K)$ in $\overline{\mathcal{C}}$ and is a subalgebra of K in $\mathcal{C}$. Since

$D_2 : {}^{A^{\text{cop}}}\overline{\mathcal{C}} \rightarrow {}_B\mathcal{C}_{\text{rat}}$ is a braided monoidal equivalence by [20, Proposition 3.10], $T(E)$ is a B -submodule of K . Finally,

$$\overline{c}_{K,K}^{B\mathcal{YD}(\mathcal{C})} \circ \Delta_K(T(E)) = \Delta_{\Gamma(K^{\text{cop}})}(T(E)) \subseteq K \otimes T(E).$$

Here $\overline{c}_{K,K}^{B\mathcal{YD}(\mathcal{C})}$ denotes the reverse braiding of $c_{K,K}^{B\mathcal{YD}(\mathcal{C})}$. Since K has a bijective antipode, $\overline{c}_{K,K}^{B\mathcal{YD}(\mathcal{C})}$ is an isomorphism by [15, Proposition 3.4.8]. Now

$$\overline{c}_{K,K}^{B\mathcal{YD}(\mathcal{C})} \circ \Delta_K(T(E)) \subseteq K \otimes T(E) \Leftrightarrow \Delta_K(T(E)) \subseteq T(E) \otimes K.$$

Consequently, $T(E) \in T_r(P)$. The inverse mapping can be established similarly. Therefore, the isomorphism T establishes a bijection between $T_l(Q^{\text{cop}})$ and $T_r(P)$. By Lemma 3.3(2), the map

$$T_r(P) \longrightarrow S_r^+(P), F \mapsto FB$$

is bijective. Therefore, the map $S_l(Q^{\text{cop}}, L) \longrightarrow S_r^+(P), E \mapsto T(E)B$ is bijective with inverse $E \mapsto T^{-1}(E \cap K)$.

The bijectivity of the multiplication map $T(E) \otimes B \rightarrow T(E)B$ for all $E \in S_l(Q^{\text{cop}}, L)$ follows directly from Lemma 3.3 (1).

(2) By a similar argument, there is a bijection

$$T_r(Q^{\text{cop}}) \longrightarrow T_l(P), E \mapsto T(E).$$

According to Lemma 3.3 (2), (3), there is a bijective map $T_r(Q^{\text{cop}}) \rightarrow S_r^+(Q^{\text{cop}})$ given by $F \mapsto FA$, and $T_l(P) = S_l(P, K)$. Since $S_r^+(Q^{\text{cop}}) = S_l^+(Q)$, the map $S_l(P, K) \rightarrow S_l^+(Q)$ is bijective.

For any $E \in S_l(P, K)$, since $T^{-1}(E) \in T_r(Q^{\text{cop}})$, the multiplication map

$$T^{-1}(E) \otimes A \rightarrow T^{-1}(E)A$$

is bijective by Lemma 3.3(1). □

3.3 One-sided coideal subalgebras of Nichols algebras

In this subsection, we fix a tuple $M = (M_1, \dots, M_\theta) \in \mathcal{F}_\theta$, where each M_i is a finite-dimensional simple Yetter-Drinfeld module in ${}^H_H\mathcal{YD}$. We endow $\mathcal{B}(M)$ with an $\mathbb{N}_0^\theta$ -grading by setting $\deg(M_i) = \alpha_i$. Consider the following maps:

$$\pi_i : \mathcal{B}(M) \rightarrow \mathcal{B}(M_i), \gamma_i : \mathcal{B}(M_i) \rightarrow \mathcal{B}(M).$$

such that $\pi_i \circ \gamma_i = \text{id}_{\mathcal{B}(M_i)}$. Let

$$K_i^M = \mathcal{B}(M)^{\text{co } \mathcal{B}(M_i)}, L_i^M = {}^{\text{co } \mathcal{B}(M_i)}\mathcal{B}(M)$$

be the right and left coinvariant spaces with respect to π_i . A direct computation shows that

$$K_i^M = \mathcal{S}_{\mathcal{B}(M)}(L_i^M). \tag{3.6}$$

For any $x \in \mathcal{B}(M)$,

$$\begin{aligned} (\pi_i \otimes \text{id})(\Delta(\mathcal{S}_{\mathcal{B}(M)})(x)) &= (\pi_i \otimes \text{id}) \circ (\mathcal{S}_{\mathcal{B}(M)} \otimes \mathcal{S}_{\mathcal{B}(M)}) \circ c_{\mathcal{B}(M), \mathcal{B}(M)}(\Delta(x)) \\ &= c_{\mathcal{B}(M), \mathcal{B}(M_i)} \circ (\mathcal{S}_{\mathcal{B}(M)} \otimes \mathcal{S}_{\mathcal{B}(M_i)}) \circ (\text{id} \otimes \pi_i)(\Delta(x)) \end{aligned}$$

Hence, the condition $x \in K_i^M$ is equivalent to $\mathcal{S}_{\mathcal{B}(M)}(x) \in L_i^M$.

Lemma 3.6. *For each $i \in \mathbb{I}$.*

(1) *Let $E \subseteq \mathcal{B}(M)$ be an $\mathbb{N}_0^\theta$ -graded right coideal subalgebra in ${}^H_H\mathcal{YD}$. Then*

$$E \subseteq L_i^M \iff M_i \not\subseteq E.$$

(2) *Let $F \subseteq \mathcal{B}(M)$ be an $\mathbb{N}_0^\theta$ -graded left coideal subalgebra in ${}^H_H\mathcal{YD}$. Then*

$$F \subseteq K_i^M \iff M_i \not\subseteq F.$$

Proof. We only prove (1), as (2) is dual. Assume $E \subseteq L_i^M$ and $M_i \subseteq E$; it follows that $M_i \subseteq L_i^M$, which forces $M_i = 0$, yielding a contradiction.

Now we assume $M_i \not\subseteq E$. Since M_i is simple, $M_i \cap E$ must be zero. Since E is an $\mathbb{N}_0^\theta$ -graded right coideal subalgebra and π_i is $\mathbb{N}_0^\theta$ -graded morphism, we deduce that $\pi_i(E)$ is a right coideal subalgebra of $\mathcal{B}(M_i)$. Assume $\pi_i(E) \neq \mathbb{k}1$. Because M_i is simple, $\pi_i(E) \cap P(\mathcal{B}(M_i)) \neq 0$. Furthermore, the simplicity of M_i and the fact that $P(\mathcal{B}(M_i)) = M_i$ jointly force $M_i \subseteq \pi_i(E)$. This contradicts $M_i \cap E = 0$. Therefore $\pi_i(E) = \mathbb{k}1$. Now for any $x \in E$,

$$(\pi_i \otimes \text{id})\Delta(x) = \pi_i(x_1) \otimes x_2 = 1 \otimes x.$$

Hence $E \subseteq L_i^M$. □

Definition 3.7. *For each $i \in \mathbb{I}$.*

(1) *We define*

$$\begin{aligned} \mathcal{K}(M) &= \{E \mid E \subseteq \mathcal{B}(M) \text{ is an } \mathbb{N}_0^\theta\text{-graded right coideal subalgebra in } {}^H_H\mathcal{YD}\}, \\ \mathcal{K}_i^+(M) &= \{E \mid E \in \mathcal{K}(M), M_i \subseteq E\}, \\ \mathcal{K}_i^-(M) &= \{E \mid E \in \mathcal{K}(M), M_i \not\subseteq E\}. \end{aligned}$$

(2) *Dually, we define*

$$\begin{aligned} \mathcal{L}(M) &= \{F \mid F \subseteq \mathcal{B}(M) \text{ is an } \mathbb{N}_0^\theta\text{-graded left coideal subalgebra in } {}^H_H\mathcal{YD}\}, \\ \mathcal{L}_i^+(M) &= \{F \mid F \in \mathcal{L}(M), M_i \subseteq F\}, \\ \mathcal{L}_i^-(M) &= \{F \mid F \in \mathcal{L}(M), M_i \not\subseteq F\}. \end{aligned}$$

From now on, we assume M admits the i -th reflection. Recall that $R_i(M) = (R_i(M_1)_1, \dots, R_i(M)_\theta)$, where

$$R_i(M)_j = \begin{cases} M_i^*, & \text{if } j = i, \\ \text{ad}(M_i)^{m_{ij}^M}(M_j), & \text{if } j \neq i. \end{cases}$$

For simplicity, we denote $R_i(M_j) = \widetilde{M}_j$ for each j .

Lemma 3.8. (1)

$$\text{ad}(M_i^*)^n(\widetilde{M}_j) = \begin{cases} 0, & \text{if } n > m_{ij}^M, \\ \text{ad}(M_i)^{m_{ij}^M - n}(M_j) & \text{if } 0 \leq n \leq m_{ij}^M. \end{cases}$$

(2) *The Nichols algebra $\mathcal{B}(R_i(M)) \cong \Omega(K_i^M) \# \mathcal{B}(M_i^*)$ is an $\mathbb{N}_0^\theta$ -graded Hopf algebra in ${}^H_H\mathcal{YD}$ with*

$$\deg(x \otimes y) = s_i^M(\deg^{\mathcal{B}(M)}(x) + \deg(y))$$

for all homogeneous elements $x \in K_i^M$ and $y \in \mathcal{B}(M_i^)$, where $\mathcal{B}(M_i^*)$ is a $\mathbb{Z}^\theta$ -graded algebra with $\deg(M_i^*) = -\alpha_i$, and $\deg^{\mathcal{B}(M)}$ denotes the degree in $\mathcal{B}(M)$.*

Proof. (1) The first statement follows from [20, Lemma 5.11].

(2) According to [20, Lemma 4.6, Lemma 4.8], $\mathcal{B}(M_i^*)$ is a $\mathbb{Z}^\theta$ -graded Hopf algebra with $\deg(M_i^*) = -\alpha_i$ and $\Omega(K_i^M)$ is an $\mathbb{Z}^\theta$ -graded Hopf algebra with $\Omega(K_i^M)(n) = K_i^M(-n)$ for $n \in \mathbb{Z}^\theta$. Under this grading, $\Omega_i(K_i^M) \# \mathcal{B}(M_i^*)$ is an $\mathbb{Z}^\theta$ -graded Hopf algebra in ${}^H_H\mathcal{YD}$. We equip $\mathcal{B}(R_i(M))$ with a new grading by shifting the degree via s_i^M . Explicitly, for $x \in K_i^M$ and $y \in \mathcal{B}(M_i^*)$, we define

$$\deg(x \# y) = s_i^M(\deg^{\mathcal{B}(M)}(x) + \deg(y)).$$

Hence $\deg(\widetilde{M}_i) = s_i^M(-\alpha_i) = \alpha_i$, and for all $j \neq i$,

$$\deg(\widetilde{M}_j) = s_i^M(\alpha_j - a_{ij}^M \alpha_i) = \alpha_j.$$

Hence $\mathcal{B}(R_i(M))$ is an $\mathbb{N}_0$ -graded Hopf algebra with $\deg(\widetilde{M}_j) = \alpha_j$ for all $j \in \mathbb{I}$ under this new grading. $\square$

We consider the action of the isomorphism T mentioned in Proposition 3.4. The Hopf algebra isomorphism T restricts to an algebra isomorphism in ${}^H_H\mathcal{YD}$:

$$T_i^M : L_i^{R_i(M)} = {}^{\text{co } \mathcal{B}(M_i^*)}(\Omega_i(K_i^M) \# \mathcal{B}(M_i^*)) \xrightarrow{\sim} K_i^M. \quad (3.7)$$

Lemma 3.9. (1) With the above notation, for all $j \neq i \in \mathbb{I}$, $0 \leq n \leq m_{ij}^M$, and $x \in \text{ad}(M_i^*)^n(\widetilde{M}_j)$,

$$\begin{aligned} T_i^M(\mathcal{S}_{\mathcal{B}(M)}^{-1}(x)) &= -x, \\ T_i^M(\mathcal{S}_{\mathcal{B}(R_i(M))}^{-1}(\text{ad}(M_i^*)^n(\widetilde{M}_j))) &= \text{ad}(M_i)^{m_{ij}^M - n}(M_j). \end{aligned}$$

(2) T_i^M is an isomorphism of $\mathbb{N}_0^\theta$ -graded objects. In particular

$$\deg(T_i^M(L_i^{R_i(M)}(\alpha))) = s_i^{R_i(M)}(\alpha),$$

where $\alpha \in \mathbb{N}_0^\theta$.

Proof. (1) We have

$$\text{ad}(M_i^*)^n(\widetilde{M}_j) = \begin{cases} 0, & \text{if } n > m_{ij}^M, \\ \text{ad}(M_i)^{m_{ij}^M - n}(M_j) & \text{if } 0 \leq n \leq m_{ij}^M. \end{cases}$$

Since $x \in \text{ad}(M_i^*)^n(\widetilde{M}_j) = \text{ad}(M_i)^{m_{ij}^M - n}(M_j) \subseteq \text{ad}(\mathcal{B}(M_i))(M_j)$, we deduce that x is primitive in K_i^M by [20, Theorem 5.5].

Let $y = \mathcal{S}_{\mathcal{B}(R_i(M))}^{-1}(x)$, we have $y \in L_i^{R_i(M)}$ by equation (3.6). While by definition of T ,

$$T_i^M(y) = \mathcal{S}_{K_i^M}^{-1} \mathcal{S}_{\mathcal{B}(R_i(M))}(y) = \mathcal{S}_{K_i^M}^{-1}(x) = -x.$$

The second equation holds because T_i^M is an isomorphism.

(2) Let $x \in \text{ad}(M_i^*)^n(\widetilde{M}_j)$. Under the $\mathbb{N}_0^\theta$ -grading of $\mathcal{B}(R_i(M))$, we have $\deg(x) = n\alpha_i + \alpha_j$. Since $\mathcal{S}_{\mathcal{B}(R_i(M))}$ preserves $\mathbb{N}_0^\theta$ -grading, $\deg(y) = n\alpha_i + \alpha_j$. By (1) we have

$$\deg(T_i^M(y)) = (m_{ij}^M - n)\alpha_i + \alpha_j = s_i^M(\deg(y)) \in \mathbb{N}_0^\theta.$$

Since M admits the i -th reflection, $R_i(M)$ admits the i -th reflection and $s_i^M = s_i^{R_i(M)}$ by [20, Corollary 5.15]. This implies $\deg(T_i^M(L_i^{R_i(M)}(\alpha))) = s_i^{R_i(M)}(\alpha)$ for each $\alpha \in \mathbb{N}_0^\theta$. $\square$

Proposition 3.10. Assume M admits the i -th reflection for some $i \in \mathbb{I}$.

(1) The map

$$t_i^M : \mathcal{K}_i^-(R_i(M)) \longrightarrow \mathcal{K}_i^+(M), \quad E \longmapsto T_i^M(E)\mathcal{B}(M_i),$$

is bijective with inverse given by $E \longmapsto (T_i^M)^{-1}(E \cap K_i^M)$. Furthermore, the following multiplication map is bijective for all $E \in \mathcal{K}_i^-(R_i(M))$.

$$T_i^M(E) \otimes \mathcal{B}(M_i) \longrightarrow T_i^M(E)\mathcal{B}(M_i).$$

(2) The map

$$\mathfrak{t}_i^M : \mathcal{L}_i^+(R_i(M)) \longrightarrow \mathcal{L}_i^-(M), \quad F \longmapsto T_i^M(F \cap L_i^{R_i(M)}),$$

is bijective with inverse given by $F \longmapsto (T_i^M)^{-1}(F)\mathcal{B}(M_i^*)$. Furthermore, the following multiplication map is bijective for all $F \in \mathcal{L}_i^-(M)$,

$$(T_i^M)^{-1}(F) \otimes \mathcal{B}(M_i^*) \longrightarrow (T_i^M)^{-1}(F)\mathcal{B}(M_i^*).$$

Proof. (1) We apply Proposition 3.5(1) by setting $P = \mathcal{B}(M)$, $Q = \mathcal{B}(R_i(M))$, $\bar{K} = L_i^{R_i(M)}$, $K = K_i^M$ and $B = \mathcal{B}(M_i)$. We obtain a bijection:

$$S_r(\mathcal{B}(R_i(M)), L_i^{R_i(M)}) \longrightarrow S_r^+(\mathcal{B}(M)), \quad E \mapsto T_i^M(E)\mathcal{B}(M_i)$$

with inverse given by $E \mapsto (T_i^M)^{-1}(E \cap K_i^M)$. Since T_i^M preserves $\mathbb{N}_0^\theta$ -grading, the bijection still holds when restricting to $\mathbb{N}_0^\theta$ -graded right coideal subalgebras. By Lemma 3.6, $E \subseteq L_i^{R_i(M)} \iff M_i \not\subseteq E$. Thus, the bijection restricts to the desired bijection

$$t_i^M : \mathcal{K}_i^-(R_i(M)) \longrightarrow \mathcal{K}_i^+(M).$$

The bijectivity of the multiplication map

$$T_i^M(E) \otimes \mathcal{B}(M_i) \longrightarrow T_i^M(E)\mathcal{B}(M_i).$$

for all $E \in \mathcal{K}_i^-(R_i(M))$ follows from Proposition 3.5(1).

(2) It follows from Proposition 3.5(2) and similar argument to (1). $\square$

4 $\mathcal{G}(M)$ is a Cartan graph

In this section, we prove our main result that the semi-Cartan graph $\mathcal{G}(M)$ is a Cartan graph. We proceed by verifying the equivalent axioms (CG3') and (CG4'). We first prove (CG3'), then study the decomposition of Nichols algebras, and finally prove (CG4').

4.1 Proof of (CG3')

For $l \in \mathbb{N}_0$ and $i_1, i_2, \dots, i_l \in \mathbb{I}$, assume that M admits the reflection sequence $(i_1, \dots, i_l)$. The goal of this subsection is to define simple Yetter-Drinfeld modules M_{β_k} corresponding to the roots, and the right coideal subalgebra $E^M(i_1, \dots, i_l)$. We first address the domains of the iterated maps involved in multi-step reflections.

For simplicity, we write

$$R_{(i_1, \dots, i_k)}(M) = R_{i_k}(\dots R_{i_1}(M))$$

inductively for $1 \leq k \leq l$. For clarity, we denote $R_{()} = M$. Now we consider the following two maps:

$$\begin{aligned} T_{i_1}^M : L_{i_1}^{R_{(i_1)}(M)} &\xrightarrow{\sim} K_{i_1}^M, \\ T_{i_2}^{R_{(i_1)}(M)} : L_{i_2}^{R_{(i_1, i_2)}(M)} &\xrightarrow{\sim} K_{i_2}^{R_{(i_1)}(M)}. \end{aligned}$$

To compose these maps, the domain must be restricted to

$$(T_{i_2}^{R_{(i_1)}(M)})^{-1}(K_{i_2}^{R_{(i_1)}(M)} \cap L_{i_1}^{R_{(i_1)}(M)}).$$

Therefore $T_{i_1}^M \circ T_{i_2}^{R_{(i_1)}(M)}$ is well-defined. We apply similar restrictions when considering the composition of t_i^M and $\mathbb{t}_i^M$.

Remark 4.1. With above assumptions, for $1 \leq k \leq l$, we define inductively

$$\begin{aligned} L_{(i_1, \dots, i_k)}^M &= (T_{i_k}^{R_{(i_1, \dots, i_{k-1})}(M)})^{-1} \left(K_{i_k}^{R_{(i_1, \dots, i_{k-1})}(M)} \cap L_{(i_1, \dots, i_{k-1})}^M \right), \\ T_{(i_1, \dots, i_k)}^M &= T_{i_1}^M \circ T_{i_2}^{R_{(i_1)}(M)} \circ \dots \circ T_{i_k}^{R_{(i_1, \dots, i_{k-1})}(M)} : L_{(i_1, \dots, i_k)}^M \longrightarrow K_{i_1}^M, \\ \mathcal{K}_{(i_1, \dots, i_k)}^-(R_{(i_1, \dots, i_k)}(M)) &= (t_{i_k}^{R_{(i_1, \dots, i_{k-1})}(M)})^{-1} \left(\mathcal{K}_{i_k}^+(R_{(i_1, \dots, i_{k-1})}(M)) \cap \mathcal{K}_{(i_1, \dots, i_{k-1})}^-(R_{(i_1, \dots, i_{k-1})}(M)) \right), \\ t_{(i_1, \dots, i_k)}^M &= t_{i_1}^M \circ \dots \circ t_{i_k}^{R_{(i_1, \dots, i_{k-1})}(M)} : \mathcal{K}_{(i_1, \dots, i_k)}^-(R_{(i_1, \dots, i_k)}(M)) \longrightarrow \mathcal{K}_{i_1}^+(M), \\ \mathcal{L}_{(i_1, \dots, i_k)}^+(R_{(i_1, \dots, i_k)}(M)) &= (\mathbb{t}_{i_k}^{R_{(i_1, \dots, i_{k-1})}(M)})^{-1} \left(\mathcal{L}_{i_k}^-(R_{(i_1, \dots, i_{k-1})}(M)) \cap \mathcal{L}_{(i_1, \dots, i_{k-1})}^+(R_{(i_1, \dots, i_{k-1})}(M)) \right), \\ \mathbb{t}_{(i_1, \dots, i_k)}^M &= \mathbb{t}_{i_1}^M \circ \dots \circ \mathbb{t}_{i_k}^{R_{(i_1, \dots, i_{k-1})}(M)} : \mathcal{L}_{(i_1, \dots, i_k)}^+(R_{(i_1, \dots, i_k)}(M)) \longrightarrow \mathcal{L}_{i_1}^-(M). \end{aligned}$$

By convention, we set

$$L_{()}^M = \mathcal{B}(M), \mathcal{K}_{()}^-(M) = \mathcal{K}(M), T_{()}^M = \text{id}, t_{()}^M = \text{id}, \mathcal{L}_{()}^+ = \mathcal{L}(M), \mathbb{t}_{()}^M = \text{id}.$$

From now on, we assume M admits all reflections. Let the reflection sequence $(i_1, \dots, i_l)$ be $[M]$ -reduced in the semi-Cartan graph $\mathcal{G}(M)$. Note that in this case, the reflection sequence $(i_k, \dots, i_l)$ is $[R_{(i_1, \dots, i_{k-1})}(M)]$ -reduced in $\mathcal{G}(R_{(i_1, \dots, i_{k-1})}(M))$ for all $1 \leq k \leq l$ by Remark 2.7 (2). For all $1 \leq k \leq l$, let

$$\beta_k = \text{id}_{[M]} s_{i_1} \cdots s_{i_{k-1}}(\alpha_{i_k}),$$

The following observation is immediate and serves as the base case for our induction. Let $l = 1$, then $\beta_1 = \alpha_{i_1}$.

- $\mathbb{k}1 \in \mathcal{K}_{i_1}^-(R_{i_1}(M))$, it follows from the fact that $M_{i_1}^* \not\subseteq \mathbb{k}1$. Then we have $t^M(\mathbb{k}1) = \mathcal{B}(M_{\beta_1})$.
- $M_{\beta_1} = M_{i_1}$ is a finite-dimensional simple object of degree $\beta_1 = \alpha_{i_1}$.

For $l > 1$, the preceding argument corresponds to the first reflection step from $R_{(i_1, \dots, i_{l-1})}(M)$ to $R_{(i_1, \dots, i_l)}(M)$. By induction, we assume the following:

- For any $2 \leq k \leq l$, if we define $\gamma_k = \text{id}_{[R_{i_1}(M)]} s_{i_2} \cdots s_{i_{k-1}}(\alpha_{i_k})$, then $\gamma_2, \dots, \gamma_l$ are pairwise distinct non-zero elements of $\mathbb{N}_0^\theta$ (note that if $l = 1$ this assumption is trivial).

- (ii) For any $2 \leq k \leq l$, $R_{(i_1, \dots, i_{k-1})}(M)_{i_k} \subseteq L_{(i_2, \dots, i_{k-1})}^{R_{i_1}(M)}$ and $\mathbb{k}1 \in \mathcal{K}_{(i_2, \dots, i_l)}^-(R_{(i_1, \dots, i_l)}(M))$. Therefore the following two objects are well-defined.

$$\begin{aligned}\tilde{M}_{\gamma_k} &:= T_{(i_2, \dots, i_{k-1})}^{R_{i_1}(M)}(R_{(i_1, \dots, i_{k-1})}(M)_{i_k}) \\ E^{R_{i_1}(M)}(i_2, \dots, i_l) &:= t_{(i_2, \dots, i_l)}^{R_{i_1}(M)}(\mathbb{k}1)\end{aligned}$$

for all $2 \leq k \leq l$. (Note that for $k = 1$, it reduces to $M_{i_1} \subseteq \mathcal{B}(M)$, which is trivial.)

- (iii) For any $2 \leq k \leq l$, $\tilde{M}_{\gamma_k} \subseteq E^{R_{i_1}(M)}(i_2, \dots, i_l)$ is a finite-dimensional subobject in ${}^H_H\mathcal{YD}$ of degree γ_k .
- (iv) The multiplication map

$$(\dots (\mathcal{B}(\tilde{M}_{\gamma_l}) \otimes \mathcal{B}(\tilde{M}_{\gamma_{l-1}})) \dots) \otimes \mathcal{B}(\tilde{M}_{\gamma_2}) \longrightarrow E^{R_{i_1}(M)}(i_2, \dots, i_l)$$

is an isomorphism of $\mathbb{N}_0^\theta$ -graded objects in ${}^H_H\mathcal{YD}$.

With the above notation and hypothesis, we establish the following proposition.

Proposition 4.2. (1) For any $1 \leq k \leq l$, $R_{(i_1, \dots, i_{k-1})}(M)_{i_k} \subseteq L_{(i_1, \dots, i_{k-1})}^M$ and $\mathbb{k}1 \in \mathcal{K}_{(i_1, \dots, i_l)}^-(R_{(i_1, \dots, i_l)}(M))$. Therefore the following two objects are well-defined.

$$\begin{aligned}M_{\beta_k} &:= T_{(i_1, \dots, i_{k-1})}^M(R_{(i_1, \dots, i_{k-1})}(M)_{i_k}), \\ E^M(i_1, \dots, i_l) &:= t_{(i_1, \dots, i_l)}^M(\mathbb{k}1)\end{aligned}$$

for all $1 \leq k \leq l$.

- (2) The elements $\beta_1, \dots, \beta_l$ are pairwise distinct non-zero elements of $\mathbb{N}_0^\theta$.
- (3) The multiplication map

$$(\dots (\mathcal{B}(M_{\beta_l}) \otimes \mathcal{B}(M_{\beta_{l-1}})) \dots) \otimes \mathcal{B}(M_{\beta_1}) \longrightarrow E^M(i_1, \dots, i_l)$$

is an isomorphism of $\mathbb{N}_0^\theta$ -graded objects in ${}^H_H\mathcal{YD}$.

- (4) For any $1 \leq k \leq l$, $M_{\beta_k} \subseteq E^M(i_1, \dots, i_l)$ is a finite-dimensional subobject in ${}^H_H\mathcal{YD}$ of degree β_k .

Proof. Since the sequence $(i_1, \dots, i_l)$ is $[M]$ -reduced, it follows from the definition that

$$\alpha_{i_1} \notin \Lambda^{R_{i_1}(M)}(i_2, \dots, i_l) = \{\gamma_2, \dots, \gamma_l\}.$$

For $2 \leq k \leq l$, assumption (ii) and (iii) say $\tilde{M}_{\gamma_k}$ has degree γ_k . By assumption (iv), we deduce that

$$\tilde{M}_{i_1} = R_{i_1}(M)_{i_1} \not\subseteq E^{R_{i_1}(M)}(i_2, \dots, i_l).$$

Therefore by definition of $\mathcal{K}_{i_1}^-(R_{i_1}(M))$, we have

$$E^{R_{i_1}(M)}(i_2, \dots, i_l) \in \mathcal{K}_{i_1}^-(R_{i_1}(M)). \quad (4.1)$$

That is, $t_{(i_2, \dots, i_l)}^{R_{i_1}(M)}(\mathbb{k}1) \in \mathcal{K}_{i_1}^-(R_{i_1}(M))$, hence $\mathbb{k}1 \in \mathcal{K}_{(i_1, \dots, i_l)}^-(R_{(i_1, \dots, i_l)}(M))$ and $E^M(i_1, \dots, i_l) := t_{(i_1, \dots, i_{k-1})}^M(\mathbb{k}1)$ is well-defined. By Lemma 3.6,

$$\tilde{M}_{i_1} = R_{i_1}(M)_{i_1} \not\subseteq E^{R_{i_1}(M)}(i_2, \dots, i_l) \Leftrightarrow E^{R_{i_1}(M)}(i_2, \dots, i_l) \subseteq L_{i_1}^{R_{i_1}(M)}.$$

By assumption (iii), $\tilde{M}_{\gamma_k} \subseteq L_{i_1}^{R_{i_1}(M)}$ for any $2 \leq k \leq l$, hence

$$T_{i_1}^M(\tilde{M}_{\gamma_k}) = T_{i_1}^M(T_{(i_2, \dots, i_{k-1})}^{R_{i_1}(M)}(R_{(i_1, \dots, i_{k-1})}(M)_{i_k})) \subseteq K_{i_1}^M,$$

which implies $R_{(i_1, \dots, i_{k-1})}(M)_{i_k} \subseteq L_{(i_1, \dots, i_{k-1})}^M$ and $M_{\beta_k} := T_{(i_1, \dots, i_{k-1})}^M(R_{(i_1, \dots, i_{k-1})}(M)_{i_k})$ is well-defined. So we have proved (1).

Since $T_{i_1}^M$ is an isomorphism of $\mathbb{N}_0^\theta$ -graded objects and $\deg(T_{i_1}^M(L_{i_1}^{R_{i_1}(M)}(\alpha))) = s_{i_1}^{R_{i_1}(M)}(\alpha)$, by Lemma 3.9 (2), then

$$\beta_k = s_{i_1}^{R_{i_1}(M)}(\gamma_k) \in \mathbb{N}_0^\theta$$

are pairwise distinct by assumption (i) for $2 \leq k \leq l$. If $\beta_1 = \beta_i$ for some $2 \leq i \leq l$, then $-\beta_1 = \gamma_i \in \mathbb{N}_0^\theta$, which yields a contradiction. This establishes (2).

Recall (4.1), $E^{R_{i_1}(M)}(i_2, \dots, i_l) \in \mathcal{K}_{i_1}^-(R_{i_1}(M))$, hence by Proposition 3.10, we have a bijective map:

$$T_{i_1}^M(E^{R_{i_1}(M)}(i_2, \dots, i_l)) \otimes \mathcal{B}(M_{i_1}) \rightarrow t_{i_1}^M(E^{R_{i_1}(M)}(i_2, \dots, i_l)) = E^M(i_1, \dots, i_l).$$

The map T itself is an algebra isomorphism. By assumption (iv), we have a bijective map

$$(\dots (T_{i_1}^M(\mathcal{B}(\tilde{M}_{\gamma_l})) \otimes T_{i_1}^M(\mathcal{B}(\tilde{M}_{\gamma_{l-1}}))) \dots) \otimes \mathcal{B}(M_{i_1}) \longrightarrow E^M(i_1, \dots, i_l).$$

By above construction, for $2 \leq k \leq l$, we have $M_{\beta_k} = T_{i_1}^M(\tilde{M}_{\gamma_k})$. Then

$$(\dots (\mathcal{B}(M_{\beta_l}) \otimes \mathcal{B}(M_{\beta_{l-1}})) \dots) \otimes \mathcal{B}(M_{\beta_1}) \longrightarrow E^M(i_1, \dots, i_l)$$

is an isomorphism of $\mathbb{N}_0^\theta$ -graded objects in ${}^H_H\mathcal{YD}$, which shows (3).

Note that $M_{\beta_1} = M_{i_1} \subseteq E^M(i_1, \dots, i_l)$ is simple. By above construction, for $2 \leq k \leq l$, we have

$$M_{\beta_k} = T_{i_1}^M(\tilde{M}_{\gamma_k}) \in E^M(i_1, \dots, i_l).$$

Since T is an isomorphism and $\tilde{M}_{\gamma_k}$ is finite-dimensional and simple in ${}^H_H\mathcal{YD}$ by assumption (iii), we deduce that for any $1 \leq k \leq l$, $M_{\beta_k} \subseteq E^M(i_1, \dots, i_l)$ is a finite-dimensional simple object in ${}^H_H\mathcal{YD}$ of degree β_k . This implies (4). $\square$

We now proceed to verify (CG3').

Theorem 4.3. *Let $M \in \mathcal{F}_\theta$, with each M_i is simple for $i \in \mathbb{I}$. Assume M admits all reflections, then for any $[X] \in \mathcal{F}_\theta(M)$ and any $[X]$ -reduced sequence κ , we have*

$$\Lambda^{[X]}(\kappa) \in \mathbb{N}_0^\theta.$$

Proof. For any $X \in \mathcal{F}_\theta(M)$, there is a sequence $(i_1, \dots, i_l)$ such that $[X] = [R_{(i_1, \dots, i_l)}(M)]$. Clearly, the tuple X admits all reflections and gives rise to a semi-Cartan graph $\mathcal{G}(X)$. For any $[X]$ -reduced sequence κ , we have $\Lambda^{[X]}(\kappa) \subseteq \mathbb{N}_0^\theta$ by Proposition 4.2 (2). This proves (CG3'). $\square$

4.2 Decomposition of Nichols algebras

Let $M \in \mathcal{F}_\theta$. Assume that M admits all reflections. Let $\kappa = (i_1, \dots, i_l) \in \mathbb{I}^l$ be an $[M]$ -reduced sequence. Recall that in Remark 4.1, we defined

$$\mathfrak{t}_{(i_1, \dots, i_k)}^M = \mathfrak{t}_{i_1}^M \circ \dots \circ \mathfrak{t}_{i_k}^{R_{(i_1, \dots, i_{k-1})}(M)} : \mathcal{L}_{(i_1, \dots, i_k)}^+(R_{(i_1, \dots, i_k)}(M)) \longrightarrow \mathcal{L}_{i_1}^-(M).$$

The goal in this subsection is to give a decomposition of $\mathcal{B}(M)$ by using $\mathfrak{t}^M$ and $E^M(\kappa)$.

Lemma 4.4. *With above assumption, we have $\mathcal{B}(R_\kappa(M)) \in \mathcal{L}_\kappa^+(R_\kappa(M))$. Furthermore, the multiplication map*

$$\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M))) \otimes E^M(\kappa) \rightarrow \mathcal{B}(M)$$

is bijective in ${}^H_H\mathcal{YD}$.

Proof. We proceed by induction on l . For $l = 1$, $M_{i_1}^* \subseteq \mathcal{B}(R_{i_1}(M))$ holds automatically and

$$\mathfrak{t}_{i_1}^M(\mathcal{B}(R_{i_1}(M))) = T_{i_1}^M(\mathcal{B}(R_{i_1}(M))) \cap \mathcal{L}_{i_1}^{R_{i_1}(M)} = K_{i_1}^M.$$

Recall $E^M(i_1) = \mathcal{B}(M_{i_1})$ in this case, hence the multiplication map $\mathfrak{t}_{i_1}^M(\mathcal{B}(R_{i_1}(M))) \otimes E^M(i_1) \rightarrow \mathcal{B}(M)$ is bijective.

Now for $l \geq 2$, we denote $M' = R_{i_1}(M)$ and $\kappa' = (i_2, \dots, i_l)$. We assume that $\mathcal{B}(R_\kappa(M)) \in \mathcal{L}_{\kappa'}^+(R_{\kappa'}(M'))$ and the multiplication map $\mathfrak{t}_{\kappa'}^{M'}(\mathcal{B}(R_\kappa(M))) \otimes E^{M'}(\kappa') \rightarrow \mathcal{B}(M')$ is bijective. From our assumptions, $\mathfrak{t}_{\kappa'}^{M'}(\mathcal{B}(R_\kappa(M))) \in \mathcal{L}_{i_2}^-(M')$. Note that $M'_{i_1} \not\subseteq E^{M'}(\kappa')$. Because the multiplication map $\mathfrak{t}_{\kappa'}^{M'}(\mathcal{B}(R_\kappa(M))) \otimes E^{M'}(\kappa') \rightarrow \mathcal{B}(M')$ is bijective, it follows that $M'_{i_1} \subseteq \mathfrak{t}_{\kappa'}^{M'}(\mathcal{B}(R_\kappa(M)))$. Hence $\mathfrak{t}_{\kappa'}^{M'}(\mathcal{B}(R_\kappa(M))) \in \mathcal{L}_{i_1}^+(M')$, $\mathfrak{t}_{i_1}^M(\mathfrak{t}_{\kappa'}^{M'}(\mathcal{B}(R_\kappa(M)))) \in \mathcal{L}_{i_1}^-(M)$ and $\mathcal{B}(R_\kappa(M)) \in \mathcal{L}_\kappa^+(R_\kappa(M))$. By Lemma 3.6 (2), the condition $\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M))) \in \mathcal{L}_{i_1}^-(M)$ is equivalent to

$$\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M))) \subseteq K_{i_1}^M.$$

By Proposition 4.2 (4) and 3.10 (1)

$$E^M(\kappa) \cong T_{i_1}^M(E^{M'}(\kappa')) \otimes \mathcal{B}(M_{i_1}).$$

To show $\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M))) \otimes E^M(\kappa) \rightarrow \mathcal{B}(M)$ is bijective, it suffices to show $\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M))) \otimes T_{i_1}^M(E^{M'}(\kappa'))$ is isomorphic to $K_{i_1}^M$ in ${}^H_H\mathcal{YD}$. Since $T_{i_1}^M$ is an algebra isomorphism, it suffices to show

$$(T_{i_1}^M)^{-1}(\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M)))) \otimes E^{M'}(\kappa') \rightarrow L_{i_1}^{M'} \quad (4.2)$$

is an isomorphism. By Proposition 4.2(2), and the definition of $T_{i_1}^M$,

$$(T_{i_1}^M)^{-1}(\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M)))) \otimes E^{M'}(\kappa') \subseteq L_{i_1}^{M'}.$$

Hence (4.2) is an injective map in ${}^H_H\mathcal{YD}$.

The multiplication map $(T_{i_1}^M)^{-1}(\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M)))) \otimes \mathcal{B}(M'_{i_1}) \rightarrow \mathfrak{t}_{\kappa'}^{M'}(\mathcal{B}(R_\kappa(M)))$ is bijective. Hence

$$((T_{i_1}^M)^{-1}(\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M)))) \otimes \mathcal{B}(M'_{i_1})) \otimes E^{M'}(\kappa') \rightarrow \mathcal{B}(M')$$

is bijective. Now fix any $\alpha \in \mathbb{N}_0^\theta$,

$$\dim \mathcal{B}(M')(\alpha) = \sum_{k=0}^{\infty} \dim((T_{i_1}^M)^{-1}(\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M)))) \otimes E^{M'}(\kappa'))(\alpha - k\alpha_{i_1}) \dim \mathcal{B}(M'_{i_1})(k\alpha_{i_1}).$$

On the other hand, for any $\alpha \in \mathbb{N}_0^\theta$, we have

$$\begin{aligned} \dim \mathcal{B}(M')(\alpha) &= \sum_{k=0}^{\infty} \dim K_{i_1}^{M'}(\alpha - k\alpha_{i_1}) \dim \mathcal{B}(M'_{i_1})(k\alpha_{i_1}) \\ &= \sum_{k=0}^{\infty} \dim L_{i_1}^{M'}(\alpha - k\alpha_{i_1}) \dim \mathcal{B}(M'_{i_1})(k\alpha_{i_1}), \end{aligned}$$

where the second equation follows by (3.6). By comparing dimensions, we deduce that for any $\alpha \in \mathbb{N}_0^\theta$, $L_{i_1}^{M'}(\alpha) = \dim((T_{i_1}^M)^{-1}(\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M)))) \otimes E^{M'}(\kappa'))(\alpha)$. Therefore (4.2) is bijective and

$$\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M))) \otimes E^M(\kappa) \rightarrow \mathcal{B}(M)$$

is an isomorphism of $\mathbb{N}_0^\theta$ -graded objects in ${}^H_H\mathcal{YD}$. $\square$

The next proposition characterizes $\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M)))$ and $E^M(\kappa)$ for a special reduced sequence $\kappa = \kappa_{ij} = (i, j, i, j, \dots)$, where $i \neq j \in \mathbb{I}$.

Proposition 4.5. *Assume that $\theta \geq 2$. Let $i, j \in \mathbb{I}$ with $i \neq j$. Assume M admits all reflections and $\overline{m}_{ij}^{[M]} < \infty$. Then*

$$E^M(\kappa) = \mathcal{B}(M_i + M_j), \quad \mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M))) = \mathcal{B}(M)^{\text{co } \mathcal{B}(M_i + M_j)}.$$

Proof. We know that $\mathcal{G}(M)$ is a semi-Cartan graph satisfying (CG3') by Theorem 4.3. Since κ only involves i and j , $\beta_k^{[M], \kappa}$ must be linear combination of α_i and α_j for all $1 \leq k \leq \overline{m}_{ij}^{[M]}$. According to Proposition 4.2(4), we have

$$E^M(\kappa) \subseteq \bigoplus_{k_1, k_2 \geq 0} \mathcal{B}(M)(k_1 \alpha_i + k_2 \alpha_j) = \mathcal{B}(M_i + M_j).$$

Conversely, by Lemma 2.10 (3), $\beta_1^{[M], \kappa} = \alpha_i$ and $\beta_{\overline{m}_{ij}^{[M]}}^{[M], \kappa} = \alpha_j$. According to Proposition 4.2(4), we have $M_i + M_j \subseteq E^M(\kappa)$, and then

$$\mathcal{B}(M_i + M_j) \subseteq E^M(\kappa),$$

since $E^M(\kappa)$ is a coideal subalgebra. Hence $E^M(\kappa) = \mathcal{B}(M_i + M_j)$.

For the second statement, we first claim that $\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M))) \subseteq \mathcal{B}(M)^{\text{co } \mathcal{B}(M_i + M_j)}$. Without loss of generality, assume $i < j$ and set $M' = (M_1, \dots, M_i + M_j, \dots, M_{j-1}, M_{j+1}, \dots, M_\theta)$. Since the multiplication map

$$\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M))) \otimes \mathcal{B}(M_i + M_j) \rightarrow \mathcal{B}(M)$$

is bijective, we have $\pi_i^{M'}(\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M)))) \cap (M_i + M_j) = 0$. Note that $\pi_i^{M'}(\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M))))$ is a nonzero left coideal subalgebra of $\mathcal{B}(M_i + M_j)$ and $\mathcal{B}(M_i + M_j)$ is a strictly graded coalgebra. Hence, $\pi_i^{M'}(\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M)))) = \mathbb{k}1$. For any $x \in \mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M)))$, this implies $\pi_i^{M'}(x) = \epsilon(\pi_i^{M'}(x)) = \epsilon(x)$. Therefore $\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M))) \subseteq \mathcal{B}(M)^{\text{co } \pi_i^{M'}} = \mathcal{B}(M)^{\text{co } \mathcal{B}(M_i + M_j)}$. Since

$$\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M))) \otimes \mathcal{B}(M_i + M_j) \cong \mathcal{B}(M) \cong \mathcal{B}(M)^{\text{co } \mathcal{B}(M_i + M_j)} \otimes \mathcal{B}(M_i + M_j).$$

We deduce that

$$\mathfrak{t}_\kappa^M(\mathcal{B}(R_\kappa(M))) = \mathcal{B}(M)^{\text{co } \mathcal{B}(M_i + M_j)}.$$

□

4.3 Proof of (CG4')

The main result of this section is the following theorem.

Theorem 4.6. *Let $M \in \mathcal{F}_\theta$. Assume that M admits all reflections, then $\mathcal{G}(M)$ satisfies (CG4') and is therefore a Cartan graph.*

For $i \neq j \in \mathbb{I}$, Let $\kappa_{ij}^{[M]}$ be the $[M]$ -reduced sequence beginning of $(i, j, i, j, \dots)$. Assuming $m := \overline{m}_{ij}^{[M]} < \infty$, we rewrite $\kappa_{ij}^{[M]} = (i_1, i_2, \dots, i_m)$. Now we denote $\kappa_{ji}^{[M]} = (i_2, \dots, i_m, i_{m+1})$, where $i_{m+1} = i_{m-1}$. By Lemma 2.10 (2), $\kappa_{ji}^{[M]}$ is $[M]$ -reduced. To consider $(r_i r_j)^m$, it is natural to consider the composition of these two reflection sequence: $\kappa_{ij}^{[M]} \circ (\kappa_{ji}^{[M]})^{-1}$.

Lemma 4.7. *Under the above notation, we consider two following reflection sequences:*

$$\begin{aligned}\kappa' &= (i_1, \dots, i_m, k), \\ \kappa'' &= (i_2, \dots, i_{m+1}, k)\end{aligned}$$

where $k \neq i, j$. Then κ' and κ'' are $[M]$ -reduced sequences.

Proof. Since $\kappa_{ij}^{[M]}$ is $[M]$ -reduced, $\beta_p^{[M], \kappa_{ij}^{[M]}} \neq \beta_q^{[M], \kappa_{ij}^{[M]}}$ for all $1 \leq p < q \leq m$. It is straightforward to see that for any $1 \leq n \leq m$, $\beta_n^{[M], \kappa'} \in \mathbb{Z}\alpha_i + \mathbb{Z}\alpha_j$ and $\beta_{m+1}^{[M], \kappa'} = \text{id}_{[M]} s_{i_1} \cdots s_{i_m}(\alpha_k) \in \alpha_k + \mathbb{Z}\alpha_i + \mathbb{Z}\alpha_j$. Hence,

$$\beta_p^{[M], \kappa'} \neq \beta_q^{[M], \kappa'}$$

for all $1 \leq p < q \leq m+1$. Therefore, by Remark 2.7 (3), κ' is $[M]$ -reduced. The proof that κ'' is $[M]$ -reduced is analogous. $\square$

Now we define $M' = R_{\kappa_{ij}^{[M]}}(M)$, and $M'' = R_{\kappa_{ji}^{[M]}}(M)$. By Proposition 4.2(2), $M'_k := R_{(i_1, \dots, i_m)}(M)_k \subseteq L_{(i_1, \dots, i_m)}^M$, and $M''_k := R_{(i_2, \dots, i_{m+1})}(M)_k \subseteq L_{(i_2, \dots, i_{m+1})}^M$. Furthermore, recall that $\mathfrak{t}_i^{\mathcal{N}} : \mathcal{L}_i^+(R_i(\mathcal{N})) \longrightarrow \mathcal{L}_i^-(\mathcal{N})$, $F \longmapsto T_i^{\mathcal{N}}(F \cap L_i^{R_i(\mathcal{N})})$. Therefore, by Proposition 4.5, we have

$$\begin{aligned}T_{(i_1, \dots, i_m)}^M(M'_k) &\subseteq T_{(i_1, \dots, i_m)}^M(\mathcal{B}(M') \cap L_{(i_1, \dots, i_m)}^M) = \mathfrak{t}_{(i_1, \dots, i_m)}^M(\mathcal{B}(M')) = \mathcal{B}(M)^{\text{co } \mathcal{B}(M_i + M_j)}, \\ T_{(i_2, \dots, i_{m+1})}^M(M''_k) &\subseteq T_{(i_2, \dots, i_{m+1})}^M(\mathcal{B}(M'') \cap L_{(i_2, \dots, i_{m+1})}^M) = \mathfrak{t}_{(i_2, \dots, i_{m+1})}^M(\mathcal{B}(M'')) = \mathcal{B}(M)^{\text{co } \mathcal{B}(M_i + M_j)}.\end{aligned}$$

With above preparations, we can now prove our main result.

Proof of Theorem 4.6. Regardless of whether $i_m = i$ or $i_m = j$, the fact that $T_{(i_1, \dots, i_m)}^M(M'_k)$, $T_{(i_2, \dots, i_{m+1})}^M(M''_k) \subseteq \mathcal{B}(M)^{\text{co } \mathcal{B}(M_i + M_j)}$ guarantee that the compositions $(T_{(i_2, \dots, i_{m+1})}^M)^{-1} \circ T_{(i_1, \dots, i_m)}^M(M'_k)$ and $(T_{(i_1, \dots, i_m)}^M)^{-1} \circ T_{(i_2, \dots, i_{m+1})}^M(M''_k)$ are well-defined. Since T preserves $\mathbb{N}_0^\theta$ -grading and irreducibility, $(T_{(i_2, \dots, i_{m+1})}^M)^{-1} \circ T_{(i_1, \dots, i_m)}^M(M'_k)$ is an $\mathbb{N}_0^\theta$ -graded subobject of $\mathcal{B}(M'')$, $(T_{(i_1, \dots, i_m)}^M)^{-1} \circ T_{(i_2, \dots, i_{m+1})}^M(M''_k)$ is an $\mathbb{N}_0^\theta$ -graded subobject of $\mathcal{B}(M')$. For all $l \in \mathbb{I}$, we have

$$s_{i_{m+1}} s_{i_m} \cdots s_{i_2} s_{i_1} \cdots s_{i_m}(\alpha_l) = (s_{i_{m+1}} s_{i_m})^m(\alpha_l),$$

$$s_{i_m} s_{i_{m+1}} \cdots s_{i_1} s_{i_2} \cdots s_{i_{m+1}}(\alpha_l) = (s_{i_m} s_{i_{m+1}})^m(\alpha_l).$$

By Lemma 2.10 (3), we have $(s_{i_{m+1}} s_{i_m})^m(\alpha_i) = \alpha_i$ and $(s_{i_{m+1}} s_{i_m})^m(\alpha_j) = \alpha_j$. On the other hand, since T preserves $\mathbb{N}_0^\theta$ -grading, we have

$$(s_{i_{m+1}} s_{i_m})^m(\alpha_k), (s_{i_m} s_{i_{m+1}})^m(\alpha_k) \in \alpha_k + \mathbb{N}_0 \alpha_i + \mathbb{N}_0 \alpha_j.$$

We assume $(s_{i_{m+1}} s_{i_m})^m(\alpha_k) = \alpha_k + a\alpha_i + b\alpha_j$ for $a, b \in \mathbb{N}_0$. Then

$$(s_{i_m} s_{i_{m+1}})^m \circ (s_{i_{m+1}} s_{i_m})^m(\alpha_k) = \alpha_k$$

implies

$$(s_{i_m} s_{i_{m+1}})^m(\alpha_k) = \alpha_k - a\alpha_i - b\alpha_j.$$

By the above observations, the degree condition forces $a = b = 0$. Hence

$$(s_{i_{m+1}} s_{i_m})^m(\alpha_k) = (s_{i_m} s_{i_{m+1}})^m(\alpha_k) = \alpha_k,$$

which implies:

$$\begin{aligned}(T_{(i_1, \dots, i_m)}^M)^{-1} \circ T_{(i_2, \dots, i_{m+1})}^M(M''_k) &\subseteq M'_k, \\ (T_{(i_2, \dots, i_{m+1})}^M)^{-1} \circ T_{(i_1, \dots, i_m)}^M(M'_k) &\subseteq M''_k.\end{aligned}$$

Therefore $M'_k \cong M''_k$ for $k \neq i, j$.

We now proceed to show: $M'_i \cong M''_i$ and $M'_j \cong M''_j$. Note that $M'_{i_m} = R_{(i_1, \dots, i_m)}(M)_{i_m} = R_{(i_1, \dots, i_{m-1})}(M)_{i_m}^*$. According to Lemma 2.10 (1), we have $\beta_m^{[M], \kappa} = \alpha_j$. Hence $M'_{i_m} \cong M_j^*$. On the other hand, $M'_{i_{m+1}} = R_{(i_1, \dots, i_m)}(M)_{i_{m+1}} = R_{(i_2, \dots, i_m)}(R_{i_1}(M))_{i_{m+1}}$. Then we have $M'_{i_{m+1}} \cong M_i^*$. Similarly,

$$\begin{aligned} M''_{i_{m+1}} &= R_{(i_2, \dots, i_{m+1})}(M)_{i_{m+1}} = R_{(i_2, \dots, i_m)}(M)_{i_{m+1}}^*, \\ M''_{i_m} &= R_{(i_2, \dots, i_{m+1})}(M)_{i_m} = R_{(i_3, \dots, i_{m+1})}(R_{i_2}(M))_{i_m}. \end{aligned}$$

Therefore $M''_{i_{m+1}} \cong M_i^*$ and $M''_{i_m} \cong M_j^*$. As a result $[M'] = [M'']$. Finally, we have

$$(r_i r_j)^m([M]) = [M], \quad \text{id}_{[M]}(s_i s_j)^m(\alpha_k) = \alpha_k$$

for all $k \in \mathbb{I} \setminus \{i, j\}$.

Now for any $[X] \in \mathcal{X}$, since M admits all reflections, there is a reflection sequence $(j_1, \dots, j_l)$ such that $[X] = [R_{(j_1, \dots, j_l)}(M)]$. It is obvious that $[X]$ admits all reflections and the above arguments hold for $[X]$. Therefore (CG4') holds for any $[X] \in \mathcal{X}$ and $\mathcal{G}(M)$ is a Cartan graph. $\square$

Corollary 4.8. *With the above assumptions. Then the pair $(\mathcal{G}(M), (\Delta^{[X], \text{re}})_{[X] \in \mathcal{X}})$ is a root system over $\mathcal{G}(M)$.*

Proof. This corollary follows from the definition of the Cartan graph and $\Delta^{[X], \text{re}}$. $\square$

5 Nichols algebras with finite Cartan graphs

In this section, we study Nichols algebras with finite Cartan graphs. We introduce tensor decomposable Nichols algebras over coquasi-Hopf algebras and use this concept to give a criterion for the finite-dimensionality of $\mathcal{B}(M)$.

5.1 Tensor decomposable Nichols algebras

Definition 5.1. *Let V be an $\mathbb{N}_0^\theta$ -graded object in ${}^H_H \mathcal{YD}$. We say V is tensor decomposable if there exist an integer $n \geq 0$, finite-dimensional and simple Yetter-Drinfeld modules $Q_1, \dots, Q_n \in {}^H_H \mathcal{YD}$ such that*

$$V \cong \bigotimes_{l=1}^n \mathcal{B}(Q_l) := (\cdots (\mathcal{B}(Q_1) \otimes \mathcal{B}(Q_2)) \otimes \cdots) \otimes \mathcal{B}(Q_n)$$

as $\mathbb{N}_0^\theta$ -graded objects in ${}^H_H \mathcal{YD}$. Moreover, the degree of Q_l satisfies $\deg(Q_l) = \beta_l$, such that $\beta_1, \dots, \beta_l$ are pairwise distinct elements in $\mathbb{N}_0^\theta \setminus \{0\}$.

For $M \in \mathcal{F}_\theta$, the Nichols algebra $\mathcal{B}(M)$ is called tensor decomposable if $\mathcal{B}(M)$ is tensor decomposable as an $\mathbb{N}_0^\theta$ -graded object in ${}^H_H \mathcal{YD}$ with the standard grading $\deg(M_i) = \alpha_i$.

For tensor decomposable objects, the following definition is useful.

Definition 5.2. *Let $s \in \text{Aut}(\mathbb{Z}^\theta)$, $\alpha \in \mathbb{Z}^\theta$ and $Q \in {}^H_H \mathcal{YD}$. Let V be a tensor decomposable $\mathbb{N}_0$ -graded object in ${}^H_H \mathcal{YD}$, we define:*

$$\begin{aligned} s([Q], \alpha) &= ([Q], s(\alpha)), \\ \Phi_+^V &= \{([Q_l], \beta_l) \mid 1 \leq l \leq n\}, \\ \Phi_-^V &= \{([Q_l^*], -\beta_l) \mid 1 \leq l \leq n\}, \\ \Phi^V &= \Phi_+^V \cup \Phi_-^V. \end{aligned}$$

Here Q_l and β_l , $1 \leq l \leq n$, are the simple objects and their degrees, respectively, in the tensor decomposition of V .

To show that this definition is well-defined, we consider the $\mathbb{N}_0$ -grading on $V = \bigoplus_{\alpha \in \mathbb{N}_0^\theta} V(\alpha)$. For any $\alpha \in \mathbb{N}_0^\theta$, $\alpha = \sum_{i=1}^\theta n_i \alpha_i$ with $n_i \geq 0$, let $|\alpha| = \sum_{i=1}^\theta n_i$. We define

$$V(n) = \bigoplus_{|\alpha|=n} V(\alpha).$$

Then $V = \bigoplus_{n \geq 0} V(n)$ is a $\mathbb{N}_0$ -grading.

Lemma 5.3. *Let U, V and W be $\mathbb{N}_0^\theta$ -graded objects in ${}^H_H\mathcal{YD}$ with finite-dimensional homogeneous components. Assume that $W(0) \cong \mathbb{k}$ in ${}^H_H\mathcal{YD}$. If*

$$U \otimes W \cong V \otimes W \quad \text{or} \quad W \otimes U \cong W \otimes V$$

as $\mathbb{N}_0^\theta$ -graded objects in ${}^H_H\mathcal{YD}$, then $U \cong V$ as $\mathbb{N}_0^\theta$ -graded objects in ${}^H_H\mathcal{YD}$.

Proof. The proof is the same as in [15, Lemma 14.4.4] and is omitted here. $\square$

Lemma 5.4. *Let V, W be a tensor decomposable $\mathbb{N}_0$ -graded object in ${}^H_H\mathcal{YD}$ with finite-dimensional homogeneous components. Assume*

$$V \cong \bigotimes_{l=1}^n \mathcal{B}(Q_l) \cong \bigotimes_{k=1}^m \mathcal{B}(P_k).$$

Then $n = m$, and there is a permutation $\sigma \in S_n$ such that $P_l \cong Q_{\sigma(l)}$ as $\mathbb{N}_0^\theta$ -graded object in ${}^H_H\mathcal{YD}$. Furthermore, $V \cong W$ if and only if $\Phi^V = \Phi^W$.

Proof. Let $\mathbb{I}_n = \{1, 2, \dots, n\}$ and $\mathbb{I}_m = \{1, 2, \dots, m\}$, $r = \min\{|\deg(Q_l)| \mid 1 \leq l \leq n\}$. Let L be the set of all l such that $1 \leq l \leq n$ and $|\deg(Q_l)| = r$. Then $\bigoplus_{l \in L} Q_l$ is the $\mathbb{N}_0$ -homogeneous component of $\bigotimes_{l=1}^n \mathcal{B}(Q_l)$ of minimal positive degree. According to $\bigotimes_{l=1}^n \mathcal{B}(Q_l) \cong \bigotimes_{k=1}^m \mathcal{B}(P_k)$, we have $r = \min\{|\deg(P_k)| \mid 1 \leq k \leq m\}$ and $\bigoplus_{l \in L} Q_l \cong \bigoplus_{k \in K} P_k$, where K is the set of all k such that $1 \leq k \leq m$ and $|\deg(P_k)| = r$. Since Q_l and P_k are finite-dimensional simple Yetter-Drinfeld modules, their endomorphism rings are local. Thus, the Krull-Schmidt theorem applies, yielding $|L| = |K|$ and there is a permutation $\sigma_r \in S_{|L|}$ such that $Q_l \cong P_{\sigma_r(k)}$.

Now we have two such decompositions since ${}^H_H\mathcal{YD}$ is braided

$$\begin{aligned} \bigotimes_{l=1}^n \mathcal{B}(Q_l) &\cong \left(\bigotimes_{l' \in \mathbb{I}_n \setminus L} \mathcal{B}(Q_{l'}) \right) \otimes \left(\bigotimes_{l'' \in L} \mathcal{B}(Q_{l''}) \right), \\ \bigotimes_{k=1}^m \mathcal{B}(P_k) &\cong \left(\bigotimes_{k' \in \mathbb{I}_m \setminus K} \mathcal{B}(P_{k'}) \right) \otimes \left(\bigotimes_{k'' \in K} \mathcal{B}(P_{k''}) \right). \end{aligned}$$

Let $n \geq 0$. Since $\bigotimes_{l'' \in L} \mathcal{B}(Q_{l''})(0)$ is the trivial object $\mathbb{k}$ and each homogeneous component of V is finite-dimensional, we have

$$\left(\bigotimes_{l' \in \mathbb{I}_n \setminus L} \mathcal{B}(Q_{l'}) \right) \cong \left(\bigotimes_{k' \in \mathbb{I}_m \setminus K} \mathcal{B}(P_{k'}) \right)$$

by Lemma 5.3. Proceeding similarly, we deduce that $n = m$ and there exists a permutation $\sigma \in S_n$ such that $P_l \cong Q_{\sigma(l)}$ as $\mathbb{N}_0$ -graded objects in ${}^H_H\mathcal{YD}$.

For the second statement, it is obvious that the condition $\Phi^V = \Phi^W$ is equivalent to $V \cong W$. $\square$

Proposition 5.5. *Let $M \in \mathcal{F}_\theta$. Assume that M admits all reflections and $\mathcal{B}(P)$ is tensor decomposable for some $[P] \in \mathcal{F}_\theta(M)$.*

- (1) *For each $X \in \mathcal{F}_\theta(M)$, $\mathcal{B}(X)$ is tensor decomposable.*
- (2) *For any $X, Y \in \mathcal{F}_\theta$ and any morphism $\omega : [X] \rightarrow [Y]$ in $\mathcal{W}(M)$. Then $\omega(\Phi^{\mathcal{B}(X)}) = \Phi^{\mathcal{B}(Y)}$.*
- (3) *For any $\alpha \in \Delta^{M, \text{re}}$, there exists $X \in {}^H_H\mathcal{YD}$ with $([X], \alpha) \in \Phi^{\mathcal{B}(M)}$.*

Proof. (1) By assumption, $\mathcal{B}(P)$ is tensor decomposable, we may write

$$\mathcal{B}(P) \cong \bigotimes_{k=0}^n \mathcal{B}(Q_k).$$

Let us fix $i \in \mathbb{I}$, since $\mathcal{B}(P_i)$ is simple of degree α_i . We can assume $Q_0 = P_i$. Let us consider simple reflection R_i . Note that $\mathcal{B}(P) \cong K_i^P \otimes \mathcal{B}(P_i)$, where $K_i^P = \mathcal{B}(P)^{\text{co } \mathcal{B}(P_i)}$. The multiplication map

$$K_i^P \otimes \mathcal{B}(P_i) \rightarrow \mathcal{B}(P) \cong \bigotimes_{k=0}^n \mathcal{B}(Q_k)$$

is bijective and $\mathbb{N}_0^\theta$ -graded. By a similar method in Lemma 5.4, $K_i^P \cong \bigotimes_{k=1}^n \mathcal{B}(Q_k)$ as $\mathbb{N}_0^\theta$ -graded objects.

Recall T_i^P defines an isomorphism

$$L_i^{R_i(P)} \cong (K_i^P)'$$

of $\mathbb{N}_0^\theta$ -graded objects in ${}^H_H\mathcal{YD}$, where $(K_i^P)' = K_i^P$ as Yetter–Drinfeld modules, and $(K_i^P)'(\alpha) = K_i^P(s_i^{R_i(M)}(\alpha))$ for all $\alpha \in \mathbb{N}_0^\theta$. Denote the degree of Q_k as β_k for $k \geq 1$. We have

$$\begin{aligned} \bigotimes_{k=1}^n \mathcal{B}'(Q_k) &\cong \bigotimes_{k=1}^n \left(\bigoplus_{l \geq 0} (\mathcal{B}'(Q_k)(s_i^{R_i(M)}(l\beta_k))) \right) \\ &= \bigotimes_{k=1}^n \left(\bigoplus_{l \geq 0} (\mathcal{B}(Q'_k)(l(s_i^{R_i(M)}(\beta_k)))) \right) \cong \bigotimes_{k=1}^n \mathcal{B}(Q'_k). \end{aligned}$$

Hence $(K_i^P)'$ and $L_i^{R_i(P)}$ are tensor decomposable. Recall $K_i^{R_i(P)} = \mathcal{S}_{\mathcal{B}(R_i(P))}(L_i^{R_i(P)})$, which preserves $\mathcal{N}_0^\theta$ -grading. The multiplication map

$$K_i^{R_i(P)} \otimes \mathcal{B}(P_i^*) \xrightarrow{\sim} \mathcal{B}(R_i(P))$$

is an isomorphism of $\mathbb{N}_0^\theta$ -graded objects in ${}^H_H\mathcal{YD}$, and hence $\mathcal{B}(R_i(P))$ is tensor decomposable. Since M admits all reflections, it is not hard to see for each $X \in \mathcal{F}_\theta(M)$, $\mathcal{B}(X)$ is tensor decomposable.

(2) Recall T_i^M preserves $\mathbb{N}_0^\theta$ -grading. we have

$$\Phi_+^{K_i^{R_i(P)}} = \Phi_+^{L_i^{R_i(P)}} = s_i^P(\Phi_+^{K_i^P}).$$

According to the isomorphism $K_i^{R_i(P)} \otimes \mathcal{B}(P_i^*) \xrightarrow{\sim} \mathcal{B}(R_i(P))$, we have

$$\Phi^{R_i(P)} = \Phi^{L_i^{R_i(P)}} \cup \{([P_i], \alpha_i)\} \cup \{([P_i^*], -\alpha_i)\} = s_i^P(\Phi^P).$$

For any $X, Y \in \mathcal{F}_\theta$ and any morphism $\omega : [X] \rightarrow [Y]$ in $\mathcal{W}(M)$, it is straightforward to verify that $\omega(\Phi^{B(X)}) = \Phi^{B(Y)}$.

(3) Let $\alpha \in \Delta^{M, \text{re}}$. Then there exists $X \in \mathcal{X}$, $\omega \in \text{Hom}([X], [M])$ such that $\alpha = \omega(\alpha_i)$. Take a tensor decomposition of $\mathcal{B}(X)$. Then for any $1 \leq l \leq n$,

$$\mathcal{B}(Q_l) = \bigoplus_{r \geq 0} \mathcal{B}(Q_l)(r\beta_l).$$

Hence

$$X_i \cong \mathcal{B}(X)(\alpha_i) \cong \bigoplus_{\substack{r_1, \dots, r_n \geq 0 \\ \sum_{1 \leq j \leq n} r_j \beta_j = \alpha_i}} \mathcal{B}(Q_l)(r_l \beta_l).$$

Since $\beta_l \in \mathbb{N}_0^\theta \setminus \{0\}$ for all l , it follows that

$$X_i \cong \bigoplus_{\substack{1 \leq j \leq n \\ \beta_j = \alpha_i}} Q_j.$$

Since M_i is irreducible, it follows that $([X_i], \alpha_i) \in \Phi^{\mathcal{B}(X)}$. Therefore $([X_i], \alpha) \in \Phi^{\mathcal{B}(M)}$. $\square$

Proposition 5.6. *Let $M \in \mathcal{F}_\theta$, the following are equivalent.*

- (1) *The tuple M admits all reflections and $\mathcal{G}(M)$ is finite.*
- (2) *The Nichols algebra $\mathcal{B}(M)$ is tensor decomposable.*

(1) $\Rightarrow$ (2) Assume that M admits all reflections and that $\mathcal{G}(M)$ is finite. Let $m = |\Delta_+^{M, \text{re}}|$. Proposition 4.2(1) implies that any $[M]$ -reduced sequence has length at most m . Suppose $(i_1, \dots, i_m)$ is a $[M]$ -reduced sequence, then for any $i \in \mathbb{I}$, $(i, i_1, \dots, i_m)$ is not a $R_i(M)$ -reduced sequence. Therefore $\alpha_i \in \Lambda^{[M]}(i_1, \dots, i_m)$ for each $i \in \mathbb{I}$. Hence for each $i \in \mathbb{I}$, $M_i = M_{\alpha_i} \subseteq E^M(i_1, \dots, i_m)$ by Proposition 4.2 (4). Since $E^M(i_1, \dots, i_m)$ is a subalgebra of $\mathcal{B}(M)$, this forces $E^M(i_1, \dots, i_m) \cong \mathcal{B}(M)$. Thus (1) implies (2).

(2) $\Rightarrow$ (1) By assumption, we know that $\Phi^{\mathcal{B}(M)}$ is a finite set. For each $i \in \mathbb{I}$, $\Phi_+^{K_i^M}$ is finite as well. Recall K_i^M is generated by $\text{ad}(\mathcal{B}(M_i))(M_j)$. Let $\psi = \{n\alpha_i + \alpha_j \mid \text{ad}_{\mathcal{B}(M_i)}(M_i)^n(M_j) \neq 0, n \geq 0\}$. Therefore for each $\alpha \in \psi$, there must exist $([M_\alpha], \alpha) \in \Phi_+^{K_i^M}$. The finiteness of $\Phi_+^{K_i^M}$ guarantees ψ is a finite set and M admits the i -th reflection. By an argument similar to that in Proposition 5.5(1) $\mathcal{B}(R_i(M))$ is tensor decomposable for each $i \in \mathbb{I}$. By induction, it follows that M admits all reflection sequence and $\mathcal{B}(X)$ is tensor decomposable for any $X \in \mathcal{F}_\theta(H)$. Moreover, $\Phi^{\mathcal{B}(X)}$ is finite since $\mathcal{B}(X)$ is tensor decomposable. Thus, by Proposition 5.5 (3), we deduce that $\Delta_+^{[X], \text{re}}$ is finite for all $X \in \mathcal{X}$. Hence (2) implies (1).

5.2 A criterion for finite-dimensionality

In this subsection, we still assume that M admits all reflections. Moreover, we require that $\mathcal{G}(M)$ is finite.

Let $P \in \mathcal{F}_\theta^H(M)$, and let $\kappa = (i_1, \dots, i_l) \in \mathbb{I}^l$ be a reduced decomposition of the longest element w_0 in $\text{Hom}(\mathcal{W}(M), [P])$. Then κ is $[P]$ -reduced by Lemma 2.11(1). Moreover, we have

$$\Delta_+^{[P], \text{re}} = \Lambda^{[P]}(\kappa) \quad (5.1)$$

by Lemma 2.11(2).

Now let $\Lambda^{[P]}(\kappa) = \{\beta_1, \dots, \beta_l\}$. According to the proof of Proposition 5.6 and 4.2(4), we have

$$\mathcal{B}(P) \cong (\dots (\mathcal{B}(P_{\beta_l}) \otimes \mathcal{B}(P_{\beta_{l-1}})) \otimes \dots) \otimes \mathcal{B}(P_{\beta_1}). \quad (5.2)$$

Lemma 5.7. *Let $Q \in \mathcal{F}_\theta(M)$, $1 \leq k \leq m$, $i \in \mathbb{I}$, and let $w : [Q] \rightarrow [P]$ be a morphism in $\mathcal{W}(M)$. Assume that $\beta_k = w(\alpha_i)$. Then $P_{\beta_k} \cong Q_i$ in ${}^H_H\mathcal{YD}$.*

Proof. By Proposition 5.5 (2), we have $\omega(\Phi^{[Q]}) = \Phi^{[P]}$. Note that $([Q_i], \alpha_i) \in \Phi^{[Q]}$, by definition

$$\omega([Q_i], \alpha_i) = ([Q_i], \beta_k) \in \Phi^{[P]}.$$

Since β_i , $1 \leq i \leq l$ are pairwise distinct, this implies $Q_i \cong P_{\beta_k}$ in ${}^H_H\mathcal{YD}$. $\square$

The next proposition is useful for our characterization of finite-dimensional Nichols algebras.

Proposition 5.8. *Let $i, j \in \mathbb{I}$, $i \neq j$, and $0 \leq t \leq -a_{ij}^{[P]}$. Then there exists $1 \leq k \leq m$ such that $\alpha_j + t\alpha_i = \beta_k$ and $\text{ad}(P_i)^t(P_j) \cong P_{\beta_k}$ in ${}^H_H\mathcal{YD}$. In particular, $\text{ad}(P_i)^t(P_j)$ is irreducible in ${}^H_H\mathcal{YD}$.*

Proof. Let $i \in \mathbb{I}$. By Proposition 4.5(3), there is an index $1 \leq h \leq m$ such that $\beta_h = \alpha_i$ and $P_{\beta_h} \cong P_i$. Since $K_i^P \otimes \mathcal{B}(P_i) \cong \mathcal{B}(P)$ and $\mathcal{B}(P)$ is tensor decomposable. It follows from Proposition 4.5(1) that

$$K_i^P \cong \bigoplus_{\substack{1 \leq k \leq m \\ k \neq h}} \mathcal{B}(P_{\beta_k}). \quad (5.3)$$

We know that the algebra K_i^P is generated by the homogeneous subspaces $(\text{ad } P_i)^t(P_j)$ of degree $\alpha_j + t\alpha_i$, $j \neq i$, $j \in \mathbb{I}$, and $0 \leq t \leq -a_{ij}^{[P]}$. Since these subspaces have pairwise distinct degrees, no other non-linear combinations of generators in K_i^P can contribute to the homogeneous space of degree $\alpha_j + t\alpha_i$. We see that for all $j \neq i$, and $0 \leq t \leq -a_{ij}^{[P]}$,

$$K_i^P(\alpha_j + t\alpha_i) = (\text{ad } P_i)^t(P_j)$$

as $\mathbb{N}_0^\theta$ -graded objects in ${}^H_H\mathcal{YD}$. On the other hand, the homogeneous part of degree $\alpha_j + t\alpha_i$ of the right hand side of (5.3) is the direct sum of all tensor products

$$\bigotimes_{\substack{1 \leq k \leq m \\ k \neq h}} (P_{\beta_k})^{n_k}, \quad n_k \geq 0 \text{ for all } k,$$

where $\sum_{1 \leq k \leq m} n_k \beta_k = \alpha_j + t\alpha_i$. For all $k \neq h$, $\beta_k \notin \mathbb{N}_0 \alpha_i$, since β_k and α_i are real roots of $[P]$. This degree equation forces $n_k = 1$ for exactly one index k , and $n_r = 0$ for all $r \neq k$. Hence $(\text{ad } P_i)^t(P_j) \cong P_{\beta_k}$ in ${}^H_H\mathcal{YD}$ for some $1 \leq k \leq m$. $\square$

Theorem 5.9. Let $M \in \mathcal{F}_\theta^H$ such that M_j is simple in ${}^H_H\mathcal{YD}$ for all $j \in \mathbb{I}$. The following are equivalent.

- (1) $\mathcal{B}(M)$ is finite-dimensional.
- (2) M admits all reflections, $\mathcal{G}(M)$ is finite and $\mathcal{B}(P_i)$ is finite-dimensional for all $P \in \mathcal{F}_\theta(M)$ and $i \in \mathbb{I}$.

Proof. (1) $\Rightarrow$ (2) For any $i \in \mathbb{I}$, the coinvariant object $\mathcal{B}(M)^{\text{co } \mathcal{B}(M_i)}$ is finite-dimensional. According to the following isomorphism

$$\Theta_i : \mathcal{B}(R_i(M)) \cong \Omega_i \left(\mathcal{B}(M)^{\text{co } \mathcal{B}(M_i)} \right) \# \mathcal{B}(M_i^*),$$

we deduce that $\dim(\mathcal{B}(R_i(M))) = \dim(\mathcal{B}(M))$ is finite-dimensional for each $i \in \mathbb{I}$. By induction, we can deduce that M admits all reflections and $\mathcal{B}(P_i)$ is finite-dimensional for all $P \in \mathcal{F}_\theta(M)$. The length of any $[M]$ -reduced sequence is bounded by $\dim \mathcal{B}(M)$. Therefore, $\mathcal{B}(M)$ is tensor decomposable by the proof of Proposition 5.6. Hence $\mathcal{G}(M)$ is finite.

(2) $\Rightarrow$ (1). Since $\mathcal{G}(M)$ is finite Cartan graph, there is a longest element $\omega \in \text{Hom}(\mathcal{W}(\mathcal{G}), M)$. The element ω corresponds to a reduced decomposition $\kappa = (i_1, \dots, i_l)$ such that $\Lambda^{[M]}(\kappa) = \{\beta_1, \dots, \beta_l\}$ and $\mathcal{B}(M) \cong (\dots (\mathcal{B}(M_{\beta_l}) \otimes \mathcal{B}(M_{\beta_{l-1}})) \otimes \dots) \otimes \mathcal{B}(M_{\beta_1})$. By Lemma 5.7 and assumption (2), we deduce that $\mathcal{B}(M)$ is finite-dimensional. $\square$

6 Cartan graphs under braided monoidal equivalences

In this section, we investigate the behavior of Cartan graphs under braided monoidal equivalences. We prove that the reflection property and the Cartan graph are invariant under such equivalences. As an application, we study Nichols algebras of diagonal type twisted by a 3-cocycle and show that their Cartan graphs are isomorphic to those over Hopf algebras.

6.1 The invariant of Cartan graphs

In this subsection, let H, M be two coquasi-Hopf algebras with bijective antipode. We fix a braided monoidal equivalence:

$$(F, \xi) : {}^H_H\mathcal{YD} \longrightarrow {}^M_M\mathcal{YD}, \quad (6.1)$$

where ξ is the tensor structure given by $\xi_{V,W} : F(V) \otimes F(W) \xrightarrow{\sim} F(V \otimes W)$, $V, W \in {}^H_H\mathcal{YD}$.

Let $X = (X_1, \dots, X_\theta) \in {}^H_H\mathcal{YD}$ be a θ -tuple, where X_i is finite-dimensional and simple for each $i \in \mathbb{I}$. It is immediate that $F(X) = (F(X_1), F(X_2), \dots, F(X_\theta))$ is a θ -tuple in ${}^M_M\mathcal{YD}$, where $F(X_i)$ is finite-dimensional and simple as well.

According to Theorem 4.6, if X ($F(X)$, respectively) admits all reflections, then $\mathcal{G}(X)$ ($\mathcal{G}(F(X))$, respectively) is a Cartan graph. Now we recall the definition of isomorphism of Cartan graphs

Definition 6.1. Let $\mathcal{G} = \mathcal{G}(I, \mathcal{X}, r, A)$ and $\mathcal{G}' = \mathcal{G}(J, \mathcal{Y}, t, B)$ be semi-Cartan graphs. A morphism $(\beta, \gamma) : \mathcal{G} \rightarrow \mathcal{G}'$ of semi-Cartan graphs is a pair (β, γ) , where $\beta : I \rightarrow J$, $\gamma : \mathcal{X} \rightarrow \mathcal{Y}$ are maps such that for all $i, j \in I$ and $X \in \mathcal{X}$,

$$\gamma(r_i(X)) = t_{\beta(i)}(\gamma(X)), \quad a_{ij}^X = b_{\beta(i)\beta(j)}^{\gamma(X)}.$$

A morphism (β, γ) is an isomorphism if and only if both maps β and γ are bijective. Furthermore, if both $\mathcal{G}$ and $\mathcal{G}'$ are Cartan graphs, we say $\mathcal{G}$ is isomorphic to $\mathcal{G}'$ as Cartan graphs.

Our main goal is as follows:

Theorem 6.2. Under the above assumptions, the tuple X admits all reflections if and only if $F(X)$ does. Under this condition, $\mathcal{G}(X)$ is isomorphic to $\mathcal{G}(F(X))$ as Cartan graphs.

The following lemma is standard, but plays an important role in our proof.

Lemma 6.3. [19, Lemma 2.16] Let H, M be two coquasi-Hopf algebras with bijective antipode, and $F : {}^H_H\mathcal{YD} \rightarrow {}^M_M\mathcal{YD}$ as braided monoidal categories. For any $V \in {}^H_H\mathcal{YD}$, we have

$$\Psi : T(F(V)) \cong F(T(V))$$

as Hopf algebras in ${}^M_M\mathcal{YD}$, which is identity on $F(V)$. Moreover, Ψ induces an isomorphism of quotient Hopf algebras

$$\Psi : \mathcal{B}(F(V)) \cong F(\mathcal{B}(V)).$$

Now we fix two simple objects $V, W \in {}^H_H\mathcal{YD}$, by [20, Lemma 5.2], $\text{ad}_H(\mathcal{B}(V))(W) = \bigoplus_{n \geq 0} \text{ad}_H(V)^n(W)$ is an object in ${}^{A(V)}_{A(V)}\mathcal{YD}$, where $A(V) = \mathcal{B}(V) \# H$. Furthermore, for each $n \geq 0$, $\text{ad}_H(V)^n(W)$ is an object in ${}^H_H\mathcal{YD}$.

Proposition 6.4. For each $n \geq 0$, we have an isomorphism of objects in ${}^M_M\mathcal{YD}$,

$$\text{ad}_M(F(V))^n(F(W)) \cong F(\text{ad}_H(V)^n(W)). \quad (6.2)$$

Proof. Let $D_0 = W$, for $n \geq 1$, we define $D_n = V \otimes D_{n-1}$ inductively. Now we define the map $\text{ad}_H^{(n)} : D_n \rightarrow T(V \oplus W)$ inductively:

$$\begin{aligned} \text{ad}_H^{(1)} &= \mu \circ (\text{id}_{V \oplus W} - c_{V,W}), \\ \text{ad}_H^{(n)} &= \mu \circ (\text{id} - c_{V, T(V \oplus W)}) \circ (\text{id}_V \otimes \text{ad}_H^{(n-1)}). \end{aligned}$$

Here μ is the multiplication in $T(V \oplus W)$ and c is the braiding of ${}^H_H\mathcal{YD}$. Therefore $\text{ad}_H^{(n)}$ is a morphism in ${}^H_H\mathcal{YD}$ for each $n \geq 0$. By definition,

$$\text{Im}(\text{ad}_H^{(n)}) = \text{ad}_H(V)^n(W).$$

We denote $E_0 = F(W)$ and $E_n = F(V) \otimes E_{n-1}$ for $n \geq 1$. We can then define a morphism in ${}^M_M\mathcal{YD}$

$$\text{ad}_M^{(n)} : E_n \rightarrow T((F(V) \oplus F(W)))$$

satisfying $\text{Im}(\text{ad}_M^{(n)}) = \text{ad}(F(V))^n(F(W))$.

By tensor structure $\xi_{V,W} : F(V) \otimes F(W) \xrightarrow{\cong} F(V \otimes W)$, we can define the following isomorphism of objects in ${}^M_M\mathcal{YD}$ inductively:

$$\begin{aligned} \Xi_n : E_n &\xrightarrow{\sim} F(D_n), \\ \Xi_n &= \zeta_{V,D_{n-1}} \circ (\text{id}_{F(V)} \otimes \Xi_{n-1}) \end{aligned}$$

with the initial step $\Xi_0 = \text{id}_{F(W)}$. We claim that the following diagram commutes.

$$\begin{array}{ccc} E_n & \xrightarrow{\text{ad}_M^{(n)}} & T(F(V \oplus W)) \\ \Xi_n \downarrow & & \downarrow \Psi \\ F(D_n) & \xrightarrow{F(\text{ad}_H^{(n)})} & F(T(V \oplus W)) \end{array}$$

This can be proved by induction on n . We denote the multiplication in $T(F(V \oplus W))$ by μ' and braiding in ${}^M_M\mathcal{YD}$ by d . For $n = 1$,

$$\begin{aligned} \Psi \circ \text{ad}_M^{(1)} &= \Psi \circ \mu' \circ (\text{id} - d_{F(V),F(W)}) \\ &= F(\mu) \circ \xi_{V,W} \circ (\Psi \otimes \Psi) - F(\mu) \circ \xi_{W,V} \circ (\Psi \otimes \Psi) \circ d_{F(V),F(W)} \\ &= F(\mu) \circ \xi_{V,W} - F(\mu) \circ \xi_{W,V} \circ d_{F(V),F(W)} \\ &= F(\mu \circ (\text{id} - c_{V,W})) \circ \xi_{V,W} \\ &= F(\text{ad}_H^{(1)}) \circ \Xi_1. \end{aligned}$$

Here the second equation uses the property that μ' is an algebra homomorphism and the fourth equation follows since ξ commutes with d .

Now we assume $\Psi \circ \text{ad}_M^{(n-1)} = F(\text{ad}_H^{(n-1)}) \circ \Xi_{n-1}$. Then

$$\begin{aligned} \Psi \circ \text{ad}_M^{(n)} &= \Psi \circ \mu' \circ (\text{id} - d_{F(V),T(F(V \oplus W))}) \circ (\text{id}_{F(V)} \otimes \text{ad}_M^{(n-1)}) \\ &= F(\mu) \circ \xi_{V,T(V \oplus W)} \circ (\text{id}_{F(V)} \otimes \Psi \circ \text{ad}_M^{(n-1)}) \\ &\quad - F(\mu) \circ \xi_{T(V \oplus W),V} \circ (\Psi \otimes \Psi) \circ d_{F(V),T(F(V \oplus W))} \circ (\text{id}_{F(V)} \otimes \text{ad}_M^{(n-1)}) \\ &= F(\mu) \circ \xi_{V,T(V \oplus W)} \circ (\text{id}_{F(V)} \otimes F(\text{ad}_H^{(n-1)}) \circ \Xi_{n-1}) \\ &\quad - F(\mu) \circ \xi_{T(V \oplus W),V} \circ d_{F(V),F(T(V \oplus W))} \circ (\text{id}_{F(V)} \otimes F(\text{ad}_H^{(n-1)}) \circ \Xi_{n-1}) \\ &= F(\mu) \circ F(\text{id}_V \otimes \text{ad}_H^{(n-1)}) \circ \xi_{V,D_{n-1}} \circ (\text{id}_{F(V)} \otimes \Xi_{n-1}) \\ &\quad - F(\mu) \circ F(c_{V,T(V \oplus W)}) \circ F(\text{id}_V \otimes \text{ad}_H^{(n-1)}) \circ \xi_{V,D_{n-1}} \circ (\text{id}_{F(V)} \otimes \Xi_{n-1}) \\ &= F(\mu \circ (\text{id} - c_{V,T(V \oplus W)})) \circ (\text{id}_V \otimes \text{ad}_H^{(n-1)}) \circ \Xi_n \\ &= F(\text{ad}_H^{(n)}) \circ \Xi_n. \end{aligned}$$

Here the second equation uses the property that μ' is an algebra homomorphism. The third equation uses the hypothesis. The fourth equation uses the naturality of ζ .

Now we have $\text{Im}(\Psi \circ \text{ad}_M^{(n)}) = \text{Im}(F(\text{ad}_H^{(n)}) \circ \Xi_n)$. Since Ψ and Ξ_n are isomorphisms and F is an equivalence of abelian categories, we have

$$\text{Im}(\text{ad}_M^{(n)}) \cong F(\text{Im}(\text{ad}_H^{(n)})).$$

This is equivalent to saying that $\text{ad}_M(F(V))^n(F(W)) \cong F(\text{ad}_H(V)^n(W))$ for all $n \geq 0$. $\square$

Now we can prove Theorem 6.2.

Proof of Theorem 6.2. If X admits the i -th reflection for some $i \in \mathbb{I}$, then for each $j \neq i$, we have $\text{ad}_H(X_i)^{-a_{ij}^X}(X_j) \neq 0$ but $\text{ad}_H(X_i)^{-a_{ij}^X+1}(X_j) = 0$ for some $a_{ij}^X \leq 0$. By Proposition 6.4, we have

$$\text{ad}_M(F(X_i))^{-a_{ij}^X}(F(X_j)) \neq 0, \text{ad}_M(F(X_i))^{-a_{ij}^X+1}(F(X_j)) = 0.$$

Obviously, the converse is true, hence X admits the i -th reflection if and only if $F(X)$ does. Since $\text{ad}_H(X_i)^{-a_{ij}^X}(X_j)$ is a simple object in $\mathcal{D}$ by [20, Lemma 5.8], it follows that

$$\text{ad}_M(F(X_i))^{-a_{ij}^X}(F(X_j)) \cong F(\text{ad}_H(X_i)^{-a_{ij}^X}(X_j))$$

is a simple object in ${}^M_M\mathcal{YD}$ as well. Meanwhile we have $F(X_i^*) = F(X_i)^*$. Hence

$$F(R_i(X)) \cong R_i(F(X)).$$

By applying Proposition 6.4 again, if $R_i(X)$ admits the l -th reflection, then for all $j \in \mathbb{I}$,

$$a_{lj}^{R_i(X)} = a_{lj}^{F(R_i(X))}.$$

Now it is straightforward to see that X admits all reflections if and only if $F(X)$ does, and for any $[Y] \in \mathcal{F}_\theta(X)$, $i, j \in \mathbb{I}$,

$$a_{ij}^{[Y]} = a_{ij}^{[F(Y)]}.$$

Now suppose X admits all reflections, then $\mathcal{G}(X)$ is given by

$$\mathcal{G}(X) = (\mathbb{I}, \mathcal{X}, r, (A^{[Y]})_{[Y] \in \mathcal{X}}),$$

where $\mathcal{X} = \{[Y] \mid Y \in \mathcal{F}_\theta(X)\}$, the map $r : \mathbb{I} \times \mathcal{X} \rightarrow \mathcal{X}$, $i \times [Y] \mapsto [R_i(Y)]$ and $A^{[Y]} = (a_{ij}^X)_{i,j \in \mathbb{I}}$ for all $[Y] \in \mathcal{X}$. Then $\mathcal{G}(F(X))$ is well-defined and can be expressed as follows,

$$\mathcal{G}(F(X)) = (\mathbb{I}, \mathcal{Y}, t, (A^{[Y]})_{[Y] \in \mathcal{Y}}),$$

where $\mathcal{Y} = \{[Y] \mid Y \in \mathcal{F}_\theta(F(X))\}$, the map $r : \mathbb{I} \times \mathcal{Y} \rightarrow \mathcal{Y}$, $i \times [Y] \mapsto [R_i(Y)]$ and $A^{[Y]} = (a_{ij}^Y)_{i,j \in \mathbb{I}}$ for all $[Y] \in \mathcal{Y}$. Let us define $\beta = \text{id}_\mathbb{I}$ and

$$\gamma : \mathcal{X} \rightarrow \mathcal{Y}, [Y] \mapsto [F(Y)].$$

It is straightforward to see that for any $Z \in \mathcal{X}$,

$$\gamma(r_i(Z)) = [F(R_i(Z))] = [R_i(F(Z))] = t_i(\gamma(Z)), \quad a_{ij}^Z = a_{ij}^{F(Z)}.$$

The map γ is injective because F is an equivalence of categories,

$$F(Y_1) \cong F(Y_2) \implies Y_1 \cong Y_2.$$

To see γ is surjective, let $[W] \in \mathcal{Y}$. Then

$$[W] = [R_{i_k} \cdots R_{i_1}(F(X))]$$

for some $(i_1, \dots, i_k) \in \mathbb{I}^k$. Defining $Y = R_{i_k} \cdots R_{i_1}(X) \in \mathcal{X}$, the established isomorphism $F(R_i(X)) \cong R_i(F(X))$ inductively guarantees $F(Y) \cong W$. Thus, $\gamma([Y]) = [W]$, making γ a bijection. Hence $\mathcal{G}(X)$ is isomorphic to $\mathcal{G}(F(X))$ as Cartan graphs. $\square$

Corollary 6.5. *With above assumptions. If the tuple X admits all reflections, then $\mathcal{W}(\mathcal{G}(X))$ and $\mathcal{W}(\mathcal{G}(F(X)))$ are equivalent as Weyl groupoids and*

$$\Delta^{Z, \text{re}} = \Delta^{\gamma(Z), \text{re}}$$

for all $Z \in \mathcal{X}$ via the isomorphism of Weyl groupoids

Proof. It follows from the definition immediately. $\square$

A special and important case of interest to us is when the braided monoidal equivalence originates from a twist. Let H be a coquasi-Hopf algebra with bijective antipode. A convolution-invertible linear map

$$J : H \otimes H \rightarrow \mathbb{k}$$

is called a twist on H if $J(h, 1) = \varepsilon(h) = J(1, h)$ for all $h \in H$. Given a twist, we can construct a new coquasi-Hopf algebra as follows: $H^J = H$ as a coalgebra and the multiplication $\circ$ on H^J is given by

$$a \circ b := J(a_1, b_1) a_2 b_2 J^{-1}(a_3, b_3)$$

for all $a, b \in H$. The associator Φ^J and the quasi-antipode $(\mathcal{S}^J, \alpha^J, \beta^J)$ are given as

$$\Phi^J(a, b, c) = J(b_1, c_1) J(a_1, b_2 c_2) \Phi(a_2, b_3, c_3) J^{-1}(a_3 b_4, c_4) J^{-1}(a_4, b_5),$$

$$\mathcal{S}^J = \mathcal{S}, \quad \alpha^J(a) = J^{-1}(\mathcal{S}(a_1), a_3) \alpha(a_2), \quad \beta^J(a) = J(a_1, \mathcal{S}(a_3)) \beta(a_2)$$

for all $a, b, c \in H$.

It is standard that

$$\text{Comod}(H) \cong \text{Comod}(H^J)$$

as tensor categories and

$$(\text{id}, \mathcal{J}) : {}^H_H \mathcal{YD} \xrightarrow{\sim} {}^{H^J}_{H^J} \mathcal{YD}, \quad V \mapsto V^J$$

is an equivalence of braided monoidal categories, where $V^J = V$ as vector spaces and $\mathcal{J}_{V, W} : V^J \otimes W^J \rightarrow V \otimes W, v \otimes w \mapsto J^{-1}(v_{-1}, w_{-1}) v_0 \otimes w_0$.

Corollary 6.6. *Let H be a coquasi-Hopf algebra with bijective antipode and J is a twist on H . Assume $X \in {}^H_H \mathcal{YD}$ admits all reflections, then $X^J \in {}^{H^J}_{H^J} \mathcal{YD}$ admits all reflections, and $\mathcal{G}(X)$ is isomorphic to $\mathcal{G}(X^J)$.*

Proof. It follows from Theorem 6.2 immediately. $\square$

6.2 Reflections of Nichols algebras of diagonal type

In this section, we investigate Nichols algebras over coquasi-Hopf algebra of diagonal type. Given G an abelian group and Φ a 3-cocycle on G , Nichols algebras over $(\mathbb{k}G, \Phi)$ of diagonal type have been studied in [16]. However, the reflection theory for these Nichols algebras has not been investigated before. We fix some notation at first.

Example 6.7. The left $\mathbb{k}G$ -comodule (V, δ) is a left-left Yetter–Drinfeld module over the coquasi-Hopf algebra $H = (\mathbb{k}G, \Phi)$ if there is a linear map $\triangleright : G \otimes V \rightarrow V$ such that for all $e, f \in G$ and $v \in {}^sV$:

$$e \triangleright (f \triangleright v) = \frac{\Phi(e, f, g)\Phi(g, e, f)}{\Phi(e, g, f)}(ef) \triangleright v, \quad (6.3)$$

$$1_G \triangleright v = v, \quad (6.4)$$

$$e \triangleright v \in {}^sV. \quad (6.5)$$

The category of all left-left Yetter–Drinfeld modules over $(\mathbb{k}G, \Phi)$ is denoted by ${}^G_G\mathcal{YD}^\Phi$. We always denote

$$\tilde{\Phi}_g(e, f) := \frac{\Phi(e, f, g)\Phi(g, e, f)}{\Phi(e, g, f)}.$$

A Direct computation shows $\tilde{\Phi}_g$ is a 2-cocycle on G . Furthermore, a special but important class of 3-cocycles is called abelian 3-cocycle. By [18, Lemma 2.10], Φ is abelian if and only if for all $x, y, z \in G$,

$$\Phi(x, y, z) = \Phi(x, z, y).$$

Now let (H, Φ) be a coquasi-Hopf algebra with bijective antipode and V be a one dimensional vector space. Let $\triangleright : H \otimes V \rightarrow V$ and $\delta : V \rightarrow H \otimes V$ be maps. Now we choose $v \in V$ and $g \in H$ and $\chi \in \text{Hom}(H, \mathbb{k})$ such that

$$\delta(v) = g \otimes v, \quad h \triangleright v = \chi(h)v \text{ for all } h \in H.$$

It is straightforward to see that V is a H -comodule if and only if $\Delta(g) = g \otimes g$ and $\varepsilon(g) = 1$. On the other hand, one may note that (2.9) holds if and only if $gh_2 \otimes \chi(h_1)v = h_1g \otimes \chi(h_2)v$. This is equivalent to

$$gh_2\chi(h_1) = h_1g\chi(h_2). \quad (6.6)$$

Now for any $h, l \in H$, V satisfies (2.7) if and only if

$$\chi(hl) = \frac{\Phi(h_2, g, l_3)}{\Phi(h_1, l_1, g)\Phi(g, h_4, l_4)}\chi(h_3)\chi(l_2). \quad (6.7)$$

This implies χ is far from being an algebra homomorphism. Nevertheless, it exhibits several useful properties. Since H has bijective antipode, if $h \in G(H)$, then $h^{-1} \in G(H)$. Moreover,

$$1 = \chi(1) = \frac{\Phi(h, g, h^{-1})}{\Phi(h, h^{-1}, g)\Phi(g, h, h^{-1})}\chi(h)\chi(h^{-1}). \quad (6.8)$$

Since elements in $G(H)$ are invertible, one deduces that $\chi(h) \neq 0$ for any $h \in G(H)$. According to (6.6), we have $gh = hg$ for all $h \in G(H)$. Now for any one-dimensional Yetter–Drinfeld module $(W = \text{span}\{w\}, \triangleright, \delta)$, given by $\delta(w) = l \otimes w$, $h \triangleright w = \zeta(h)w$, where $\zeta \in \text{Hom}(H, \mathbb{k})$ satisfying (6.6). If h is group-like, the equation reduces to

$$\zeta(gh)\chi(h) = \zeta(hg)\chi(h).$$

Since $hg = gh$ and $\zeta(g) \neq 0$, we can deduce that $\zeta\chi(h) = \chi\zeta(h)$ for all $h \in G(H)$.

Lemma 6.8. Let $V \in {}^H_H\mathcal{YD}$ be a Yetter-Drinfeld module of diagonal type. Denote

$$\text{Supp } V := \{v_{-1} \mid \delta(v) = v_{-1} \otimes v_0, v \in V\}.$$

Then $\text{Supp } V$ generates an abelian group G , and $\Phi|_G$ is an abelian 3-cocycle on G .

Proof. $\text{Supp } V$ generates an abelian group G , and $\Phi|_G$ is a 3-cocycle on G follows from the above analysis. By [18, Lemma 2.10], V is diagonal type if and only if $\Phi|_G$ is an abelian 3-cocycle on G . $\square$

Let $M = (M_1, \dots, M_\theta) \in \mathcal{F}_\theta$ be a tuple of Yetter-Drinfeld modules of diagonal type. For all $j \in \mathbb{I}$, let x_j be a basis of M_j , and let $g_j \in H$, $\chi_j \in \text{Hom}(H, \mathbb{k})$ such that

$$\delta_{M_j}(x_j) = g_j \otimes x_j, \quad h \cdot x_j = \chi_j(h)x_j$$

for all $h \in H$. Then g_j is an invertible group-like element for all $j \in \mathbb{I}$. For all $j, k \in \mathbb{I}$ let $q_{jk} \in \mathbb{k}^\times$ such that

$$c_{M_j, M_k}(x_j \otimes x_k) = q_{jk} x_k \otimes x_j.$$

Therefore $G := G_M = \langle g_1, \dots, g_\theta \rangle$ and Φ is an abelian 3-cocycle on G and $M \in {}^G_G\mathcal{YD}^\Phi$.

For $n \geq 3$, we will use the notation

$$x_i^n := ((x_i \cdot x_i) \cdots) \cdot x_i.$$

to represent iterated multiplication.

Lemma 6.9. For each $i \in \mathbb{I}$, $x_i^n = 0$ if and only if $(n)!_{q_{ii}} = 0$.

Proof. We first consider $n = 3$ as a base case. By direct computation:

$$\begin{aligned} \Delta(x_i^3) &= (x_i^2 \otimes 1 + 1 \otimes x_i^2 + (1 + q_{ii})x_i \otimes x_i)(x_i \otimes 1 + 1 \otimes x_i) \\ &= x_i^3 \otimes 1 + 1 \otimes x_i^3 + (1 + q_{ii} + q_{ii}^2)x_i^2 \otimes x_i + \frac{1}{\Phi(g_i, g_i, g_i)}(1 + q_{ii} + q_{ii}^2)x_i \otimes x_i^2. \end{aligned}$$

It is easy to see that $x_i^3 = 0$ if and only if $(3)!_{q_{ii}} = 0$, since $\Phi(g_i, g_i, g_i) \neq 0$.

We claim that

$$\Delta(x_i^n) = \sum_{j=0}^n \binom{n}{n-j}_{q_{ii}} \prod_{k=1}^{j-1} \frac{1}{\Phi(g_i^{n-j}, g_i^k, g_i)} x_i^{n-j} \otimes x_i^j. \quad (6.9)$$

Here, the associators are trivial when $j = 0$ and $j = 1$. We prove this claim by induction.

$$\begin{aligned} \Delta(x_i^{n+1}) &= \Delta(x_i^n) \Delta(x_i) \\ &= \left(\sum_{j=0}^n \binom{n}{n-j}_{q_{ii}} \prod_{k=1}^{j-1} \frac{1}{\Phi(g_i^{n-j}, g_i^k, g_i)} x_i^{n-j} \otimes x_i^j \right) (1 \otimes x_i + x_i \otimes 1) \end{aligned}$$

Let us focus on coefficient of the term $x_i^{n-j+1} \otimes x_i^j$, which is given by

$$\begin{aligned} &\binom{n}{n-j+1}_{q_{ii}} \prod_{k=1}^{j-2} \frac{1}{\Phi(g_i^{n-j+1}, g_i^k, g_i)} \frac{1}{\Phi(g_i^{n-j+1}, g_i^{j-1}, g_i)} \\ &+ \binom{n}{n-j}_{q_{ii}} \prod_{k=1}^{j-1} \frac{1}{\Phi(g_i^{n-j}, g_i^k, g_i)} \frac{\Phi(g_i^{n-j}, g_i, g_i^j)}{\Phi(g_i^{n-j}, g_i^j, g_i)} \prod_{k=1}^{j-1} \frac{1}{\widetilde{\Phi}_{g_i}(g_i^k, g_i)} q_{ii}^j \end{aligned}$$

A Direct computation shows

$$\Phi(g_i^{n-j}, g_i^k, g_i) = \Phi(g_i^{n-j}, g_i, g_i^k) = \frac{\Phi(g_i^{n-j+1}, g_i^k, g_i) \Phi(g_i^{n-j}, g_i, g_i^{k+1})}{\Phi(g_i^{n-j}, g_i^{k+1}, g_i) \Phi(g_i, g_i^k, g_i)}$$

Therefore

$$\prod_{k=1}^{j-1} \frac{1}{\Phi(g_i^{n-j}, g_i^k, g_i)} \frac{\Phi(g_i^{n-j}, g_i, g_i^j)}{\Phi(g_i^{n-j}, g_i^j, g_i)} \prod_{k=1}^{j-1} \frac{1}{\tilde{\Phi}_{g_i}(g_i^k, g_i)} = \prod_{k=1}^{j-1} \frac{1}{\Phi(g_i^{n-j+1}, g_i^k, g_i)}.$$

On the other hand,

$$\binom{n}{n-j+1}_{q_{ii}} + \binom{n}{n-j}_{q_{ii}} q_{ii}^j = \binom{n+1}{n-j+1}_{q_{ii}}.$$

The above calculation implies

$$\Delta(x_i^{n+1}) = \sum_{j=0}^n \binom{n+1}{n-j+1}_{q_{ii}} \prod_{k=1}^{j-1} \frac{1}{\Phi(g_i^{n-j+1}, g_i^k, g_i)} x_i^{n-j+1} \otimes x_i^j.$$

Since Φ is invertible, $\Delta(x^n) = 0$ if and only if $\binom{n}{n-j}_{q_{ii}} = 0$ for all $0 \leq j \leq n$. This is equivalent to $(n)!_{q_{ii}} = 0$. $\square$

Lemma 6.10. For any $i, j \in \mathbb{I}$, let $n \in \mathbb{N}_0$, then

$$\begin{aligned} \delta(\text{ad}(x_i)^n(x_j)) &= g_i^n g_j \otimes \text{ad}(x_i)^n(x_j), \\ g \triangleright \text{ad}(x_i)^n(x_j) &= \prod_{k=0}^{n-1} \tilde{\Phi}_g(g_i, g_i^k g_j) \chi_i^n(g) \chi_j(g) \text{ad}(x_i)^n(x_j). \end{aligned}$$

Proof. Both equations follow from induction on n and the fact that G is abelian. $\square$

Lemma 6.11. For any $i, j \in \mathbb{I}$. The condition $\text{ad}(x_i)^n(x_j) \neq 0$, but $\text{ad}(x_i)^{n+1}(x_j) = 0$ is equivalent to

$$(n+1)_{q_{ii}}(q_{ii}^n q_{ij} q_{ji} - 1) = 0, \quad (k)_{q_{ii}}(q_{ii}^{k-1} q_{ij} q_{ji} - 1) \neq 0, \text{ for all } 0 \leq k \leq n. \quad (6.10)$$

Proof. **We claim that**

$$\begin{aligned} \Delta(\text{ad}(x_i)^n(x_j)) &= \text{ad}(x_i)^n(x_j) \otimes 1 + 1 \otimes \text{ad}(x_i)^n(x_j) \\ &\quad + \sum_{k=1}^n \binom{n}{k}_{q_{ii}} \prod_{l=n-k}^{n-1} (1 - q_{ii}^l q_{ij} q_{ji}) \prod_{q=1}^k \Phi(g_i^{q-1}, g_i, g_i^{n-q} g_j) x_i^k \otimes \text{ad}(x_i)^{n-k}(x_j). \end{aligned} \quad (6.11)$$

The case $n = 1$ is trivial, yielding

$$\Delta(\text{ad}(x_i)(x_j)) = \text{ad}(x_i)(x_j) \otimes 1 + 1 \otimes \text{ad}(x_i)(x_j) + (1 - q_{ij} q_{ji}) x_i \otimes x_j.$$

For $n = 2$, direct computation shows

$$\begin{aligned} \Delta(\text{ad}(x_i)^2(x_j)) &= \Delta(x_i) \Delta(\text{ad}(x_i)(x_j)) - \Delta(g_i \triangleright (\text{ad}(x_i)(x_j))) \Delta(x_i) \\ &= \text{ad}(x_i)^2(x_j) \otimes 1 + 1 \otimes \text{ad}(x_i)^2(x_j) + (1 - q_{ii}^2 q_{ij} q_{ji} + q_{ii} - q_{ii} q_{ij} q_{ji}) x_i \otimes \text{ad}(x_i)(x_j) \\ &\quad + \Phi(g_i, g_i, g_j) (1 - q_{ij} q_{ji}) (1 - q_{ii} q_{ij} q_{ji}) x_i^2 \otimes x_j \\ &= \text{ad}(x_i)^2(x_j) \otimes 1 + 1 \otimes \text{ad}(x_i)^2(x_j) + (1 + q_{ii}) (1 - q_{ii} q_{ij} q_{ji}) x_i \otimes \text{ad}(x_i)(x_j) \\ &\quad + \Phi(g_i, g_i, g_j) (1 - q_{ij} q_{ji}) (1 - q_{ii} q_{ij} q_{ji}) x_i^2 \otimes x_j. \end{aligned}$$

Now we assume (6.11) holds for $\text{ad}(x_i)^n(x_j)$. Then

$$\begin{aligned}
\Delta(\text{ad}(x_i)^{n+1}(x_j)) &= \Delta(x_i)\Delta(\text{ad}(x_i)^n(x_j)) - \Delta(g_i \triangleright (\text{ad}(x_i)^n(x_j)))\Delta(x_i) \\
&= \Delta(x_i)\Delta(\text{ad}(x_i)^n(x_j)) - \prod_{p=0}^{n-1} \tilde{\Phi}_{g_i}(g_i, g_i^{n-1-p}g_j)q_{ii}^nq_{ij}\Delta(\text{ad}(x_i)^n(x_j))\Delta(x_i) \\
&= \text{ad}(x_i)^{n+1}(x_j) \otimes 1 + 1 \otimes \text{ad}(x_i)^{n+1}(x_j) \\
&\quad + x_i \otimes \text{ad}(x_i)^n(x_j) - \prod_{p=0}^{n-1} \tilde{\Phi}_{g_i}(g_i, g_i^{n-1-p}g_j)q_{ii}^nq_{ij}(g_i^n g_j \triangleright x_i) \otimes \text{ad}(x_i)^n(x_j) \\
&\quad + \prod_{p=0}^{n-1} \tilde{\Phi}_{g_i}(g_i, g_i^{n-1-p}g_j)q_{ii}^nq_{ij} \text{ad}(x_i)^n(x_j) \otimes x_i - \prod_{p=0}^{n-1} \tilde{\Phi}_{g_i}(g_i, g_i^{n-1-p}g_j)q_{ii}^nq_{ij} \text{ad}(x_i)^n(x_j) \otimes x_i \\
&\quad + (1 \otimes x_i + x_i \otimes 1) \left(\sum_{k=1}^n \binom{n}{k} \prod_{q_{ii} \mid l=n-k}^{n-1} (1 - q_{ii}^l q_{ij} q_{ji}) \prod_{q=1}^k \Phi(g_i^{q-1}, g_i, g_i^{n-q}g_j) x_i^k \otimes \text{ad}(x_i)^{n-k}(x_j) \right) \\
&\quad - \prod_{p=0}^{n-1} \tilde{\Phi}_{g_i}(g_i, g_i^{n-1-p}g_j)q_{ii}^nq_{ij} \left(\sum_{k=1}^n \binom{n}{k} \prod_{q_{ii} \mid l=n-k}^{n-1} (1 - q_{ii}^l q_{ij} q_{ji}) \prod_{q=1}^k \Phi(g_i^{q-1}, g_i, g_i^{n-q}g_j) \right. \\
&\quad \left. x_i^k \otimes \text{ad}(x_i)^{n-k}(x_j) \right) \Delta(x_i).
\end{aligned}$$

We focus on the coefficient of $x_i^{k+1} \otimes \text{ad}(x_i)^{n-k}(x_j)$, where $0 \leq k \leq n$. For $k = 0$, this term equals:

$$\begin{aligned}
&(1 - q_{ii}^n q_{ij} q_{ji}) x_i \otimes \text{ad}(x_i)^n(x_j) + \frac{\Phi(g_i, g_i, g_i^{n-1}g_j)}{\Phi(g_i, g_i, g_i^{n-1}g_j)} \binom{n}{1}_{q_{ii}} (1 - q_{ii}^{n-1} q_{ij} q_{ji}) q_{ii} x_i \otimes x_i \text{ad}(x_i)^{n-1}(x_j) \\
&- \prod_{p=0}^{n-1} \tilde{\Phi}_{g_i}(g_i, g_i^{n-1-p}g_j)q_{ii}^nq_{ij} \binom{n}{1}_{q_{ii}} (1 - q_{ii}^{n-1} q_{ij} q_{ji}) \frac{1}{\Phi(g_i, g_i^{n-1}g_j, g_i)} x_i \otimes \text{ad}(x_i)^{n-1}(x_j) x_i \\
&= (1 - q_{ii}^{2n} q_{ij} q_{ji} + \binom{n}{1}_{q_{ii}} (1 - q_{ii}^{n-1} q_{ij} q_{ji}) q_{ii}) x_i \otimes \text{ad}(x_i)^n(x_j) \\
&= \binom{n+1}{1}_{q_{ii}} (1 - q_{ii}^n q_{ij} q_{ji}) x_i \otimes \text{ad}(x_i)^n(x_j)
\end{aligned}$$

Now for $1 \leq k \leq n-1$, if we denote the coefficient $\prod_{p=0}^{n-1} \tilde{\Phi}_{g_i}(g_i, g_i^{n-1-p}g_j)q_{ii}^nq_{ij}$ by γ , then the term $x_i^{k+1} \otimes \text{ad}(x_i)^{n-k}(x_j)$ is given by

$$\begin{aligned}
&(x_i \otimes 1) \left(\binom{n}{k}_{q_{ii}} \prod_{q_{ii} \mid l=n-k}^{n-1} (1 - q_{ii}^l q_{ij} q_{ji}) \prod_{q=1}^k \Phi(g_i^{q-1}, g_i, g_i^{n-q}g_j) x_i^k \otimes \text{ad}(x_i)^{n-k}(x_j) \right) \\
&+ (1 \otimes x_i) \left(\binom{n}{k+1}_{q_{ii}} \prod_{q_{ii} \mid l=n-k-1}^{n-1} (1 - q_{ii}^l q_{ij} q_{ji}) \prod_{q=1}^{k+1} \Phi(g_i^{q-1}, g_i, g_i^{n-q}g_j) x_i^{k+1} \otimes \text{ad}(x_i)^{n-k-1}(x_j) \right) \\
&- \gamma \left(\binom{n}{k}_{q_{ii}} \prod_{q_{ii} \mid l=n-k}^{n-1} (1 - q_{ii}^l q_{ij} q_{ji}) \prod_{q=1}^k \Phi(g_i^{q-1}, g_i, g_i^{n-q}g_j) x_i^k \otimes \text{ad}(x_i)^{n-k}(x_j) \right) (x_i \otimes 1) \\
&- \gamma \left(\binom{n}{k+1}_{q_{ii}} \prod_{q_{ii} \mid l=n-k-1}^{n-1} (1 - q_{ii}^l q_{ij} q_{ji}) \prod_{q=1}^{k+1} \Phi(g_i^{q-1}, g_i, g_i^{n-q}g_j) x_i^{k+1} \otimes \text{ad}(x_i)^{n-k-1}(x_j) \right) (1 \otimes x_i)
\end{aligned}$$

Let A_1, A_2, A_3, A_4 denote these four terms, respectively. The associators $\text{Asso}(A_i)$ are as fol-

lows:

$$\begin{aligned}
\text{Asso}(A_1) &= \prod_{q=1}^k \Phi(g_i^{q-1}, g_i, g_i^{n-q} g_j) \Phi(g_i, g_i^k, g_i^{n-k} g_j) \prod_{r=1}^{k-1} \Phi(g_i, g_i^r, g_i) = \prod_{q=1}^{k+1} \Phi(g_i^{q-1}, g_i, g_i^{n+1-q} g_j) \\
\text{Asso}(A_2) &= \prod_{q=1}^{k+1} \Phi(g_i^{q-1}, g_i, g_i^{n-q} g_j) \frac{\Phi(g_i, g_i^{k+1}, g_i^{n-k-1} g_j)}{\Phi(g_i^{k+1}, g_i, g_i^{n-k-1} g_j)} \prod_{r=1}^k \tilde{\Phi}_{g_i}(g_i^r, g_i) = \prod_{q=1}^{k+1} \Phi(g_i^{q-1}, g_i, g_i^{n+1-q} g_j) \\
\text{Asso}(A_3) &= \prod_{p=0}^{n-1} \tilde{\Phi}_{g_i}(g_i, g_i^{n-1-p} g_j) \prod_{q=1}^k \Phi(g_i^{q-1}, g_i, g_i^{n-q} g_j) \prod_{r=1}^{n-k} \tilde{\Phi}_{g_i}^{-1}(g_i, g_i^{n-k-r} g_j) \\
&= \prod_{q=1}^{k+1} \Phi(g_i^{q-1}, g_i, g_i^{n+1-q} g_j) \\
\text{Asso}(A_4) &= \prod_{p=0}^{n-1} \tilde{\Phi}_{g_i}(g_i, g_i^{n-1-p} g_j) \prod_{q=1}^{k+1} \Phi(g_i^{q-1}, g_i, g_i^{n-q} g_j) \frac{1}{\Phi(g_i^{k+1}, g_i^{n-k-1} g_j, g_i)} \\
&= \prod_{r=1}^{n-k} \tilde{\Phi}_{g_i}^{-1}(g_i, g_i^{n-k-r} g_j) \prod_{q=1}^{k+1} \Phi(g_i^{q-1}, g_i, g_i^{n+1-q} g_j)
\end{aligned}$$

Combining these terms yields

$$\begin{aligned}
&\prod_{q=1}^{k+1} \Phi(g_i^{q-1}, g_i, g_i^{n+1-q} g_j) \left[\binom{n}{k} \prod_{q_{ii} \ l=n-k}^{n-1} (1 - q_{ii}^l q_{ij} q_{ji}) \right. \\
&\quad \left. + \binom{n}{k+1} \prod_{q_{ii} \ l=n-k-1}^{n-1} (1 - q_{ii}^l q_{ij} q_{ji}) \right] x_i^{k+1} \otimes \text{ad}(x_i)^{n-k}(x_j) \\
&= \prod_{q=1}^{k+1} \Phi(g_i^{q-1}, g_i, g_i^{n+1-q} g_j) \binom{n+1}{k+1} \prod_{q_{ii} \ l=n-k}^n (1 - q_{ii}^l q_{ij} q_{ji}) x_i^{k+1} \otimes \text{ad}(x_i)^{n-k}(x_j).
\end{aligned}$$

For $k = n$, a similar computation shows the term is

$$\prod_{l=0}^n (1 - q_{ii}^l q_{ij} q_{ji}) \Phi(g_i^l, g_i, g_i^{n-l} g_j) x_i^{n+1} \otimes x_j.$$

This completes the proof of the claim (6.11).

Now according to (6.11), it is straightforward to see that the condition $\text{ad}(x_i)^n(x_j) \neq 0$, $\text{ad}(x_i)^{n+1}(x_j) = 0$ is equivalent to

$$(n+1)_{q_{ii}}(q_{ii}^n q_{ij} q_{ji} - 1) = 0, \quad (k)_{q_{ii}}(q_{ii}^{k-1} q_{ij} q_{ji} - 1) \neq 0, \text{ for all } 0 \leq k \leq n.$$

□

Corollary 6.12. *Let $M = (M_1, \dots, M_\theta)$ be the θ -tuple as above. Then M admits the i -th reflection if and only if there is $n \in \mathbb{N}_0$ such that for each $j \neq i$,*

$$(n+1)_{q_{ii}}(q_{ii}^n q_{ij} q_{ji} - 1) = 0.$$

Proof. This follows from Lemma 6.11 immediately. □

Lemma 6.13. *Let $i, j \in \mathbb{I}$. Assume that M admits the i -th reflection.*

- (1) *If $j \neq i$, $a_{ij}^M = -\min\{m \in \mathbb{N}_0 \mid (m+1)_{q_{ii}}(q_{ii}^m q_{ij} q_{ji} - 1) = 0\}$.*
- (2)

$$R_i(M)_j = \begin{cases} \mathbb{k}x_i^*, & \text{if } j = i, \\ \mathbb{k}(\text{ad } x_i)^{-a_{ij}^M}(x_j), & \text{if } j \neq i. \end{cases}$$

Proof. (1) and (2) follows from Lemma 6.11 and definition of reflection immediately. $\square$

Now we choose basis for $R_i(M)_j$. We define

$$y_j := \begin{cases} x_i^*, & \text{if } j = i, \\ (\text{ad } x_i)^{-a_{ij}^M}(x_j), & \text{if } j \neq i. \end{cases}$$

Here we choose x_i^* such that $x_i^*(x_i) = 1$.

Lemma 6.14. *With the above notation and assume that M admits the i -th reflection. Then*

$$\delta(y_j) = g_j g_i^{-a_{ij}^M} \otimes y_j \quad (6.12)$$

for all $j \in \mathbb{I}$. Meanwhile, for any $g \in G$,

$$g \triangleright y_j = \begin{cases} \chi_i(g)^{-1} \Phi(g, g_i^{-1}, g_i)^{-1} y_i, & \text{if } j = i, \\ \prod_{k=0}^{-a_{ij}^M-1} \tilde{\Phi}_g(g_i, g_i^k g_j) \chi_i^{-a_{ij}}(g) \chi_j(g) y_j, & \text{if } j \neq i. \end{cases} \quad (6.13)$$

Proof. For $j \neq i$, since G is abelian, the statement follows from Lemma 6.10.

For $j = i$, note that the evaluation map $\text{ev} : \mathbb{k} y_i \otimes \mathbb{k} x_i \rightarrow \mathbb{k}$ is given by

$$\text{ev}(y_i \otimes x_i) = y_i(\alpha(g_i)) x_i.$$

In $(\mathbb{k}G, \Phi)$, $\alpha(g) = 1$ for all $g \in G$. Hence $\text{ev}(y_i \otimes x_i) = 1$. Since ev is a morphism in ${}^G_C \mathcal{YD}^\Phi$, it follows that

$$g \triangleright \text{ev}(y_i \otimes x_i) = 1.$$

This implies $\delta(y_i) = g_i^{-1} \otimes y_i$. Meanwhile, we have

$$\tilde{\Phi}_g(g_i^{-1}, g_i) < h \triangleright y_i, h \triangleright x_i > = 1.$$

We denote $g \triangleright y_i = \zeta(g) y_i$, then $\zeta(g) \chi(g) \tilde{\Phi}_g(g_i^{-1}, g_i) = 1$. Hence

$$\zeta(g) = \chi_i(g)^{-1} \tilde{\Phi}_g^{-1}(g_i^{-1}, g_i) = \chi_i(g)^{-1} \Phi(g, g_i^{-1}, g_i)^{-1}.$$

For further statement, Equation (6.8) implies

$$\chi_i(g) \chi_i(g^{-1}) \tilde{\Phi}_{g_i}^{-1}(g, g^{-1}) = 1,$$

we obtain that

$$\zeta(g) = \tilde{\Phi}_{g_i}^{-1}(g, g^{-1}) \tilde{\Phi}_g^{-1}(g_i^{-1}, g_i) \chi_i(g^{-1}) = \Phi(g_i, g, g^{-1})^{-1} \Phi(g, g_i^{-1}, g_i)^{-1} \chi_i(g^{-1}).$$

$\square$

Lemma 6.15. *Let $i \in \mathbb{I}$ and M admits the i -th reflection.*

(1) *Assume the braiding matrix of $R_i(M)$ is $(q'_{jk})_{j,k \in \mathbb{I}}$, then for $j, k \neq i$,*

$$q'_{ii} = q_{ii},$$

$$q'_{ji} = \prod_{l=0}^{-a_{ij}^M-1} \tilde{\Phi}_{g_i^{-1}}^{-1}(g_j g_i^l, g_i) \zeta(g_j) \zeta^{-a_{ij}^M}(g_i),$$

$$q'_{ij} = \prod_{l=0}^{-a_{ij}^M-1} \tilde{\Phi}_{g_i^{-1}}^{-1}(g_i, g_i^l g_j) \chi_i^{-a_{ij}^M}(g_i^{-1}) \chi_j(g_i^{-1}),$$

$$q'_{jk} = \prod_{l=0}^{-a_{ik}^M-1} \tilde{\Phi}_{g_i^{-1}}^{-1}(g_i, g_i^l g_k) \prod_{m=0}^{-a_{ij}^M-1} (\tilde{\Phi}_{g_i}^{-1}(g_j g_i^m, g_i) \chi_i(g_i)^{-a_{ij}^M} \chi_i(g_j))^{-a_{ik}^M} \tilde{\Phi}_{g_k}^{-1}(g_j g_i^m, g_i) \chi_k(g_i)^{-a_{ij}^M} \chi_k(g_j).$$

(2) The above elements $(q'_{jk})_{j,k \in \mathbb{I}}$ satisfy the following relations.

$$\begin{aligned} q'_{jj} &= q_{jj} q_{ij}^{-a_{ij}^M} q_{ji}^{-a_{ij}^M} (a_{ij}^M)^2, \\ q'_{ij} q'_{ji} &= q_{ij}^{-1} q_{ji}^{-1} q_{ii}^{2a_{ij}^M}, \\ q'_{jk} q'_{jk} &= (q_{jk} q_{kj}) (q_{ik} q_{ki})^{-a_{ij}^M} (q_{ij} q_{ji})^{-a_{ik}^M} q_{ii}^{2a_{ij}^M a_{ik}^M}. \end{aligned}$$

Proof. The proof proceeds by direct computation:

(1) We know that $q'_{ii} = \zeta(g_i^{-1})$, while

$$\zeta(g_i^{-1}) = \Phi(g_i, g_i^{-1}, g_i)^{-1} \Phi(g_i^{-1}, g_i^{-1}, g_i)^{-1} \chi_i(g_i) = q_{ii}.$$

For $j, k \neq i$,

$$\begin{aligned} q'_{ji} &= \zeta(g_i^{-a_{ij}^M} g_j) \stackrel{(6.7)}{=} \prod_{l=0}^{-a_{ij}^M-1} \tilde{\Phi}_{g_i^{-1}}^{-1}(g_j g_i^l, g_i) \zeta(g_j) \zeta^{-a_{ij}^M}(g_i) \\ q'_{ij} &\stackrel{(6.13)}{=} \prod_{l=0}^{-a_{ij}^M-1} \tilde{\Phi}_{g_i^{-1}}^{-1}(g_i, g_i^l g_j) \chi_i^{-a_{ij}^M}(g_i^{-1}) \chi_j(g_i^{-1}) \\ q'_{jk} &\stackrel{(6.13)}{=} \prod_{l=0}^{-a_{ik}^M-1} \tilde{\Phi}_{g_i^{-a_{ij}^M} g_j}^{-1}(g_i, g_i^l g_k) \chi_i^{-a_{ik}^M}(g_i^{-a_{ij}^M} g_j) \chi_k(g_i^{-a_{ij}^M} g_j) \\ &\stackrel{(6.7)}{=} \prod_{l=0}^{-a_{ik}^M-1} \tilde{\Phi}_{g_i^{-a_{ij}^M} g_j}^{-1}(g_i, g_i^l g_k) \prod_{m=0}^{-a_{ij}^M-1} (\tilde{\Phi}_{g_i}^{-1}(g_j g_i^m, g_i) \chi_i(g_i)^{-a_{ij}^M} \chi_j(g_j))^{-a_{ik}^M} \tilde{\Phi}_{g_k}^{-1}(g_j g_i^m, g_i) \chi_k(g_i)^{-a_{ij}^M} \chi_k(g_j) \end{aligned}$$

(2) Since $a_{ii}^M = 2$, it is straightforward to see $q'_{ii} = q_{ii}^{1-2a_{ii}^M+(a_{ii}^M)^2}$. For $j \neq i$,

$$\begin{aligned} q'_{jj} &= \prod_{l=0}^{-a_{ij}^M-1} \tilde{\Phi}_{g_i^{-a_{ij}^M} g_j}^{-1}(g_i, g_i^l g_j) \prod_{m=0}^{-a_{ij}^M-1} (\tilde{\Phi}_{g_i}^{-1}(g_j g_i^m, g_i) \chi_i(g_i)^{-a_{ij}^M} \chi_i(g_j))^{-a_{ij}^M} \tilde{\Phi}_{g_j}^{-1}(g_j g_i^m, g_i) \chi_j(g_i)^{-a_{ij}^M} \chi_j(g_j) \\ &= q_{jj} q_{ij}^{-a_{ij}^M} q_{ji}^{-a_{ij}^M} (a_{ij}^M)^2, \end{aligned}$$

and

$$\begin{aligned} q'_{ij} q'_{ji} &= \prod_{l=0}^{-a_{ij}^M-1} \tilde{\Phi}_{g_i^{-1}}^{-1}(g_i, g_i^l g_j) \chi_i^{-a_{ij}^M}(g_i^{-1}) \chi_j(g_i^{-1}) \prod_{l=0}^{-a_{ij}^M-1} \tilde{\Phi}_{g_i^{-1}}^{-1}(g_j g_i^l, g_i) \zeta(g_j) \zeta^{-a_{ij}^M}(g_i) \\ &= \chi_i^{-a_{ij}^M}(g_i^{-1}) \chi_j(g_i^{-1}) \zeta(g_j) \zeta^{-a_{ij}^M}(g_i) \\ &= \chi_i^{-a_{ij}^M}(g_i^{-1}) \chi_j(g_i^{-1}) \chi_i(g_j)^{-1} \Phi(g_j, g_i^{-1}, g_i)^{-1} (\chi_i(g_i)^{-1} \Phi(g_i, g_i^{-1}, g_i)^{-1})^{-a_{ij}^M} \\ &= \chi_i^{-a_{ij}^M}(g_i^{-1}) \chi_j(g_i^{-1}) \chi_i(g_j)^{-1} \Phi(g_j, g_i^{-1}, g_i)^{-1} (\chi_i(g_i)^{-1} \chi_i(g_i)^{-1} \chi_i(g_i^{-1})^{-1})^{-a_{ij}^M} \\ &= \chi_j(g_i^{-1}) \chi_i(g_j)^{-1} \Phi(g_j, g_i^{-1}, g_i)^{-1} \chi_i(g_i)^{2a_{ij}^M} \\ &= \chi_j(g_i)^{-1} \chi_i(g_j)^{-1} \chi_i(g_i)^{2a_{ij}^M} = q_{ij}^{-1} q_{ji}^{-1} q_{ii}^{2a_{ij}^M}. \end{aligned}$$

Now for $j, k \neq i$,

$$\begin{aligned}
q'_{jk}q'_{kj} &= \prod_{l=0}^{-a_{ik}^M-1} \tilde{\Phi}_{g_i^{-a_{ij}^M}g_j}^{-a_{ij}^M-1}(g_i, g_i^l g_k) \prod_{m=0}^{-a_{ij}^M-1} (\tilde{\Phi}_{g_i}^{-1}(g_j g_i^m, g_i) q_{ii}^{-a_{ij}^M} q_{ji})^{-a_{ik}^M} \tilde{\Phi}_{g_k}^{-1}(g_j g_i^m, g_i) q_{ik}^{-a_{ij}^M} q_{jk} \\
&\quad \prod_{l=0}^{-a_{ij}^M-1} \tilde{\Phi}_{g_i^{-a_{ik}^M}g_k}^{-a_{ik}^M-1}(g_i, g_i^l g_j) \prod_{m=0}^{-a_{ik}^M-1} (\tilde{\Phi}_{g_i}^{-1}(g_k g_i^m, g_i) q_{ii}^{-a_{ik}^M} q_{ki})^{-a_{ij}^M} \tilde{\Phi}_{g_j}^{-1}(g_k g_i^m, g_i) q_{ij}^{-a_{ik}^M} q_{kj} \\
&= (q_{ii}^{-a_{ij}^M} q_{ji})^{-a_{ik}^M} q_{ik}^{-a_{ij}^M} q_{jk} (q_{ii}^{-a_{ik}^M} q_{ki})^{-a_{ij}^M} q_{ij}^{-a_{ik}^M} q_{kj} \\
&= (q_{jk} q_{kj}) (q_{ik} q_{ki})^{-a_{ij}^M} (q_{ij} q_{ji})^{-a_{ik}^M} q_{ii}^{2a_{ij}^M a_{ik}^M}.
\end{aligned}$$

□

Definition 6.16. Let (V, c) be a finite-dimensional braided vector space of diagonal type and let $q = (q_{ij})_{i,j \in \mathbb{I}}$ be a braiding matrix of V . We say that (V, c) is of **Cartan type**, if there exists a Cartan matrix $A = (a_{ij})_{i,j \in \mathbb{I}}$ such that for all $i, j \in \mathbb{I}$,

$$q_{ij}q_{ji} = q_{ii}^{a_{ij}}, \quad \text{where } 0 \leq -a_{ij} < \text{ord}(q_{ii}) \text{ if } i \neq j. \quad (6.14)$$

Lemma 6.17. Assume that $M = (M_1, \dots, M_\theta) \in {}_{\mathbb{G}}\mathcal{YD}^\Phi$ is of Cartan type with Cartan matrix A . Then M admits the i -th reflection for all $i \in \mathbb{I}$ and $a_{ij}^M = a_{ij}$ for all $i, j \in \mathbb{I}$.

Proof. Let $i \in \mathbb{I}$. Recall that if $j \neq i$, $a_{ij}^M = -\min\{m \in \mathbb{N}_0 \mid (m+1)_{q_{ii}}(q_{ii}^m q_{ij} q_{ji} - 1) = 0\}$. If $a_{ij} = 0$, then it follows from the definition that $a_{ij}^M = 0$. Now if $-a_{ij} \geq 1$, it is obvious that $-a_{ij}^M \leq -a_{ij}$. Suppose $-a_{ij}^M < -a_{ij}$. Since $q_{ij}q_{ji} = q_{ii}^{a_{ij}}$ and $0 \leq -a_{ij} < \text{ord}(q_{ii})$, the condition $-a_{ij}^M < -a_{ij}$ implies $q_{ii}^{-a_{ij}^M} q_{ij} q_{ji} - 1 = q_{ii}^{-a_{ij}^M} q_{ii}^{a_{ij}} - 1 \neq 0$. Therefore, to satisfy the minimality condition, we must have:

$$(-a_{ij}^M + 1)_{q_{ii}} = 0 \implies q_{ii}^{-a_{ij}^M+1} = 1.$$

Consequently, we obtain $(q_{ij}q_{ji})^{-a_{ij}^M+1} = (q_{ii}^{a_{ij}})^{-a_{ij}^M+1} = 1$ for any $j \in \mathbb{I} \setminus \{i\}$. This forces $-a_{ij}^M + 1$ to be a multiple of $\text{ord}(q_{ii})$. Since $\text{ord}(q_{ii}) > -a_{ij}$, it follows that:

$$-a_{ij}^M + 1 \geq \text{ord}(q_{ii}) > -a_{ij}.$$

This contradicts the assumption $-a_{ij}^M < -a_{ij}$. Thus, $a_{ij}^M = a_{ij}$ for all $i, j \in \mathbb{I}$. □

Lemma 6.18. Assume that $M = (M_1, \dots, M_\theta) \in {}_{\mathbb{G}}\mathcal{YD}^\Phi$ is of Cartan type with Cartan matrix A . Let $i \in \mathbb{I}$, then the following hold.

(1) The labels of the generalized Dynkin diagram of $R_i(M)$ are

$$q'_{jj} = q_{jj}, \quad q'_{jk}q'_{kj} = q_{jk}q_{kj}$$

for all $j, k \in \mathbb{I}$.

(2) The tuple $R_i(M)$ is of Cartan type with Cartan matrix A .

Proof. Since M admits i -th reflection for all $i \in \mathbb{I}$ by Lemma 6.17, the tuple $R_i(M)$ is well-defined. Since $q_{ii}^{a_{ij}^M} = q_{ij}q_{ji}$, the first statement follows from Lemma 6.15(2). While the second statement follows from the definition of Cartan type. □

Theorem 6.19. Assume that $M = (M_1, \dots, M_\theta) \in {}^G\mathcal{YD}^\Phi$ is of Cartan type. Then the following hold.

- (1) M admits all reflections.
- (2) Let $M' \in \mathcal{F}_\theta(M)$. Then M' is of Cartan type.
- (3) The Cartan graph of M is standard, that is $A^{[X]} = A^{[Y]}$ for all $[X], [Y] \in \mathcal{X}$.
- (4) The Cartan graph of M is finite if and only if A^M is of finite type.
- (5) The Nichols algebra $\mathcal{B}(M)$ is finite-dimensional if and only if A^M is of finite type and for all $i \in \mathbb{I}$ there exists $m \in \mathbb{N}_0$ such that $(m+1)_{q_{ii}} = 0$.

Proof. Statements (1), (2), and (3) follow from repeated applications of Lemma 6.17. Since $\mathcal{G}(M)$ is standard, it follows from the definition of Weyl groupoid that

$$\mathcal{W}(\mathcal{G}(M)) = W(A^M),$$

where $W(A)$ denotes the Weyl group of the Cartan matrix A^M . Hence, $\mathcal{G}(M)$ is finite if and only if $W(A)$ is finite, which is equivalent to A^M being of finite type. This proves (4).

Now, the finite-dimensionality of $\mathcal{B}(M)$ is equivalent to the conditions that M admits all reflections, $\mathcal{G}(M)$ is finite and $\mathcal{B}(P_i)$ is finite-dimensional for all $P \in \mathcal{F}(M)$ by Theorem 5.9. While $\mathcal{B}(P_i)$ is finite-dimensional for all $P \in \mathcal{F}(M)$, $i \in \mathbb{I}$ if and only if $(m+1)_{q_{ii}} = 0$ for some $m \in \mathbb{N}_0$. This proves (5). $\square$

6.3 On Nichols algebras of diagonal type having isomorphic Cartan graphs

In this subsection, we will show that the semi-Cartan graph of Nichols algebra of diagonal type over coquasi-Hopf algebra is always isomorphic to one over Hopf algebra.

Without loss of generality, we assume that

$$G = Z_{m_1} \times Z_{m_2} \cdots \times Z_{m_n} = \langle g_1 \rangle \times \langle g_2 \rangle \cdots \times \langle g_n \rangle$$

with $m_i \mid m_{i+1}$ for $1 \leq i \leq n-1$. We denote it by G for simplicity. Denote $\mathcal{A}$ the set of all $\mathbb{N}_0$ -sequences:

$$\underline{a} = (a_1, a_2, \dots, a_l, \dots, a_n, a_{12}, a_{13}, \dots, a_{st}, \dots, a_{n-1,n}, a_{123}, \dots, a_{rst}, \dots, a_{n-2,n-1,n})$$

such that $0 \leq a_l < m_l$, $0 \leq a_{st} < (m_s, m_t)$, $0 \leq a_{rst} < (m_r, m_s, m_t)$, with indices ordered lexicographically. Let g_i be the generator of Z_{m_i} , $1 \leq i \leq n$. For each $\underline{a} \in \mathcal{A}$, define

$$\Phi_{\underline{a}} \left(g_1^{i_1} \cdots g_n^{i_n}, g_1^{j_1} \cdots g_n^{j_n}, g_1^{k_1} \cdots g_n^{k_n} \right) = \prod_{l=1}^n \zeta_{m_l}^{a_l i_l \left[\frac{j_l + k_l}{m_l} \right]} \prod_{1 \leq s < t \leq n} \zeta_{m_s}^{a_{st} k_s \left[\frac{i_t + j_t}{m_t} \right]} \prod_{1 \leq r < s < t \leq n} \zeta_{(m_r, m_s, m_t)}^{a_{rst} k_r j_s i_t} \quad (6.15)$$

Here ζ_m denotes a primitive m -th root of unity.

Lemma 6.20. [16 Proposition 3.8] $\{\omega_{\underline{a}} \mid \underline{a} \in \mathcal{A}\}$ forms a complete set of representatives of the normalized 3-cocycles over G up to 3-cohomology.

Lemma 6.21. [16, Corollary 3.13, Proposition 3.14] Let G be a finite abelian group and ω a normalized 3-cocycle over G as in (6.15), then

$$\begin{aligned} \Phi_{\underline{a}} \text{ is an abelian cocycle} &\iff a_{rst} = 0 \text{ for all } 1 \leq r < s < t \leq n \\ &\iff \text{Every Yetter-Drinfeld module over } (\mathbb{k}G, \Phi_{\underline{a}}) \text{ is of diagonal type.} \end{aligned}$$

From now on, we always assume $\Phi_{\underline{a}}$ is an abelian 3-cocycle. Let

$$G \cong Z_{m_1^2} \times Z_{m_2^2} \times \cdots \times Z_{m_n^2} = \langle g_1 \rangle \times \langle g_2 \rangle \times \cdots \times \langle g_n \rangle.$$

There is a group surjection $\pi : \mathbb{G} \longrightarrow G$, $\mathfrak{g}_i \mapsto g_i$, for $1 \leq i \leq n$. Note that the projection π admits a section $\iota : G \rightarrow \mathbb{G}$, $g_i \mapsto \mathfrak{g}_i$. We may pull back the 3-cocycles over G and obtain a 3-cocycles over $\mathbb{G}$. That is, the map

$$\pi^*(\Phi) : \mathbb{G} \times \mathbb{G} \times \mathbb{G} \longrightarrow \mathbb{k}^\times, \quad (g, h, k) \mapsto \Phi(\pi(g), \pi(h), \pi(k)).$$

is a 3-cocycle over $\mathbb{G}$. In fact, $\pi^*(\Phi)$ is a coboundary. If we define $J_{\underline{a}}$ via

$$J_{\underline{a}} : \mathbb{G} \times \mathbb{G} \longrightarrow \mathbb{k}^\times, \quad (\mathfrak{g}_1^{x_1} \cdots \mathfrak{g}_n^{x_n}, \mathfrak{g}_1^{y_1} \cdots \mathfrak{g}_n^{y_n}) \mapsto \prod_{l=1}^n \zeta_{m_l^2}^{a_l x_l (y_l - y'_l)} \prod_{1 \leq s < t \leq n} \zeta_{m_s m_t}^{a_{st} x_t (y_s - y'_s)},$$

then $\pi^*(\Phi_{\underline{a}}) = \partial(J_{\underline{a}})$.

In [16, Section 4], the authors provided a method to classify such Nichols algebras.

$$\begin{array}{ccc} \text{Given } \mathcal{B}(M) \in {}_{\mathbb{G}}\mathcal{YD}^{\Phi_{\underline{a}}} & & \mathcal{B}(M')^{J_{\underline{a}}} \cong \mathcal{B}(M) \in {}_{\mathbb{G}}\mathcal{YD}^{\Phi_{\underline{a}}} \\ \downarrow & & \uparrow \text{If [16, (4.8)] holds} \\ \mathcal{B}(M) \cong \mathcal{B}(\tilde{M}) \in {}_{\mathbb{G}}\mathcal{YD}^{\pi^*(\Phi_{\underline{a}})} & & \mathcal{B}(M')^{J_{\underline{a}}} \in {}_{\mathbb{G}}\mathcal{YD}^{\partial(J_{\underline{a}})} \\ \downarrow & & \uparrow \\ \mathcal{B}(\tilde{M}) \text{ is twisted equivalent to } \mathcal{B}(\tilde{M})^{J_{\underline{a}}^{-1}} \in {}_{\mathbb{G}}\mathcal{YD} & & \text{Given Nichols algebra } \mathcal{B}(M') \in {}_{\mathbb{G}}\mathcal{YD} \end{array}$$

Figure I

Figure II

Lemma 6.22. [16, Lemma 4.6, Proposition 4.7] *The triple $(M, \rho, \blacktriangleright)$, denoted simply by $\tilde{M}$ is an object in ${}_{\mathbb{G}}\mathcal{YD}^{\pi^*(\Phi_{\underline{a}})}$ in the following way:*

$$\begin{aligned} \rho : M_i &\rightarrow \mathbb{k}\mathbb{G} \otimes M_i, & \rho &= (\iota \otimes \text{id})\delta, \\ \blacktriangleright : \mathbb{k}\mathbb{G} \otimes M_i &\rightarrow M_i, & g \blacktriangleright x_i &= \pi(g) \triangleright x_i. \end{aligned} \tag{6.16}$$

The Nichols algebra $\mathcal{B}(\tilde{M}) \in {}_{\mathbb{G}}\mathcal{YD}^{\pi^*(\Phi_{\underline{a}})}$ is isomorphic to $\mathcal{B}(M)$. Moreover, $\mathcal{B}(\tilde{M})$ is twist equivalent to a Nichols algebra over Hopf algebra.

Proposition 6.23. *If $M \in {}_{\mathbb{G}}\mathcal{YD}^{\Phi_{\underline{a}}}$ admits all reflections, then $\tilde{M} \in {}_{\mathbb{G}}\mathcal{YD}^{\pi^*(\Phi_{\underline{a}})}$ admits all reflections. Furthermore, $\mathcal{G}(M)$ is isomorphic to $\mathcal{G}(\tilde{M})$.*

Proof. If we denote $\iota(g_i) = \mathfrak{g}_i$, it is straightforward to see that for all $i, j \in \mathbb{I}$,

$$\begin{aligned} \rho(x_i) &= \mathfrak{g}_i \otimes x_i, \\ \mathfrak{g}_j \blacktriangleright x_i &= \pi(\mathfrak{g}_j) \triangleright x_i = q_{ij} x_i. \end{aligned}$$

That is, M and $\tilde{M}$ has the same braiding matrix. Recall in ${}_{\mathbb{G}}\mathcal{YD}^{\Phi}$,

$$\begin{aligned} \Delta_{\mathcal{B}(M)}(\text{ad}_{\mathcal{B}(M)}(x_i)^n(x_j)) &= \text{ad}(x_i)^n(x_j) \otimes 1 + 1 \otimes \text{ad}(x_i)^n(x_j) \\ &+ \sum_{k=1}^n \binom{n}{k} \prod_{q_{ii} \mid l=n-k}^{n-1} (1 - q_{ii}^l q_{ij} q_{ji}) \prod_{q=1}^k \Phi_{\underline{a}}(g_i^{q-1}, g_i, g_i^{n-q} g_j) x_i^k \otimes \text{ad}(x_i)^{n-k}(x_j). \end{aligned}$$

While, in ${}_{\mathbb{G}}\mathcal{YD}^{\pi^*(\Phi_{\underline{a}})}$,

$$\begin{aligned} \Delta_{\mathcal{B}(\tilde{M})}(\text{ad}_{\mathcal{B}(\tilde{M})}(x_i)^n(x_j)) &= \text{ad}(x_i)^n(x_j) \otimes 1 + 1 \otimes \text{ad}(x_i)^n(x_j) \\ &+ \sum_{k=1}^n \binom{n}{k} \prod_{q_{ii} \mid l=n-k}^{n-1} (1 - q_{ii}^l q_{ij} q_{ji}) \prod_{q=1}^k \pi^*(\Phi_{\underline{a}})(g_i^{q-1}, \mathfrak{g}_i, \mathfrak{g}_i^{n-q} \mathfrak{g}_j) x_i^k \otimes \text{ad}(x_i)^{n-k}(x_j). \end{aligned}$$

Therefore, it is straightforward to see that M admits the i -th reflection if and only if $\tilde{M}$ does, and $a_{ij}^M = a_{ij}^{\tilde{M}}$.

Now we consider the following tuples: $R_i(M) \in {}_{\mathbb{G}}\mathcal{YD}^{\Phi}$ and $R_i(\tilde{M}) \in {}_{\mathbb{G}}\mathcal{YD}^{\pi^*(\Phi_{\underline{a}})}$. We denote their braiding matrix by $(q'_{ij})_{i,j \in \mathbb{I}}$ and $(q''_{ij})_{i,j \in \mathbb{I}}$ respectively. By Lemma 6.15,

$$\begin{aligned} q'_{jj} &= q''_{jj}, \\ q'_{jk}q'_{kj} &= q''_{jk}q''_{kj} \end{aligned}$$

for all $j, k \in \mathbb{I}$.

By (6.11), for any $j \in \mathbb{I}$, we know that $R_i(M)$ admits j -th reflection if and only if there is an integer $m_{jk}^{R_j(M)}$ for each $k \neq j$, such that

$$(m_{jk}^{R_j(M)} + 1)_{q'_{jj}} ((q'_{jj})^{m_{jk}^{R_j(M)}} q'_{jk}q'_{kj} - 1) = 0.$$

Based on the preceding analysis, this is equivalent to saying that $R_i(\tilde{M})$ admits j -th reflection. Furthermore, $a_{jk}^{R_i(M)} = a_{jk}^{R_i(\tilde{M})}$. By iterated reflections, we deduce that $\tilde{M} \in {}_{\mathbb{G}}\mathcal{YD}^{\pi^*(\Phi_{\underline{a}})}$ admits all reflections.

Now we denote $\mathcal{X} = \{[X] \mid X \in \mathcal{F}_{\theta}(M)\}$ and $\mathcal{Y} = \{[Y] \mid Y \in \mathcal{F}_{\theta}(\tilde{M})\}$. Define

$$\gamma : \mathcal{X} \rightarrow \mathcal{Y}, [X] \mapsto [\tilde{X}],$$

where $\tilde{X}$ is given by (6.16). It is immediate that

$$(\text{id}_{\mathbb{I}}, \gamma) : \mathcal{G}(M) \longrightarrow \mathcal{G}(\tilde{M})$$

is an isomorphism of Cartan graphs. □

Theorem 6.24. *Let $M = (M_1, \dots, M_{\theta}) \in {}_{\mathbb{G}}\mathcal{YD}^{\Phi_{\underline{a}}}$ be a Yetter-Drinfeld module of diagonal type. Assume M admits all reflections, then $\tilde{M}^{I_{\underline{a}}^{-1}} \in {}_{\mathbb{G}}\mathcal{YD}$ admits all reflections. Furthermore $\mathcal{G}(M)$ is isomorphic to $\mathcal{G}(\tilde{M}^{I_{\underline{a}}^{-1}})$.*

Proof. It follows from Proposition 6.23 and Corollary 6.6. □

Corollary 6.25. *The following two sets of isomorphism classes of Cartan graphs are equal:*

$$\begin{aligned} \mathcal{P} &= \left\{ [\mathcal{G}(M)] \left| \begin{array}{l} M = (M_1, \dots, M_{\theta}) \in {}_{\mathbb{G}}\mathcal{YD}^{\Phi}, \theta \geq 2, M \text{ admits all reflections,} \\ G \text{ is an abelian group, } \Phi \text{ is an abelian 3-cocycle} \end{array} \right. \right\}, \\ \mathcal{Q} &= \left\{ [\mathcal{G}(V)] \left| \begin{array}{l} M = (V_1, \dots, V_{\tau}) \in {}_{\mathbb{G}}\mathcal{YD}, \tau \geq 2, V \text{ admits all reflections} \\ G \text{ is an abelian group,} \end{array} \right. \right\}. \end{aligned}$$

Proof. Since Hopf algebras are always coquasi-Hopf algebras, we have $\mathcal{Q} \subseteq \mathcal{P}$. The reverse inclusion follows from Theorem 6.24. □

It is well-known that any Nichols algebras of diagonal type over Hopf algebras can define generalized Dynkin diagram. The definition now can be established in our settings.

Definition 6.26. *Let V be a θ -dimensional braided vector space of diagonal type. Let $(x_i)_{i \in \mathbb{I}}$ be a basis of V and let $q = (q_{ij})_{i,j \in \mathbb{I}}$ be a matrix of non-zero scalars in $\mathbb{k}$ with $c_{V,V}(x_i \otimes x_j) = q_{ij}x_j \otimes x_i$ for all $i, j \in \mathbb{I}$. The **generalized Dynkin diagram** of V with respect to the basis $(x_i)_{i \in \mathbb{I}}$ is a labeled graph $\mathcal{D}$ with θ vertices. The vertices of $\mathcal{D}$ correspond to the integers $i \in \mathbb{I}$ and are labeled by q_{ii} . For any*

$1 \leq i < j \leq \theta$, there is an edge between vertex i and vertex j if and only if $q_{ij}q_{ji} \neq 1$. In this case, $q_{ij}q_{ji}$ is the label of this edge.

The Dynkin diagram of $M = (M_1, \dots, M_\theta) \in \mathcal{F}_\theta$ with $\dim M_j = 1$ for all $j \in \mathbb{I}$ is the Dynkin diagram of the braided vector space $M_1 \oplus \dots \oplus M_\theta$ with respect to the basis $(x_i)_{i \in \mathbb{I}}$, where $0 \neq x_i \in M_i$ for all i .

Corollary 6.27. Let $M = (M_1, \dots, M_\theta) \in {}^G_C \mathcal{YD}^{\Phi_a}$ be a Yetter-Drinfeld module of diagonal type. Then $\mathcal{B}(M)$ has a finite root system if and only if the generalized Dynkin diagram of M lies in the classification result of [12].

Proof. Recall the fact that any Nichols algebra of diagonal type over a Hopf algebra has a finite root system if and only if the generalized Dynkin diagram lies in the table in [12]. Now our result follows from Theorem 6.24. $\square$

The last result is a quasi-version of recent results [6].

Corollary 6.28. Let $M = (M_1, \dots, M_\theta) \in {}^G_C \mathcal{YD}^{\Phi_a}$ be a Yetter-Drinfeld module of diagonal type, then $\mathcal{B}(M)$ is finite GK-dimensional if and only if the corresponding root system is finite.

Proof. If $\mathcal{B}(M)$ is finite GK-dimensional, but the corresponding root system $\Delta_{X \in \mathcal{X}}^{X, \text{re}}$ is infinite. Then it is obvious that $\mathcal{G}(M)$ is a infinite Cartan graph. By Theorem 6.24, we can construct a Yetter-Drinfeld module $M' \in {}^G_C \mathcal{YD}$, such that $\mathcal{B}(M) \cong \mathcal{B}(M')$ and $\mathcal{G}(M) \cong \mathcal{G}(M')$. Then $\mathcal{G}(M)$ is infinite Cartan graph and $\mathcal{B}(M')$ is infinite GK-dimensional by [6]. This contradicts to $\mathcal{B}(M)$ is finite GK-dimensional. Conversely, one can prove that $\mathcal{B}(M)$ has a finite root system implies $\mathcal{B}(M)$ is finite GK-dimensional. $\square$

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